# HYDRODYNAMICS OF FRACTAL AGGREGATES

*M.Tech thesis submitted*
*in partial fulfilment of the requirements*
*for the degree of*

**Master of Technology**

By

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# Approval Sheet

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# Abstract

Condensed matter often occurs in the form of aggregates, exhibiting fractal structures in a statistical sense. Their hydrodynamic behavior plays an important role in various applications in industry and environment. Under the circumstances of rare nuclear accidents of high consequence, which might lead to the release of radioactive aerosols into environment~~,~~ the hydrodynamics of the aerosols needs to be studied. Nuclear aerosols coagulate to form aggregates which are fractal in nature. Different shapes arising as a result of aggregation leads to behaviour that cannot be predicted based on regular spherical structure. A quantity known as the mobility diameter has been a topic of interest over the past several decades, especially from the perspective of eliciting the scaling laws with respect to aggregate size and fractal dimensions. While previous studies have demonstrated proportionality of this quantity with respect to the radius of gyration, a systematic elucidation of this quantity across a range fractal dimensions and its relationship with fluid penetration into the aggregate has been lacking. This thesis attempts to fill this gap. For this purpose, well-known Stokesian dynamic approach, which takes into account all orders of monomer-monomer interactions to construct the resistance matrix of a rigid cluster has been employed in this thesis. Fractal dimensions ranging from 1.76 to 3, and aggregates containing 10 to 10000 particles are considered in this study. Both deterministic fractal (Vicsek) and statistical fractals formed by various methods such as Monte Carlo, DLA and DLCCA are considered. While confirming the asymptotic cluster-size independence of the hydrodynamic ratio (mobility radius divided by the radius of gyration), the study reveals monotonically increasing dependencies of hydrodynamic ratio on increasing fractal dimension. To explore the underlying cause of this monotonic dependence, correlation with fluid penetration into the cluster is sought. The degree of penetration of the velocity of the fluid into the aggregate core is quantified via the concept of “penetration depth” ($\delta$). Most interesting, the penetration depth is found to have a monotonic dependence on fractal dimension and the aggregation process. It is found that higher hydrodynamic ratio is associated with lower penetration depth. This finding providing a deeper connection between the occurrence of fluid-aggregate resistance, vis-a-vis-velocity penetration. On the whole, the present study has established the concept of hydrodynamic ratio on a quantitative footing over a wide range of fractal dimensions through a single mathematical framework and developed a deeper understanding of these relationships by linking them to velocity penetration.

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# Chapter 1

# Introduction

Understanding the mechanisms of transport of particles suspended in a fluid is of crucial importance in aerosol systems, geosciences and chemical technology. In the former case, the increasing concern on the role played by particulate matter on human health, bioaerosols in the context of COVID-19 and global warming is driving large number of studies on characterizing these particulates. Similarly, whether it be the transport of contaminants in ground water, or the constant need to improve the performance of technologies using disperse phase systems in manufacturing, material processing and catalysis requires increasingly sophisticated approaches to elucidate fluid-particle interactions. The occurrence of particles in the form of aggregates is a special case of interest. Small particles from various processes stick together to form aggregates. When particles are dispersed in a continuous phase, the particles are more prone to aggregation as they have more surface area. Aggregates are generally porous and are extensively used for catalysis. Quite often such aggregates are formed as a part of evolution of the colloid phase due to the irreversible sticking from the collisions to which the dispersed particles are subjected to due to thermal agitations. They are formed by coagulation of organic or inorganic particles in industries or laboratories, and can also be generated by simulations. The aggregates such as soot particles are ubiquitous in aerosols formed from combustion sources. They are important in areas of research where flocculation is observed with the structures appear rarified. These objects were initially termed as “random” or “fluffy”.

Euclidean geometry which is used to describe objects in a well-defined space cannot explain real objects with a more complex structure. Mandelbrot coined the term ‘fractal’(from the Latin word *fractus* meaning fragmented) observed natural patterns – such as the shapes of coastlines, mountains and clouds where the components of Euclidean geometry with the lines, curves and other shapes are inept. He stated that “*Fractals are geometrical shapes that, contrary to those of Euclid, are not regular at all. First, they are irregular all over. Secondly, they have the same degree of irregularity on all scales. A fractal object looks the same when examined from far away or nearby-it is self-similar. As you approach it, however, you find that small pieces of the whole, which seemed from a distance to be formless blobs, become well-defined objects whose shape is roughly that of the previously examined whole”*.

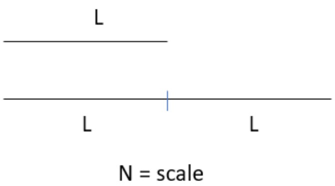
Figure 1-1 shows a more complex structure but when looked upon closely, it is observed that they follow a more similar pattern at different scales. The individual part of the cauliflower looks same but on a smaller scale. This property is self-similarity and fractals should exhibit this property on all scales. Natural fractals are not necessarily self-similar at all much smaller scales.

*Figure 1-1:Romanesco cauliflower - A natural fractal [from wired.com]*

Fractal dimension (or *fractional dimension*) is an index to characterize fractal patterns that quantifies their complexity. Definition of fractal dimension as given by Mandelbrot et. al (1983) "*Fractal dimension is a statistical quantity that gives us an indication of how completely a fractal appears to fill space, as one zooms down to finer and finer scales*."

Dimension of an object gives us the number of parameters needed to define the position of the object. A point has zero dimension, a line has one dimension, a square has two dimensions and a cube has three dimensions.

1. Line: When a straight line is scaled by a factor of 2, we have two copies of the same line.




2. Square/Rectangle: When a rectangle of side length L and breadth *b* is scaled by a factor of 2, we have four copies of the original rectangle. The same applies for the case of a square.

N = $\text{scale}^2$

3. Cube: When cube is scaled by the same factor of 2, we have 8 copies of the smaller cube. N = $scale^3$

For geometric shapes, we notice that N = $scale^D$. Now we take a look at a famous fractal called Sierpinski triangle.

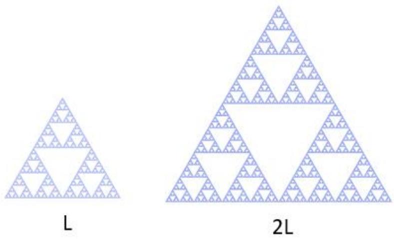



*Figure 1-2: Sierpinski triangle. Before scaling(left) and after scaling by 2(right)*

This is obtained when a single triangle is cut of several triangular parts and when looked closely, the structure repeats itself. If we scale the smaller triangle by a factor of 2 (length and height), we get three copies of the smaller triangle. Dimension of a Sierpinski triangle is determined as $2^D$ = 3 and D = $\frac{\ln 3}{\ln 2} \approx 1.585$. This dimension is not an integer. This fractional dimension is the reason for the name fractals. In fact, fractals are shapes with a non-integer dimension.

While Mandelbrot's study established the concept of fractals on a mathematical footing, their application to natural objects is rather statistical than deterministic in nature. The experimental investigation of Forrest and Witten (1979) using metal vapour aerosols first established the statistical nature of the concept of fractality of the aggregates. Since then, the fractal approach formed an inalienable part of study of material aggregates both from a morphological and kinetic perspective. Fractal aggregates are built with identical primary particles, where the primary particles are considered to be hard spheres.

## 1.1 Hydrodynamic studies

Two important parameters can characterize these fractals: Radius of gyration ($R_g$) and fractal dimension ($D_f$). Fractal aggregates always follow the scaling relationship given by Forest and Witten (1981):

$$N = k_f \left(\frac{R_g}{a}\right)^{D_f} \qquad 1.1$$

where a is the primary particle radius and $k_f$ is the fractal pre-factor.

Radius of gyration is obtained using ${R_g}^2 = \frac{1}{N}\sum_{i=1}^{N}(r_i - r_{cm})^2$ where $r_i$ indicates the position of sphere *i* and $r_{cm}$ indicates the center of mass of N spheres.

Mobility radius, $R_m$ can be obtained by using $R_m = \frac{F_{ag}}{6\pi\mu\widehat{U}}$ where $F_{ag}$ and $U_{ag}$ are the force acting on the aggregate and the velocity of the aggregate. In some context in aerosol, mobility relationship expressed via dynamic shape factor [DeCarlo (2004)].

Historically Kirkwood-Riseman(1948) theory and porous sphere models have been used to understand the hydrodynamic properties of fractal aggregates. Kirkwood-Riseman(1948) theory is based on the principle that velocity of the individual particles in a cluster is affected by the presence of other particles in the same cluster, collectively causing disturbances in the velocity of the fluid both within and outside of the cluster. The disturbances were initially accounted by the simplest Oseen(1927) Tensor while other researchers worked with Rotne-Prager-Yamakawa(RPY) tensor [Rotne and Prager (1969) & Yamakawa (1970)].

Approach of the porous sphere model is to consider the fractal aggregate to be a porous sphere in the low Reynolds number region. Stokes equation governs the flow surrounding the fractal aggregate while the flow within the cluster is governed by either Darcy's law or Brinkman equation [Nyugen (2007)]. Darcy's equation is given by $-\frac{\mu}{\kappa}\mathrm{u} = \nabla p$. The flow equations in the interior regions contains the parameter, permeability ($\kappa$) which is a function of volume fraction of particles [Brinkman (1947)]. This permeability is critical to find the drag force acting on an aggregate. Even though Darcy's model is simple, it fails to match the flow inside and outside the porous sphere [Starov et al. (2001)]. Brinkman(1949) equation is given by $\mu\nabla^2\mathrm{u} - \frac{\mu}{\kappa}\mathrm{u} = \nabla p$. Brinkman's modification provides an equation that interpolates between the Stokes equation and Darcy's law. While Brinkman's equation is better suited to handle homogeneous permeable structures as shown by Beavers and Joseph(1967), fractal aggregates have a structure with varied permeability. Models used by Veerapaneni and Wiesner(1996) considered permeability to be proportional to the radial distance (increases radially) from the center of mass to the outer radius. Permeability in the models of Kim and Yuan(2005) increases quadratically from the center ($\kappa \propto r^2$). Both Kirkwood-Riseman theory and porous sphere models are useful to study the hydrodynamics related to fractal aggregates.

Using a mean-field approximation, Wiltzus(1987) predicted $R_m \propto R_g$ and was the first to show the proportionality based on experiments for large aggregates. Static and dynamic light scattering experiments were used to obtain $R_g$ and $R_m$ for aggregates dispersed in liquid media. Subsequently, De Gennes et al.(1979) showed that this proportionality of $R_m$ and $R_g$ is applicable in the Kirkwood-Riseman approximation. From a practical perspective, the ratio $\frac{R_m}{R_g}$ referred to as the hydrodynamic ratio is the quantity of crucial interest. It is apparent from the linear correlations between the two that the hydrodynamic ratio obtained by the application of Kirkwood-Riseman theory is a function of fractal dimension only [Van Sarloos (1987)], regardless of the cluster mass, N. However, this is an asymptotic result valid for large clusters, and the work of Chen et al.(1988) with DLA and bond percolation aggregates points at the dependence of the hydrodynamic ratio with N. Specifically, $\frac{R_m}{R_g}$ increases for DLA as N increases and then reaches an asymptotic value. However, there appears to be no universality of trend as evidenced by the study of Warren(1994), which showed an increase of $\frac{R_m}{R_g}$ with N for DLA for which the density is the highest in the centre, whereas a decreasing trend leading to quick saturation, was observed for DLCCA aggregates. But they all suggest the values are constant since the changes are very small. In other words, there appears to be certain nonuniversality with respect to the transient behavior of the hydrodynamic ratio, being specifically dependent upon the type of the aggregate.

Gmachowski(1996) with porous sphere model using a modified Brinkman equation shows that the hydrodynamic ratio increases as the fractal dimension increases and reaches a maximum value when $D_f = 3$. Similar conclusions about the dependency with fractal dimension were reached by numerical simulations of Fellay et al.(2013), Chopard et al.(2006) and Lattuada et al.(2010).

While variabilities with different methods and theories, varieties of findings in previous works could also be because of different approaches by different researchers. This brings a need in examining the factors across a wide range of $D_f$ and N and fractal types using a single methodological framework. In view of this, it desired to examine further bridge across wide types of fractals.

The response of aggregates to the flow of fluids, characterized by their hydrodynamic resistance is a fundamental aspect that has received significant attention in the past several decades. In catalysis it is important to understand how the reactants distribute within the fractal aggregates; in aerosols, it is crucial to establish laws for settling rates and other transport characteristics. While historically several mean field approaches have been attempted to describe the inherent many body interactions within the cluster, the most satisfactory approach is provided by solving the Navier-Stokes equation by accounting for fluid-particle interaction of individual monomers. Furthermore, creeping flow approximations described by Stokes equation can be used for colloidal aggregates since the particle size is very small leading to creeping flow conditions. The computational cost related to most of the methods involving hydrodynamic interactions is usually high which leads us to Stokesian Dynamics. It offers a relationship between the hydrodynamic forces acting on the particles and its velocities. The relation between the geometric and hydrodynamic parameter for different fractal dimensions and cluster size forms the crux of the study.

### 1.2 Permeability studies

The basic distinguishing feature of hydrodynamics of aggregates from that of spherical objects is the question of fluid penetration into the interior regions of the objects. It is known that fluid penetrates into the aggregate, but the extent of penetration for different fractals can be used understand further about such structures. Porous sphere models used permeability to obtain hydrodynamic ratio modelling the aggregate to be a permeable sphere. Similar study can be conducted with Stokesian dynamics by the velocity changes within the structure. Kim et al. (2002) used Stokesian Dynamics to obtain fluid velocity indirectly by the use of "tracer" particle flowing through the aggregate such that it does not overlap any of the constituent particles. Average velocity was obtained in three different directions and the results obtained shows excellent agreement with Brinkman(1948) equation in conjunction with Davies(1953) permeability expression for the case of low fractal dimensions. Also, Davies permeability expression gives velocity matching with Kim et al.(2002) result near the ends of the cluster. This was justified by Kim et al.(2000) suggesting that internally fluid penetration will be minimum as volume fraction is large for $D_f < 3$.

From the Brinkman extension of the mean-field theory[Van Sarloos (1987)], the collective effect of the monomers within the aggregate is to introduce velocity screening, which implies an exponential form for the decay of its profile as one moves towards the centre from the surface. Debye et al.(1948) uses a 'shielding length' to study the exponential decay of velocity for the case of polymer beads. Hwang et al.(2000) obtained 'penetration depth' for measuring the distance that the fluid can penetrate into a porous medium. Similarly, we introduce another "penetration depth" based on the different flow profiles across fractals. The same parameter derived from other permeability models suggest that fluid penetration varies with the size of the cluster and decreases as the size increases.

### 1.3 Motivation

Nuclear reactors are designed to be safe through an approach known as Defense-in -Depth (D-i-D) involving various safety features. As a result, the occurrence of nuclear reactor accidents are extremely rare events. However, if it takes place for reasons beyond D-i-D, a severe accident can be of very high consequence from environmental and health risk perspectives. Release of nuclear aerosols into the atmosphere can have various health, environmental and geopolitical implications [Dickinson (2014)]. These aerosols are irregular in structure and cannot be modelled as spherical particles. To address these situations, equivalent spherical descriptions are used wherein the effects of irregularity are clubbed into shape factors. For example, for a fractal particle the migration velocity induced by external force fields such as gravity or electric field will have a shape factor multiplying the volume (or mass) equivalent diameter of the particle, which conveys how the fractality has modified the drag force over a sphere having the same mass as the aggregate. Such shape and structure effects are crucial for developing predictive models to understand aerosol behavior in containments where they are held-up when an accident occurs and from which they may get released to environment if a breach occurs in the containment. Extent of fractality (compact, rarefied and highly rarefied) would depend on concentration and chemical nature of materials release, and process conditions in the containment and accounting for the fractal nature of aggregates will be a valuable input in aerosol safety codes. These aerosols can get released in environment having either wet or dry conditions. If the aerosols are released under wet conditions, surface area minimization will occur resulting in making the structure close to a spherical one. Whereas under dry conditions, surface area will not be minimized and this is the area that needs to be addressed.

Fractal nature of aggregation is also found with soot formed from combustion processes [Samson et al. (1987)], removal of impurities in the form of slag to obtain clean metal products [Huadong et al. (2009)], nanoparticles added to enhance the fluid properties [Jing et al. (2019)], protein [Feder et al. (1984)] and much more. Fundamental issues of the influence of shape and structure on the dynamics is central to these studies.

*Chapter 2* presents the various aggregation models used for this study such as DLA, Tunable CCA and DLCCA. Finally, the parameters associated to the various fractals generated are summarized.

*Chapter 3* starts with a few methods used to simulate suspensions where lubrication interactions are difficult to incorporate followed by the introduction to Stokesian Dynamics. This starts with the derivation of Oseen Tensor followed by multipole expansion. Then Faxen's law leads us to the force-velocity relationship through the resistance matrix.

*Chapter 4* discusses the Stokesian Dynamics algorithm and drag coefficient results of Durlofsky et al. (1987) are verified. Methodology involved for the case of rigid aggregates are also shown.

*Chapter 5* presents the geometrical characteristics of all the fractals generated and investigates the role of cluster size and the fractal dimension on hydrodynamic ratio. The results obtained are compared with the literature values. Effect of coordination number on the forces experienced by the fractals are also discussed.

*Chapter 6* presents the screening effects observed for different fractal dimensions in the form of the parameter "penetration depth" and its dependency on the number of particles. It is compared with various permeability models where the aggregate is modelled as a porous sphere.

*Chapter 7* introduces the various potential which can be used where the particles move freely under the influence of spring-type forces and DLVO-type forces. Its application is discussed with few test cases.

*Chapter 8* concludes the hydrodynamic results for the case of rigid aggregates.

# Chapter 2

# Fractal Aggregates

## 2.1 Fractal characteristics

Fractal dimension can be obtained by fitting the number of monomers present in shells within various radial distances from the center of each aggregate. However, the sparse distribution of monomers near the end makes it difficult to obtain the required parameters at levels of desired accuracies. Hence, radius of gyration was opted as the most robust size parameter for arriving at the scaling laws.

### 2.1.1 Radius of gyration

One of the parameters to quantify how mass is distributed within an object is the radius of gyration. It is obtained by taking square root of mass of inertia divided by the mass of the object. For rigid bodies, it can be calculated by taking the mass average of square of the distance of the *i*$^{th}$ object from the center of mass $(r_i - r_{cm})^2$ .

$$R_g{}^2 = \frac{\sum_i m_i \, (r_i - r_{cm})^2}{\sum_i m_i} \qquad 2.1$$

Radius of gyration for an aggregate can be explained as the mean square of the distance between the center of spheres and the center of mass of the aggregate.

$$R_g{}^2 = \frac{1}{N}\sum_{i=1}^{N}(r_i - r_{cm})^2 \qquad 2.2$$

where $r_i$ and $r_{cm}$ are the position of the *i*$^{th}$ sphere center and the center of mass containing N spheres. Center of mass is obtained by $r_{cm} = \frac{1}{N}\sum_{i=1}^{N} r_i$. These spheres are assumed to be of the same size and having the same mass. All the spheres considered in this study have unit radius and are separated by an interparticle distance of 2.01.

The radius of gyration can also be given by

$$R_g{}^2 = \frac{1}{N}\sum_{i=1}^{N}(r_i - r_{cm})^2 + a^2 \qquad 2.3$$

This definition is used in generating one of the fractals. Only when the aggregate comprises fewer spheres is there a difference between the two definitions. When N = 1, the gyration radius using equation 2.2 will be zero despite the fact that it will have a finite size.

Hence adding the radius of primary particle in equation 2.3 gives a unit radius. As N is large, the difference between $R_g$ obtained using both equation 2.2 and equation 2.3 is negligible.

### 2.1.2 Hydrodynamic radius

To quantify the hydrodynamic behavior of fractal aggregates, hydrodynamic radius or mobility radius is used. It is the radius of an equivalent sphere that experiences the same drag force when it has the same velocity as that of an aggregate when moving through the same fluid. Drag force experienced by the sphere for creeping flow cases are given by

$$F = 6\pi\mu u R_m \qquad 2.4$$

In case of a fractal aggregate, the hydrodynamic/mobility radius can be obtained by

$$R_m = \frac{F_{ag}}{6\pi\mu\hat{U}} \qquad 2.5$$

where $F_{ag}$ and $\hat{U}$ are the force and velocity of the aggregate with respect to fluid. A unit force is applied to N particles in each direction separately, and the aggregate velocity for each direction is calculated. Aggregate velocity can be obtained by the change in center of mass with time.

$$\hat{U} = \frac{(r_{cm})_{t=1} - (r_{cm})_{t=0}}{\Delta t} \qquad 2.6$$

where $r_{cm} = \sqrt{x_{cm}^2 + y_{cm}^2 + z_{cm}^2}$

It may also be calculated by assigning unit velocity to N particles in the x,y and z directions, and then calculating the total force acting in those directions. The mobility radius is calculated in all three directions and averaged.

### 2.1.3 Scaling law

An aggregate growing in size has a mass '$M$' proportional to the radius '$r$' to the power $D$ (dimensionality factor). D is also called the mass fractal dimension. This is the mass scaling principle concerned with fractal aggregates. In other words, the number of copies of an object is determined by relating it with the scale and the power of the scale [Sorenson (2011)]

$$M(r) \sim r^D \qquad 2.7$$

If this relation is satisfied, the structure is also self-similar. This means that a small part of the structure looks similar to the complete structure.

Equation 2.7 can be written with number of primary particles, N as mass

$$N(r) = cr^D \qquad 2.8$$

The scaling law given by Forrest and Witten (1981) relating N to the radius of gyration

$$N = k_f \left(\frac{R_g}{a}\right)^{D_f} \qquad 2.9$$

where a is the primary particle radius, $k_f$ is the fractal pre-factor, $D_f$ is the fractal dimension and $R_g$ is the radius of gyration. Fractal aggregates in nature or those simulated will be in the range $1.5 \leqslant D_f \leqslant 3$. With increasing fractal dimension, there are more particles inside the same radius. As a result, we may conclude that increasing fractal dimension correlates with aggregate compactness. When the fractal dimension is smaller than 2, the fractals are more transparent and often tend to appear more elongated as well.

*Methods to determine fractal dimension*

1. Clusters of sizes ranging from N=100 to 10000 can be used to calculate the fractal dimension using equation 2.9. To achieve a more precise fractal dimension, this is done for two or more different clusters.
2. *Spherical binning*: Here a single aggregate is considered and the number of spheres within a particular region is used to determine $D_f$ using equation 2.8.

## 2.2 Aggregation models

In this study three dimensional aggregates are composed of N monomeric spherical particle. In experimental systems, such as aerosol aggregations, colloidal clusters, etc., depending on the concentration of monomers, different mechanisms may be governing the process of aggregate formation. One of the mechanisms is found at low monomer concentration where particle – cluster aggregation occurs. An example of this mechanism is Diffusion Limited Aggregation, where monomers diffuse one-by-one to the aggregated cluster and become incorporated in it, resulting in growth of the structure. Witten and Sander introduced Diffusion Limited Aggregation (DLA) that results in producing a complex “dendritic” structure. The DLA model can be used as a basis for aggregates in various diffusion limited processes and colloidal systems where flocs are formed. The other type of aggregation occurs when monomer concentration is higher facilitating formation of multiple clusters which in turn collide and gets aggregated with each other forming larger clusters. The cluster–cluster aggregation model describes porous structure like silica gels and allophane aggregates. Most combustion generated aggregates are said to come under this category.

*2.2.1 DLA*

Diffusion Limited Aggregation (DLA) model was developed by Witten and Sander (1981) that results in producing a complex dendritic structure. Off-lattice aggregates have been focused since the work of Witten and Sander [Meakin (1999)]. Aggregation of smoke particles were initially studied using DLA. A seed particle is placed at the center and cannot move. Another particle is fired from a randomly selected point on the boundary circle. Since simulation of a continuously moving particle is difficult, the new particle is displaced stepwise with a certain distance. If this moving particle goes out of the jump circle, another new particle is fired from a new point on the boundary circle. This particle after a few random walks sticks with the first particle or goes out of the jump circle. This process is repeated several times. In order for the newly fired particle to stick to particles in the cluster, as it gets closer the displacement is altered depending on the position of nearby particles to avoid overlapping. As more and more particles are added, particles stick to the end of the branches and is difficult for them to penetrate and reach the sites near the seed particle.

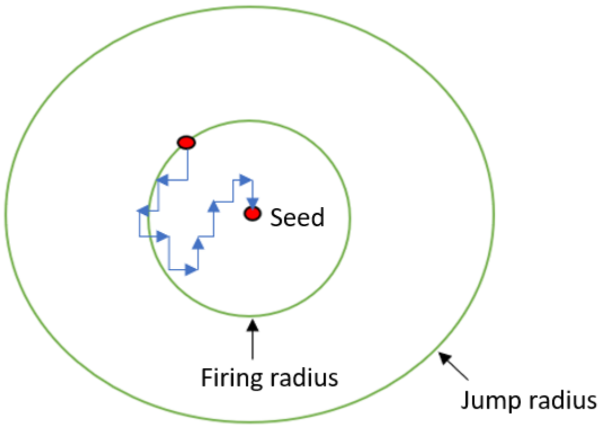



*Figure 2-1:Schematic diagram of a particle having a random walk to the seed particle*

Fig 2-2(left) shows the structure obtain using this DLA model. The determined fractal dimension is 2.47 and these fractals are off-lattice models. Fractal dimension of DLA observed in literature is generally 2.5 [Tolman and Meakin (1988)]. Sticking probability of particle(s) in this model is unity and if it is decreased, we can get Reaction Limited Aggregates (RLA) with fractal dimension going upto 3 [Richardson (1973)].

Conditions and variables used along with the numbers generated:

| | |
|---|---|
| **Seed particle coordinate** | (0,0,0) |
| **Initial firing radius** | 200 |

| | |
|---|---|
| **Jump circle** | 100 x firing radius |
| **Random walk step size** | -*Initial*: 10<br>-Changes based on position of other particles in the cluster |
| **Particle radius** | 1 |
| **Interparticle distance** | 2.01 |
| **Number of monomers** | 100 to 3000,10000(3 different clusters),<br>4000 to 9000(single cluster) |

*Table 2-1: Detail of fractals generated using DLA*

Both the firing radius and jumping radius increases as more particles are being added.

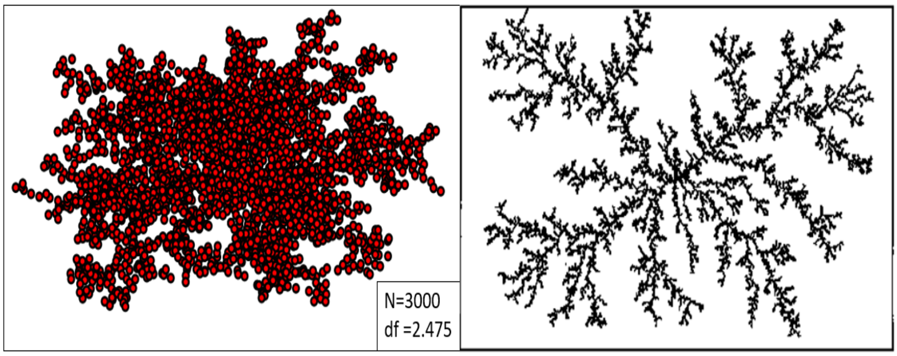



*Figure 2-2:DLA fractals obtained with N=3000 (left). Clear picture of how the structure will look like (Right) [Picture taken from: pas.rochester.edu]*

### *2.2.2 Tunable Monte Carlo fractals*

The coordinates of the fractals are generated using a C++ code using the tunable cluster-cluster aggregation method. The cluster-cluster algorithm begins with clusters generated using Sequential algorithm (SA), which is a tunable particle-cluster aggregation method. In the Sequential Algorithm, spherical particles are added to the cluster individually in an iterative manner. [Filippov (2000)] Each iteration ensures that the scaling law (equation 2.9) is satisfied. Let us say we have N – 1 spheres already and the position of the next sphere is given by

$$(r_N - r_{N-1}^0)^2 = \frac{N^2a^2}{N-1}\left(\frac{N}{k_f}\right)^{\frac{2}{D_f}} - \frac{Na^2}{N-1} - Na^2\left(\frac{N-1}{k_f}\right)^{\frac{2}{D_f}} \qquad 2.10$$

where $r_{N-1}^0$ is the centre of mass of previous N – 1 spheres and a is the radius of the primary spherical particle. The new sphere should be attached such that there is no overlapping at the surface of the sphere and there is at least one contact point with the previous N-1 spheres. This is repeated for the next sphere. Aggregates generated using Sequential algorithm (containing 5-10 spheres) are combined in a pairwise manner such that there is no overlapping and the aggregates have at least one contact point. Two different clusters keep combining at each level and at the last level, there is one large cluster. The scaling law is also satisfied at each level. The number of spheres in the aggregate can be given by

$$N_{agg} = N_{spherule} * 2^{No\ of\ levels} \quad 2.11$$

This CCA ensures placing the center of two clusters to random points and the distance between the centers of these two clusters are given by

$$\Gamma^2 = \frac{a^2(N_1 + N_2)}{N_1 N_2}\left(\frac{N_1 + N_2}{k_f}\right)^{\frac{2}{D_f}} - \frac{N_1 + N_2}{N_2}{R_1}^2 - \frac{N_1 + N_2}{N_1}{R_2}^2 \quad 2.12$$

$N_1$ and $N_2$ are the number of spheres in each cluster and $R_1$ and $R_2$ are the radius of gyration of the two clusters being combined.

It is noted that the aggregate generated using this algorithm is not symmetrical. The overlapping of spheres is avoided by having a minimum interparticle distance r = 2a.

*Fig 2-3* shows the variation of the structure as fractal dimension increases. It can be observed that fractal dimension of 1.76 has a more elongated structure and 2.46 has a more compact form.

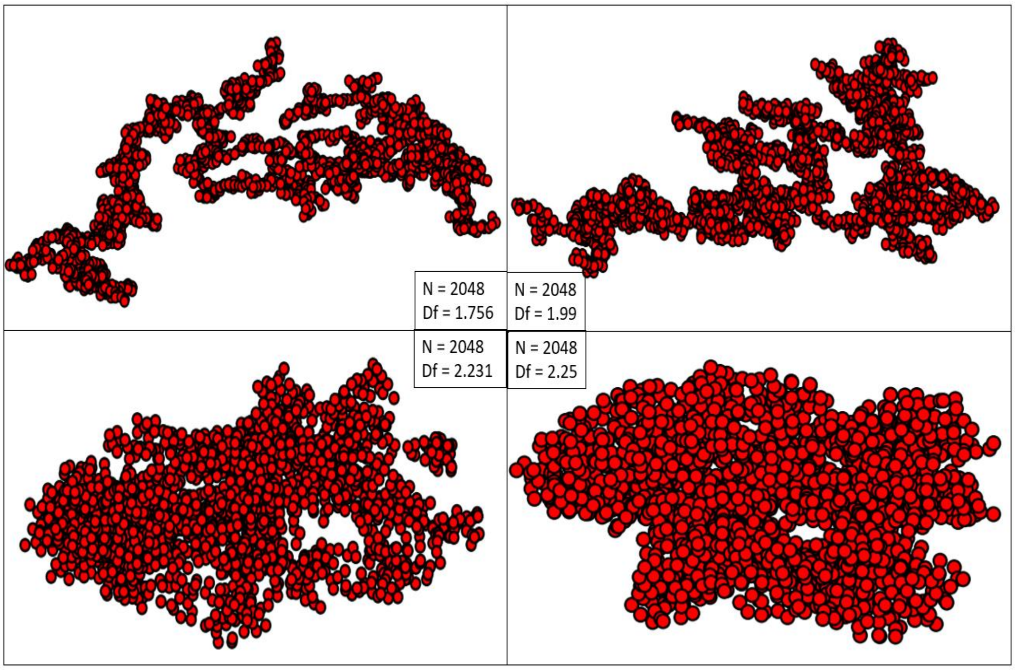



*Figure 2-3:Aggregates generated using this Tunable CCA algorithm. Number of particles in a cluster is 2048. Clusters with fractal dimensions of 1.756(top left), 1.99(top right), 2.23(bottom left) and 2.45(bottom right)*

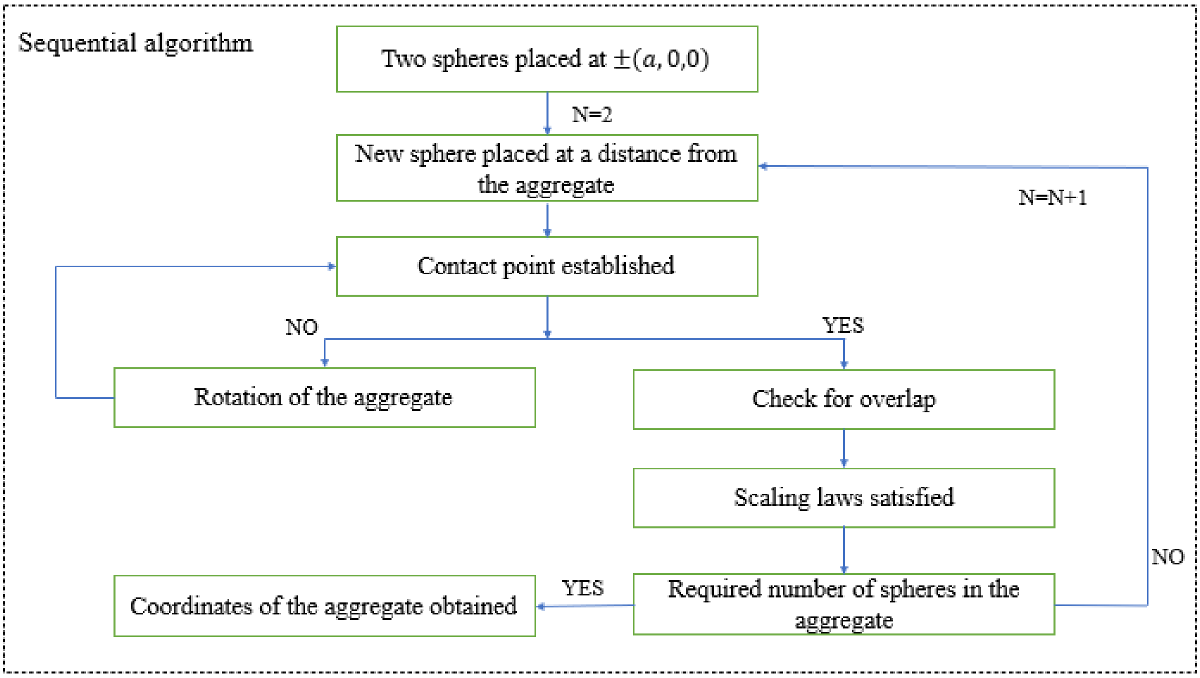



*Figure 2-4:Flowchart for Sequential Algorithm*

The sum of fractal pre-factor and the fractal dimension should be less than 3.5 to avoid the infinite looping of the algorithm. This is because avoiding overlapping of spheres is difficult for closely packed spheres since there are closer neighbors.

| | |
|---|---|
| ***Pre-factor*** | *1* |
| ***Fractal dimension*** | 1.76, 2, 2.25, 2.49 |
| ***Radius of sphere*** | 1 |
| ***Error tolerance for interparticle distance*** | $10^{-3}$ |
| ***Number of spheres in initial cluster using SA*** | 4 to 10 |
| ***Number of levels of coagulation*** | 2 to 10 |
| $$\boldsymbol{N = N_{spherule} * 2^{No\ of\ levels}}$$ | |
| ***Interparticle distance*** | 2 rescaled to 2.01 |

*Table 2-2: Details of fractals generated using Tunable CCA*

Fractal dimension was tuned to get the values mentioned in the table above. Fractals were generated with $R_g$ being determined by equation 2.3 according to the algorithm but we estimate $R_g$ universally for all aggregates with equation 2.2. Hence $D_f$ observed in this study will be slightly different from the table values above.

| | | | | |
|---|---|---|---|---|
| $D_f$(equation 2.3) | 1.76 | 2 | 2.25 | 2.49 |
| $D_f$(equation 2.2) | 1.756 | 1.991 | 2.231 | 2.456 |

*Table 2-3:Fractal dimensions observed by getting radius of gyration by different ways*

### *2.2.3 Diffusion Limited Cluster-Cluster Aggregation*

The DLA model can be used as a basis for aggregates in various diffusion limited processes and colloidal systems where flocs are formed. But since growth of the dendritic structure is over a single immobile particle and only one particle diffuses at any time near the cluster, this is not possible in many colloidal systems. Meakin et al. (1983) introduced a new model where not only single particles but also clusters are allowed to diffuse and stick with other clusters or single particles. Small clusters diffuse and aggregate resulting in the formation of a larger one. In this model, all equal sized spheres are dispersed randomly in a box and they move in random Brownian motion resulting in collision and forming clusters. These clusters diffuse in the box in random direction and form a larger cluster when they stick together with other clusters. This goes on until all the clusters result in one large aggregate. This will be the only remaining aggregate in the box. Aggregates formed using this CCA model follow fractal geometry.

Number of particles N is determined from the particle concentration and the side-length, L of the cubic box where the particles are distributed.

$$N = C * L^3 \qquad 2.13$$

After the particles are randomly distributed in the box, a particle or a cluster is chosen at random. A random number, *x* is generated in the range [0,1] and the particle or cluster starts moving if *x* is less than the probability to move. The probability for a particle or a cluster to move is given by

$$P_{move} = \frac{D_i}{D_{max}} \qquad 2.14$$

Here $D_i = D_1 \times S^{\gamma}$ is the diffusion coefficient of the cluster of i particles.

$D_1 = \frac{k}{f}$ x T is the diffusion coefficient of a single spherical particle and $D_{max}$ is the maximum diffusion coefficient of any cluster. *k* is the Boltzmann constant; *f* is the coefficient of friction and T is the temperature of the system. This Brownian motion of particles is appropriate if the interparticle interactions are smaller than $k_B T$ [Meakin (1999)].

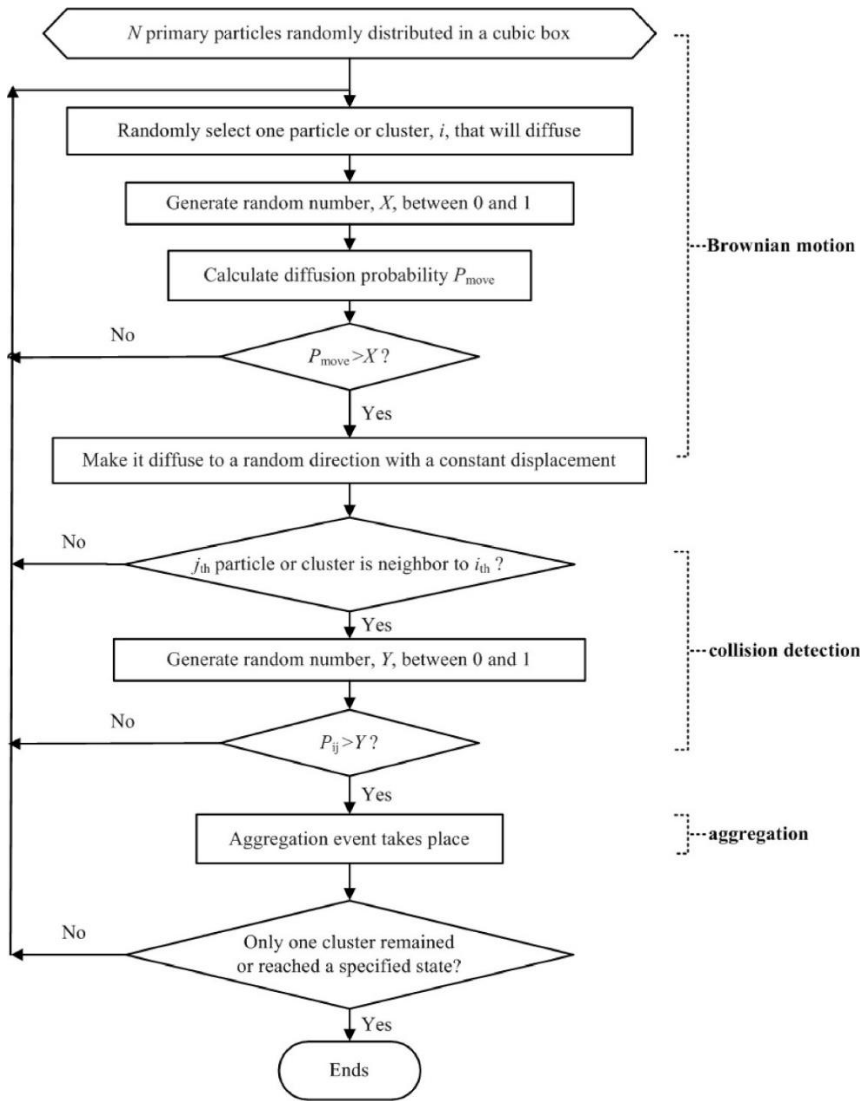



*Figure 2-5:Algorithm flowchart for CCA as given by Xiong et al. (2014)*

$S$ is the mass of the cluster given by the number of spheres in the cluster. $\gamma$ is the diffusivity exponent which can be negative, positive and even zero. Different values of $\gamma$ will lead to different results as it has an effect on the diffusion coefficient. The cluster chosen can make a random walk in any direction and if there are no particles or clusters nearby, another cluster is chosen. If there happens to be a collision between two clusters, the sticking probability of the clusters each containing i and j particles is

$$P_{ij} = P_1 \times (i \times j)^{\sigma} \qquad 2.15$$

$P_1$ is the sticking probability of single particles and $\sigma$ is the sticking probability exponent.

Another random number y is generated in the range [0,1] and if the random number is less than $P_{ij}$, aggregation is considered. These steps are repeated till the box is left with only one aggregate [ Xiong (2014)].

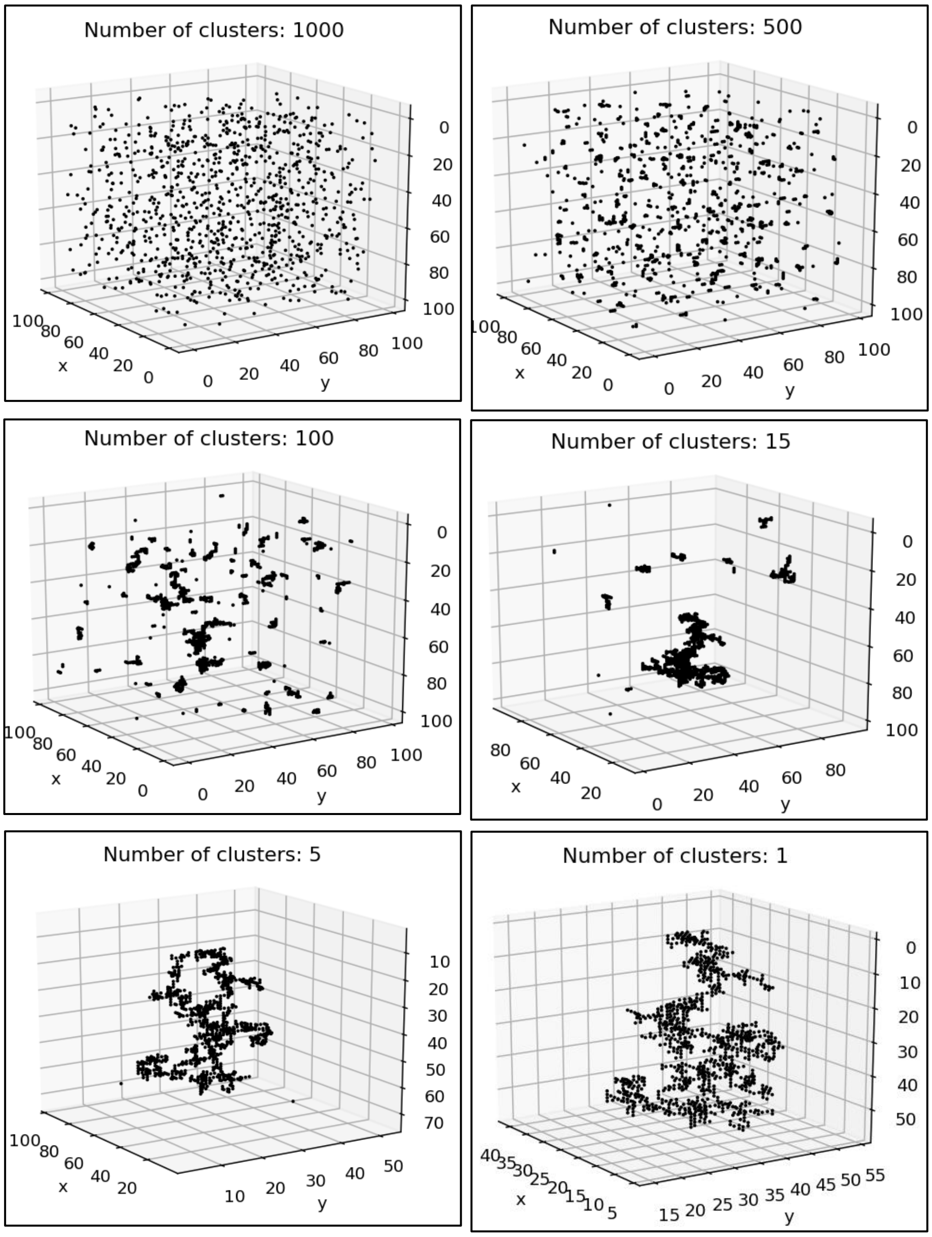



*Figure 2-6:Evolution of an aggregate of 1000 particles in CCA algorithm. σ=0, γ=0 and $P_1$=1. Number of clusters: 1000 (top left), 500 (top right),100 (middle left),15 (middle right),5 (bottom left) and 1 (Bottom right)*

The CCA fractal aggregates are classified into two based on the sticking probability: Diffusion Limited Cluster-Cluster Aggregation (DLCCA) and Reaction Limited Aggregation (RLCCA). The sticking probability is 1 for the case of DLCCA and less than 1 for RLCCA. Sticking probability of less than 1 means that the colliding particles stick less rather they bounce off or repel each other. This results in aggregates with higher fractal dimension because the particles will be able to stick to other clusters deep inside making it more compact. Thus, RLCCA aggregates have higher fractal dimension than the aggregates from DLCCA [Nyugen (2017)].

Fig 2-7 shows the fractals obtained as the initial concentration is changed from C=0.001 to C=0.01. The below table shows how aggregation changes with respect to σ and γ.

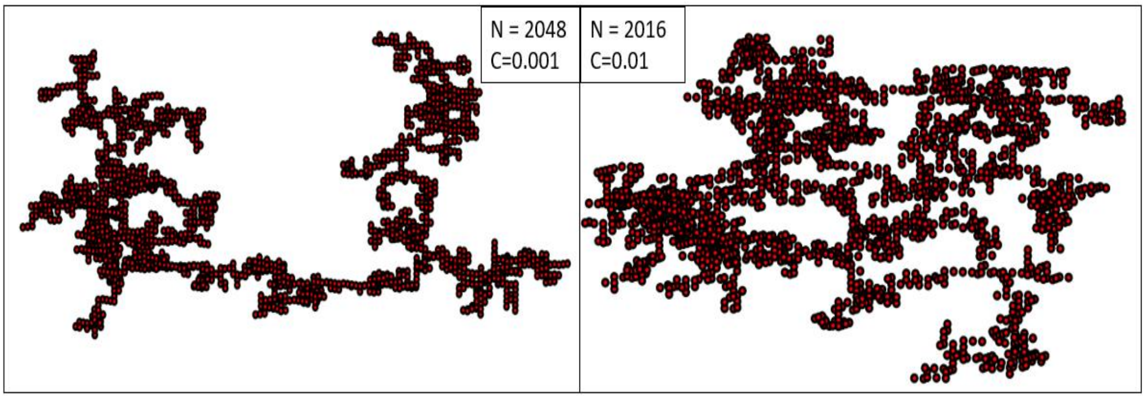



*Figure 2-7:CCA aggregates with initial concentration C=0.001 and N=2048(left) and C=0.01 and N=2016(right)*

| Parameter | Effects |
|---|---|
| $\gamma < 0$ | • Small clusters diffuse faster than the larger ones.<br>• Small clusters have more probability for aggregation.<br>• Absence of single particles or small clusters near the end of the process. |
| $\gamma = 0$ | • Diffusion coefficient of large and small clusters are the same. |
| $\gamma > 0$ | • Large clusters will diffuse faster than the smaller ones.<br>• Except the big cluster, mostly single particles are present. |
| $\sigma < 0$ | • Sticking probability decreases as the cluster size increases.<br>• Aggregate will have a denser structure with low sticking probability. |
| $\sigma = 0$ | • Sticking probability independent of cluster size. |
| $\sigma > 0$ | • Sticking probability increases with increase in the cluster size.<br>• Larger cluster have high chance of aggregation with other clusters. |

*Table 2-4: Effect of the variation of diffusivity exponent and sticking exponent*

The input parameters that can be changed in the user interface and the corresponding values chosen are given below.

| Parameters | C | L | $\sigma$ | $\gamma$ | $P_1$ | T (K) |
|---|---|---|---|---|---|---|
| Values chosen | 0.01 | 22 to 100 | | -0.5, 0, 0.5 | | |
| | 0.001 | 50 to 215 | 0 | 0 | 1 | 298 |

*Table 2-5: Details of fractals generated using CCA algorithm*

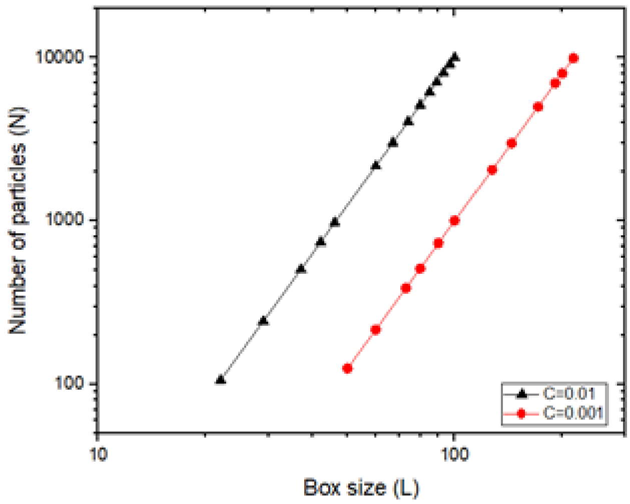



*Figure 2-8a: Box size and the corresponding number of particles for different concentrations C=0.01(▲) and C=0.001(●)*

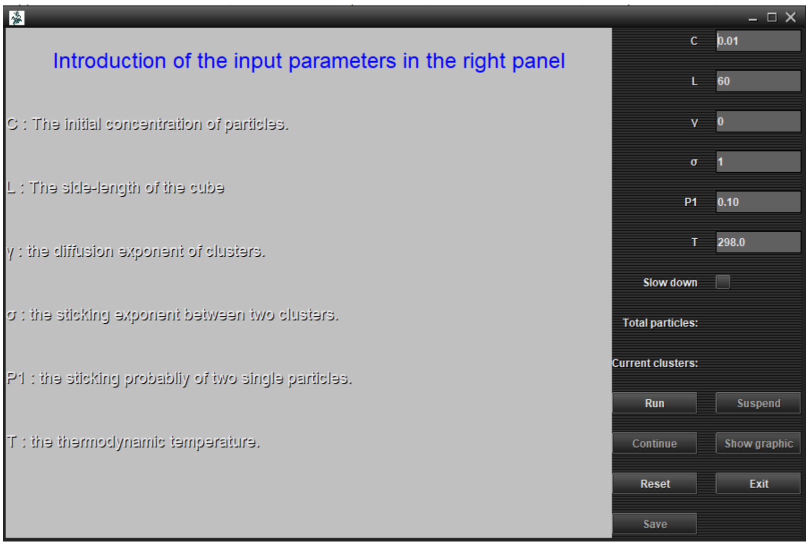



*Figure 2-9:GUI used to generate DLCCA & RLCCA clusters*

Literature suggests DLCCA having a fractal dimension of 1.78 [Meakin and Jullien (1988)] and RLCCA having fractal dimensions of 1.99 [Meakin (1989)].

*2.2.5 Other fractals*

*a. Vicsek*

It is isotropic and at different scales, the fractal looks self-similar. For each iteration *x*, the number of spheres in the fractal will be $7^x$. This is the most common mathematical fractal.

As can be seen from Fig 2-10, at the first iteration there are seven spheres. Six spheres in x,y and z directions above and below the central sphere. This structure is repeated to all direction to get $7^x$ spheres for ‘x’ iterations. All the spheres are of unit radius and inter-particle spacing is 2.01.

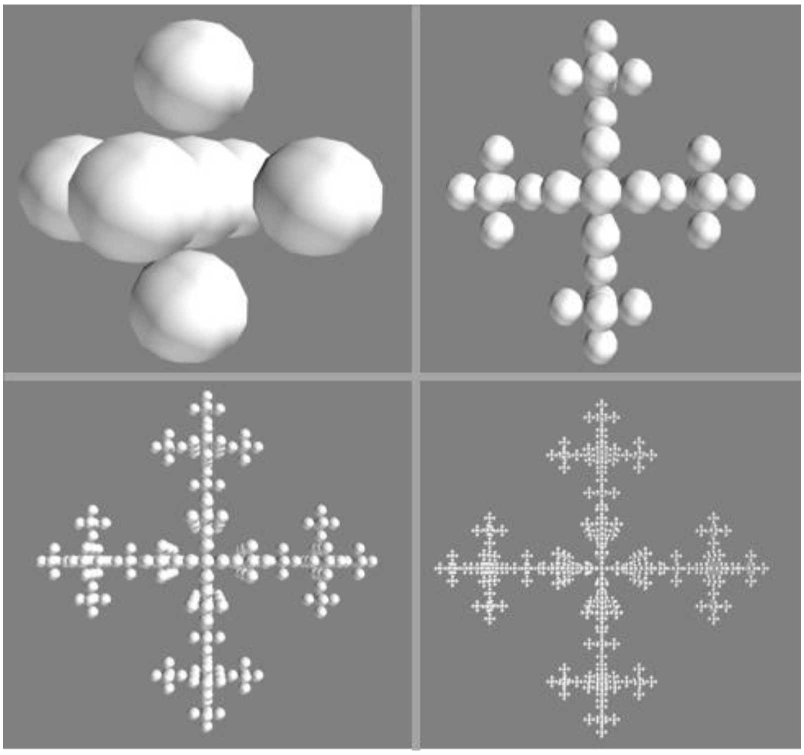


*Figure 2-10: Vicsek fractals ranging from N=7 to N=2401*

*b. Simple cubic unit*

The hydrodynamics of a denser structure with a fractal dimension of 3 needs to be studied. These spherical aggregates have constant interparticle spacing of 2.01 and an average volume fraction of particles as 0.52.

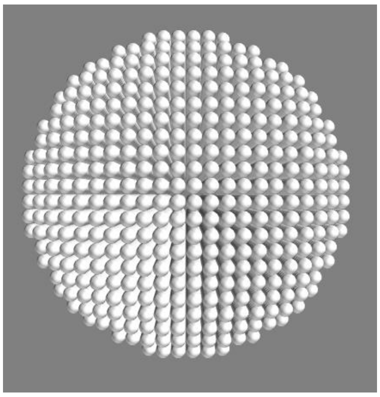


*Figure 2-11:Simple cubic unit with 5520 spheres*

## 2.3 Cluster geometry

First, we establish the adequacy and accuracy of the fractals generated by various methods. This is achieved by estimating the errors on the average fractal dimensions obtained and its comparison with literature values. Fractal dimension of DLA is close to the exact value of 2.5 with 0.65% accuracy. Maximum error in fractal dimension is observed for DLCCA. Both the fractal dimension and pre-factor are obtained using regression of N versus $R_g/a$ in the range of N=20 to N=10000. Tunable CCA has less error because of the fact that the rules for generating the clusters are constrained to satisfy the scaling relations at stages. Pre-factor for Tunable CCA will be unity if the radius of gyration is determined using equation 2.3. All primary particles are of unit radius, *a*.

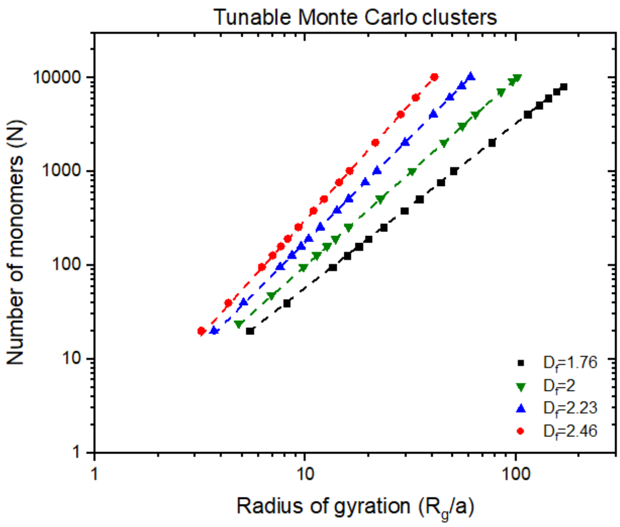



*Figure 2-12: Power law fits for Tunable Monte Carlo Clusters, number of monomers in a fractal, N vs the radius of gyration scaled with particle radius,* $R_g/a$ ■ $- D_f = 1.76$, ▼ $-D_f = 1.99$, ▲ $-D_f = 2.23$, ● $- D_f = 2.46$).

Fig 2-12 shows the variation of radius of gyration varies with the cluster size, N for Tunable Monte Carlo clusters with primary particle radius of unity. The slope and intercept for the log-log plots of N vs $R_g$ gives the fractal dimension and pre-factor respectively. The radius of gyration is obtained for Tunable Monte Carlo clusters, DLA, DLCCA and Simple cubic unit of various cluster sizes N using equation 2.2. Interparticle spacing for all the fractals is 2.01.

The table below shows few values of radius of gyration as N is changing.

| N | 256 | 512 | 1024 | 2048 | 4096 | 10240 |
|---|---|---|---|---|---|---|
| $R_g$ | 9.27 | 12.27 | 16.24 | 21.47 | 28.37 | 41 |

*Table 2-6:Rg vs N for Tunable Monte Carlo clusters (Fractal dimension of 2.46±0.01)*

| Fractal | | Fractal dimension | Prefactor |
|---|---|---|---|
| Tunable Monte Carlo clusters | | 1.76 ± 0.002 | 1.00 ± 0.01 |
| | | 1.99 ± 0.002 | 1.010 ± 0.01 |
| | | 2.23± 0.004 | 1.020 ± 0.005 |
| | | 2.46 ± 0.009 | 1.05 ± 0.01 |
| DLA | | 2.47 ± 0.02 | 0.48 ± 0.03 |
| DLCCA | C = $10^{-3}$ $\gamma = 0$ | 1.84 ± 0.05 | 1.32 ± 0.25 |
| CCA | C = $10^{-2}$ $\gamma = -0.5$ | 2.3 ± 0.05 | 0.37 ± 0.08 |
| | C = $10^{-2}$ $\gamma = 0$ | 2.20 ± 0.05 | 0.58 ±0.09 |
| | C = $10^{-2}$ $\gamma = 0.5$ | 2.12 ± 0.09 | 0.83 ± 0.23 |
| Vicsek | | 1.75 ± 0.01 | 2.27 ± 0.07 |
| Simple Cubic Unit | | 3.01 ± 0.003 | 1.09 ± 0.01 |

*Table 2-7: Fractal dimension and pre-factor (3 significant digits) observed for various fractals*

### *2.3.1 Observations*

1. Simple cubic unit is tightly packed as the number of particles almost scales close to $R^3$.
2. DLCCA/CCA:
   a. As the initial concentration increases, the fractal dimension also increases.
   b. When $\gamma$ varies from -0.5 to 0.5, it can be seen that the fractal dimension is decreasing while the pre-factor increases. This change of $\gamma$ from -0.5 to 0.5 brings about change from small clusters diffusing faster to large clusters diffusing faster.
3. *Fractals with similar fractal dimension but different pre-factor:* The effect of pre-factor for DLA ($D_f = 2.47$ & $k_f = 0.48$) and Tunable Monte Carlo cluster ($D_f = 2.46$ & $k_f = 1.05$) are shown in figure 2-13. DLA having a pre-factor small compared to a Tunable Cluster has particles distributed widely.

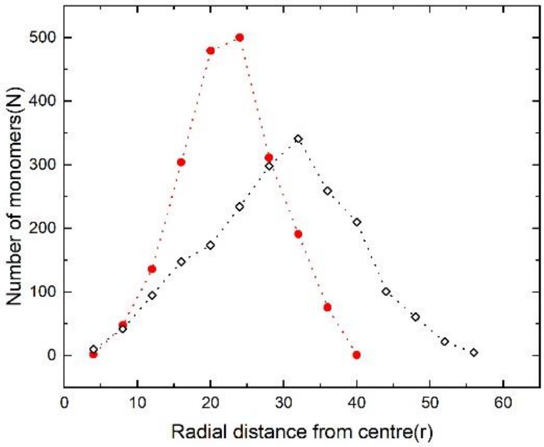



*Figure 2-13: Particle distribution for Tunable CCA(●) and DLA(◇) for N=2000.*

Scaling principle is clearly conserved for all the fractals generated. Further hydrodynamic studies are conducted on these fractals to study the effect of fractal dimension and the number of primary particles in the cluster.

### Appendix 2A: N vs $R_g$ plots

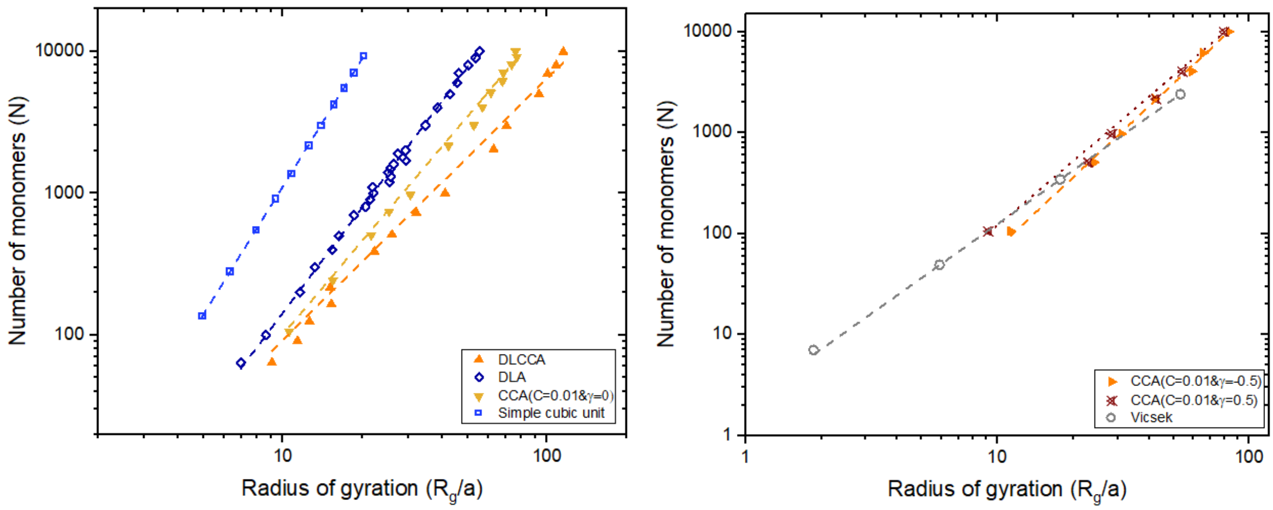



*Figure 2A-1: Power law fits for number of monomers in a fractal, N vs the radius of gyration scaled with particle radius,* $R_g/a$ *for fractals (*□ $-simple\ cubic\ unti(D_f = 3)$, ◇ $-DLA(D_f = 2.47)$,▼ $-CCA(D_f = 2.2)$,▲ $-DLCCA(D_f = 1.84)$, ○ $-Vicsek(D_f = 1.75)$,▶ $-CCA(C = 0.01\&\gamma = -0.5)$, ⊲ $-CCA(C = 0.01\&\gamma = 0.5)$

# Chapter 3

# Hydrodynamic Calculations Using Stokesian Dynamics

### 3.1 Introduction

Particles suspended in a fluid medium occurs commonly in a wide variety of settings. They are found in nature and as well as in industrial processes. The presence of particles in the medium alters the properties of the suspension such as the viscosity, yield and normal stresses. Viscosity, μ is the measure of fluid resistance to flow. In order to measure viscosity of a fluid, it may be placed between parallel plates and sheared. If the fluid is Newtonian, the stress-strain rate relationship would be linear ($\sigma = \mu\dot{\gamma}$) where sigma denotes the stress and $\dot{\gamma}$ is the strain rate.

When particles are suspended in the fluid, this linear relationship cannot be used. With particles, Einstein(1906) shows that effective viscosity, η depends linearly on the volume fraction of the particle, $\phi$:

$$\eta = \left(1 + \frac{5}{2}\phi\right)\mu \qquad 3.1a$$

But this expression is only valid for low volume fractions ($\phi \leq 0.03$) where the velocity field around a sphere is not greatly influenced by other spheres. Effect of force on an individual sphere by the presence of another sphere means that there are hydrodynamic interactions between the spheres. These interactions lead to include an additional term in equation 3.1 proportional to $\emptyset^2$[Batchelor (1977)].

$$\eta = \mu(1 + 2.5\phi + 6.2\phi^2) \qquad 3.1b$$

[Add the Batchelo equation here and his papers reference] There are better models for higher volume fractions. Most commonly used is the Krieger-Dougherty equation:

$$\eta = \mu\left[1 - \frac{\phi}{\phi_m}\right]^{-2\cdot 5\phi_m} \qquad 3.1c$$

where, $\phi_m$ is the maximum volume fraction, chosen to be 0.64 corresponding to random closed packing in the case of hard spheres. As $\phi$ increases to larger values, viscosity becomes more sensitive to volume fraction $\phi$. At low volume fractions, many different models lead us back to Einstein (1906) equation.

When particles of two different sizes are suspended, effective viscosity can be lower than suspensions containing particles of the same size [Larson (1999)] This can be explained the presence of small particles in the space between the larger particles.

It is observed that as the shear stress increases, effective viscosity decreases which means there is shear thinning in the suspension. As the shear stress increases further, effective viscosity goes through a minimum and then increases if the suspension is shear thickening. This is for the case of dense suspensions. When $\phi \geq 0.3$, effective viscosity becomes more sensitive to shear rate. This sensitivity occurs when the increase in shear rate affects the equilibrium for the distribution of interparticle spacing. When the shear rate is increases, the particles are driven closer to each other resulting in the formation of clusters. When such clusters deform, the thin layer separating the particles experience large lubrication forces [Larson (1999)].

Rheology of concentrated suspensions can be too difficult to predict and is a more challenging problem theoretically. Computer simulations are essential to study concentrated suspension.

Early work of suspended particles considered either a very dilute system or a small number of particles immersed in a simple flow. Many body interactions are important as concentration increases. Lubrication interactions become necessary for suspensions of even moderate concentration [Townsend (2017)]

The primary problem in creating an accurate model for suspensions in a low Reynolds number flow is the inclusion of lubrication forces while keeping the computational expense affordable. Lubrication interactions become progressively significant as the particle concentration increases. This is on the grounds that the increased number of particles within the given space brings about a greater number of close interactions. Failure to include lubrication interactions may result in physically unrealistic results [Brickell (2012)]. Few methods used for simulating suspensions are briefly summarised below.

*3.1.1 Collocation technique*

Ganatos et al. (1978) introduces a collocation technique. A minimum of four collocation points will be considered for a sphere and each point has three unknowns. It is made sure that boundary conditions at these points are satisfied by expanding the solution of Stokes equation. More number of collocation points would be needed to satisfy the boundary conditions when two spheres are in close contact with each other. In order to reduce the difficulties involved with the computations, particles with higher order of symmetry are considered [Durlofsky (1987)].

*3.1.2 Boundary integral equation method*

In this method we are solving an integral equation discretised over the particle surfaces *[Weinbaum et al. (1990)]*. However, for particles in close contact, a greater number of surface elements would be required to solve and is computationally expensive. Even with parallel computing, the expenses are hard to overcome for a large system with increased number of particles. This method is especially helpful while dealing with irregular shaped particles. While dealing with particles having some symmetry, integrating over the entire surface of the particles leads to unnecessary expense. [Brickell (2012)]

*3.1.3 Multipole method*

The integral used in the boundary integral method is expanded and we truncate the expansion at the desired accuracy. There is a computational advantage for low order expansions [Cichocki and Jones (2000)]. All the terms of the expansion are necessary to include lubrication interactions [Brickell (2012)]

*3.1.4 Dissipative Particle Dynamics*

Dissipative Particle Dynamics (DPD) can also be used. DPD was developed by Hoogerbrugge and Koelman (1992) to replace Lattice Gas Automata. This method can be used to simulate for larger length and time steps compared to Molecular Dynamics. DPD fails to include lubrication interactions making it less attractive [Brickell (2012)]. Similar to LBM, these interactions can also be included “by-hand”. However, upon addition of pairwise interactions, DPD is slower in comparison with LBM.

*3.1.5 Lattice-Boltzmann method*

LBM is also an alternative to conventional numerical methods for the Navier–Stokes equation. This method was developed by Ladd (1994). The Navier–Stokes solver needs iterative procedures to obtain a converged solution. But LBM does not solve the Navier–Stokes equation directly, thus making it computationally fast. This method originated from Lattice Gas Automata (LGA). Fluid molecules are represented with Boolean elements in LGA. However, large-scale simulations using LGA is not efficient. LGA was replaced to eliminate white noise and lack of Galilean invariance. Lattice-Boltzmann method solves the Boltzmann equation. The usage of particle distribution function is employed in LBM. Two steps are involved: the propagation step and the collision step. LBM can also be applied to infinite domains [Chen & Doolen (1998)]. Main disadvantage of this method is it does not include lubrication interactions directly. However, lubrication interactions can be added “by-hand” but making the process computationally expensive.

The velocity disturbance caused by the particles decays very slowly at $r^{-1}$. Greater force is needed to push two particles as the distance between the particle surface decreases. Lubrication interactions are not included directly in any of the above methods. Along with the lubrication interactions, truncating the multipole expansion and computing the mobility tensor which is an important element to describing the hydrodynamic interactions has made Stokesian Dynamics a better option. This is not possible in any of the other methods mentioned. Hydrodynamic close-range interactions which are accounted in the resistance matrix has been formulated by Jeffrey & Onishi (1984).

## 3.2 Stokesian Dynamics

Stokesian Dynamics was at first proposed by Durlofsky, Brady and Bossis(1987) and is based on the pairwise multipole expansions and Faxen's laws to determine the relation between the forces and velocities acting on individual particles as a result of hydrodynamic interactions. Hydrodynamic interactions are divided into far-field contributions and near-field contributions. At the point when two particles suspended in a viscous fluid approach one another, the movement of each particle is impacted by the other, even without non-hydrodynamic interparticle interactions. Motion of one particle generates a velocity field which influences the motion on another particle.

The far-field interactions are calculated by expanding the moments of the force density about every particle. Lubrication interactions separately account for the near-field contributions.

### *3.2.1 Governing equations for Stokes flow*

The flow of a Newtonian fluid is given by the Navier-Stokes equation:

$$\rho\left(\frac{\partial \mathrm{u}}{\partial \mathrm{t}}+(\mathrm{u}.\nabla)\mathrm{u}\right)=-\nabla \mathrm{P}+\mu\nabla^2\mathrm{u} \qquad 3.2a$$

and the continuity equation

$$\nabla.(\rho u)=0 \qquad 3.2b$$

where u is the fluid velocity, $\rho$ is the density, P is the fluid pressure and $\mu$ is the fluid viscosity.

Introducing non-dimensional variables

$$\tilde{r}=\frac{r}{L},\tilde{u}=\frac{u}{U},\tilde{t}=\frac{t}{t_p},\tilde{P}=\frac{PL}{\mu U},\tilde{\nabla}=L\nabla,\tilde{\nabla}^2=L^2\nabla^2$$

Where L is the characteristic length and U is the characteristic velocity.

$$\frac{\rho uL}{\mu}\left(\frac{\partial\tilde{u}}{\partial\tilde{t}}+\left(\tilde{u}\cdot\tilde{\nabla}\right)\tilde{u}\right)=-\tilde{\nabla}\tilde{P}+\tilde{\nabla}^2\tilde{u} \qquad 3.3$$

Reynolds number associated, $Re=\frac{\rho uL}{\mu}$

When Re<<1, we obtain the Stokes equation

$$0 = -\tilde{\nabla}\tilde{P} + \tilde{\nabla}^2\tilde{u} \qquad 3.4$$

Stokes equation for a Newtonian and incompressible fluid becomes

$$-\nabla P = \mu\nabla^2 u \qquad 3.5a$$

and the continuity equation becomes

$$\nabla . u = 0 \qquad 3.5b$$

From this, it is clear that there is no time dependence in Stokes flow. Boundary conditions such as the particle configuration, external forces and torque helps to determine the instantaneous flow. If the boundary conditions change with respect to time, then the solution will be a function of time. They can be therefore termed as quasi-static flow [Deen (1998)]. Stokes equations are linear and allow us to use the principle of superposition [Frungieri (2018)].

### 3.3 Formulation

Bossis and Brady (1987) considered two distinct methods for ascertaining particle interactions. When the translational and rotational velocities on particles are known, the corresponding forces and torques acting can be determined by using the Grand Resistance Matrix, R. Similarly, when the forces and torques acting on particles are known, translational and rotational velocities can be determined from the Grand Mobility Matrix, M. This problem formulation is called the Mobility problem. These quantities are linearly related for Stokes flow problems (Re << 1). Green's functions can also be used to solve the far-field interactions because of this linearity.

Resistance problem is given by

$$\begin{pmatrix} F^\alpha \\ L^\alpha \\ S^\alpha \end{pmatrix} = R \begin{pmatrix} U^\alpha - u^\infty \\ \Omega^\alpha - \Omega^\infty \\ -E^\infty \end{pmatrix} \qquad 3.6a$$

U$-u^\infty$ $and\ \Omega - \Omega^\infty$ are vectors of length 3N for N particles. $-E^\infty$ is a vector of length 9N. F and L are vectors of length 3N. S is the stresslet of length 9N [Brady & Bossis (1987)].

Similarly, the Mobility problem is written as

$$\begin{pmatrix} U^\alpha - u^\infty \\ \Omega^\alpha - \Omega^\infty \\ -E^\infty \end{pmatrix} = M \begin{pmatrix} F^\alpha \\ L^\alpha \\ S^\alpha \end{pmatrix} \qquad 3.6b$$

Initially Stokesian Dynamics included only the force and torque. As higher moments from the moment expansion are included, more terms can be taken in the vectors on both sides.

Stokesian dynamics works well with spherical particles. This is because force moments of higher order can be included easily for spherical particles. That is not the same case for irregular shaped particles [Swan (2011)].

Properties of the grand resistance matrix, R and grand mobility matrix, M:

1.Dependent on the configuration of particles

2.Symmetric nature and

3.Positive definite.

The inversion of Grand mobility matrix is possible since it is positive definite. Coupling of force and translational velocity between two particles scales as $r^{-1}$, where $r$ is the distance between the two particles. Similarly, the force moments of order say m and velocity moments of order n can be coupled and scaled as $r^{-(1+m+n)}$ [Swan (2011)]. In order to obtain the resistance matrix, the inversion of mobility matrix alone is not sufficient since it fails to include the lubrication interaction [Durlofsky (1987)]. Inverting the mobility matrix is computationally expensive, adding lubrication interaction is very easy comparatively.

As discussed earlier, interactions are divided into the corresponding near-field and far-field components. Far-field component of resistance matrix is obtained by inverting the far-field mobility matrix, $M^{\infty}$.This mobility matrix is obtained for all particle pairs using Faxen's laws. To include lubrication interactions, resistance matrix $R_{2B}$ is found for two-body interaction between all pairs within a critical radius r* from a particle centre. In order to avoid double counting, we remove mobility interaction $M^{\infty}$ for each particle inside r*. Inverting this gives us ${R_{2B}}^{\infty}$ [Brady (1987)].

$$R = (M^{\infty})^{-1} + R_{2B} - {R_{2B}}^{\infty} \qquad 3.7$$

Map of Stokesian Dynamics [Brickell (2012)] that gives an overview of the method of Stokesian Dynamics is given in Fig 3-1.

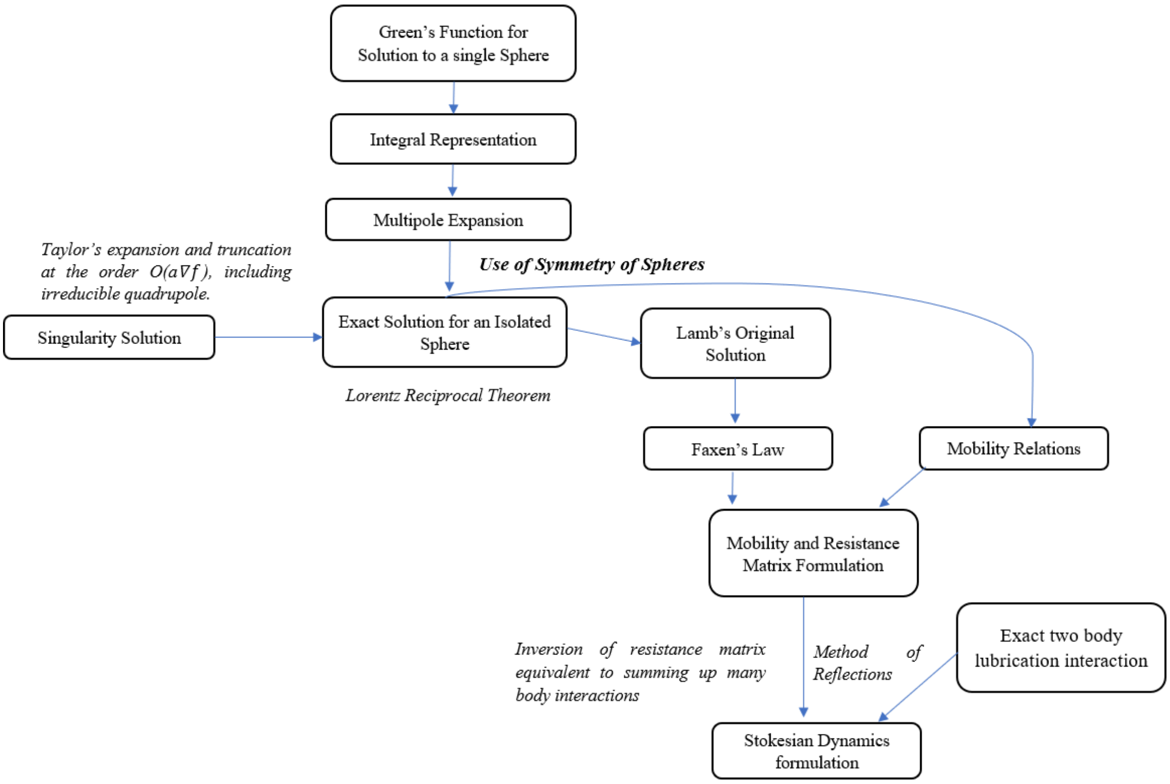



*Figure 3-1: Map of Stokesian Dynamics as adapted from or reproduced from [Brickell (2012)]*

### 3.4 Green's function

The primary Green's function of Stokes flow is the Stokeslet, which is associated with a singular point force embedded in a Stokes flow. Other singularity solutions are obtained from its derivatives. To consider the effect of large number of particles, first one could consider the effect of a single spherical particle in a fluid. Green's function gives us the velocity field in a viscous fluid by the presence of a point force at the centre of particle assuming the particle radius as zero [Lisicki (2013)].

For Re<<1, Stokes equation is

$$-\nabla P + \mu \nabla^2 u = 0 \qquad 3.8$$

When forced with a delta function,

$$-\nabla P + f\,\delta(x - x_o) + \mu \nabla^2 u = 0 \qquad 3.9$$

$f\,\delta(x - x_o)$ is the forcing term where f is an arbitrary constant.

General solution for velocity to equation 3.9 is

$$u_i = \frac{1}{8\pi\mu} J_{ij}(x - x_o) f_j \qquad 3.10$$

Here, the function $J_{ij}$ is to be determined and known as Green's function. It depends on the current position, x and also the position of point force, $x_o$. The delta function is

$$\delta(\hat{x}) = -\nabla^2\left(\frac{1}{4\pi r}\right) \tag{3.11}$$

Balancing the dimensions of the pressure term with those of the delta function.

$$P = -\frac{1}{4\pi}\,f.\nabla\left(\frac{1}{r}\right) \tag{3.12}$$

The velocity field can be written in the form of

$$u = \frac{1}{\mu}f.(\nabla\nabla - I\nabla^2)H \tag{3.13}$$

H satisfies the forced biharmonic equation.

$$H = -\frac{r}{8\pi} \tag{3.14}$$

We get an expression

$$u_j = \frac{1}{8\pi\mu}f_i\left(\frac{\delta_{ij}}{r} + \frac{\hat{x}_i\hat{x}_j}{r^3}\right) \tag{3.15}$$

Thus, the Green's function is obtained as $J_{ij} = \frac{\delta_{ij}}{r} + \frac{\hat{x}_i\hat{x}_j}{r^3}$ 3.16

**3.5 Multipole expansion**

Multipole expansion gives us the disturbance field in the fluid created by the presence of particles. Inclusion of all terms from the expansion for the mobility matrix ensures lubrication interactions are added [Townsend (2018)]. Velocity field anywhere in the suspension of N particles each with surface $S_\alpha$can be given by

$$u_i(x) = u_i^\infty(x) - \sum_{\alpha=1}^{N}\int_{S_\alpha} u_i^\alpha \, dS_y \tag{3.17}$$

Velocity field induced by a point force at centre of particle $\alpha$, $u_i^\alpha$, is

$$u_i^\alpha = \frac{1}{8\pi\mu}J_{ij}f_j \tag{3.18}$$

$$u_i(x) = u_i^\infty(x) - \frac{1}{8\pi\mu}\sum_{\alpha=1}^{N}\int_{S_\alpha} J_{ij}\,(x-y)f_j \, dS_y \tag{3.19}$$

where $J_{ij}$ is the Oseen's tensor.

Disturbance created by a rigid particle can be represented by a distribution of point forces imparted to the fluid on the surface of the particle and the disturbance created by an aggregate can be given by sum of the individual particle disturbances. At any surface point, $\mathrm{f_j}(y) = \sigma_{jk}(y)n_k$ Where $n_k$ is the normal to the surface.

Negative sign indicates that force acting on the fluid decreases its velocity.

It is possible to solve the integral representation by dividing the particle into elements and get a solution. This can be computationally expensive especially when the particles are close together. For N particles, number of unknowns to be determined by dividing each particle into M elements will be (3M+6) N. This includes force component in three directions and their corresponding translational and rotational velocities. Minimum number of elements possible for a particle in 3-dimension will be 12. This is the number of nearest neighbours for the particle. When M=12, number of unknowns will be 42N. This is the case in 3-dimension. Hence, 42N equations need to be solved. The number of unknowns for particles in 2-dimension will be (2M+3) N. The minimum number of elements possible for a particle in 2-dimension will be 5 and minimum number of unknowns will be 13N. A simulation with large number of particles renders this method costly [Durlofsky (1987)].

$\sum_{\alpha=1}^{N}\left(\int_{S\alpha} \mathrm{J_{ij}}\,(x-y)\mathrm{f_j}dS_y\right)$ can be written as

$$\sum_{\alpha=1}^{N}\int \mathrm{J_{ij}}\,(x-x^{\alpha})\mathrm{f_j}dS_y + \int (x_k^{\alpha} - y_k).\nabla_k \mathrm{J_{ij}}\,(x-x^{\alpha})\,\mathrm{f_j}\,dS_y$$
$$+\frac{1}{2}\int (x_k^{\alpha} - y_k)(x_l^{\alpha} - y_l).\nabla_k\nabla_l \mathrm{J_{ij}}\,(x-x^{\alpha})\,\mathrm{f_j}\,dS_y \qquad 3.20$$
$$+\frac{1}{6}\int (x_k^{\alpha} - y_k)(x_l^{\alpha} - y_l)(x_m^{\alpha} - y_m).\nabla_k\nabla_l\nabla_m \mathrm{J_{ij}}\,(x-x^{\alpha})\,\mathrm{f_j}\,dS_y$$
$$+\cdots$$

Thus, expansion of the Green's function as a Taylor series around $x^{\alpha}$, the centre of particle $\alpha$, yields equation 3.20. We have expanded the Green's function as moments of force distribution. The $n^{\text{th}}$ moment is given by

$$Q_{i\ldots j}^{n}=\int_{S^{\alpha}}\prod_{i\ldots}^{n}(y_i - x_i^{\alpha})\,f_j(y)dS_y \qquad 3.21$$

and the velocity at any point x in the fluid can be written as

$$u_i(x) = u_i^{\infty}(x) - \frac{1}{8\pi\mu}\sum_{\alpha=1}^{N}\Big[\mathrm{J_{ij}}\,(x - x^{\alpha})Q_j^{o} + \nabla_k \mathrm{J_{ij}}\,(x - x^{\alpha})Q_{kj}^{1} + \frac{1}{2}\nabla_k\nabla_l \mathrm{J_{ij}}\,(x - x^{\alpha})Q_{klj}^{2} + \frac{1}{6}\nabla_k\nabla_l\nabla_m \mathrm{J_{ij}}\,(x - x^{\alpha})Q_{klmj}^{3} + \cdots\Big] \qquad 3.22$$

We can write the second and third moments in terms of lower order moments.

$$u_i(x) = u_i^{\infty}(x) - \frac{1}{8\pi\mu}\sum_{\alpha=1}^{N}\Big[\mathrm{J_{ij}}\,(x - x^{\alpha})Q_j^{o} + \nabla_k \mathrm{J_{ij}}\,(x - x^{\alpha})Q_{kj}^{1} + \frac{1}{6}a^2\nabla^2 \mathrm{J_{ij}}\,(x - x^{\alpha})Q_j^{0} + \frac{1}{10}a^2\nabla^2\big[\nabla_m J_{ij} Q_{mj}^{1} + \nabla_l J_{ij} Q_{lj}^{1} + \nabla_k J_{ij} Q_{kj}^{1}\big]\Big] \qquad 3.23$$

The point force distributions also correspond to the force, torque and stresslet acting on the particle.

The force acting on the fluid is given by zeroth moment.

$$Q_j^{o} = -\int_{S\alpha} f_j \, dS_y \; = \; -F_j{}^{\alpha} \qquad 3.24a$$

$$Q_{kj}^{1} = -\int_{S\alpha} (x_k^{\alpha} - \, y_k) f_j \, dS_y \qquad 3.24b$$

First order moment is split into its symmetric and antisymmetric parts.

$$L_i = \int_{S\alpha} \varepsilon_{ijk}(x_J^{\alpha} - \, y_J) f_K \, dS_y \qquad 3.25$$

$$S_{ij} = \frac{1}{2}\int_{S\alpha}\Big[(x_i^{\alpha} - \, y_i) f_K + (x_j^{\alpha} - \, y_j) f_K - \frac{2}{3}\delta_{ij}(x_k^{\alpha} - \, y_k) f_K\Big] \, dS_y \qquad 3.26$$

Antisymmetric part corresponds to the torque exerted by the particle on the fluid. Symmetric part corresponds to the stresslet. Stresslet does not include the trace elements.

$$Q_{kj}^{1} = -\int_{S\alpha} (x_k^{\alpha} - \, y_k) f_j \, dS_y \; = \; P^{\alpha}\delta_{kj} + S_{kj}^{\alpha} + \frac{1}{2}\epsilon_{jkl} L_j^{\alpha} \qquad 3.27$$

We neglect the Pressure term as the effect of pressure is less or negligible for suspensions.

$$u_i(x) = u_i^{\infty}(x) - \frac{1}{8\pi\mu}\sum_{\alpha}\left[-\left\{1+\frac{1}{6}a^2\nabla^2\right\}J_{ij}(x-x^{\alpha})F_j^{\alpha} + \frac{1}{2}\varepsilon_{jkl}\nabla_k J_{il}(x-x^{\alpha})L_j^{\alpha} + \left\{1+\frac{1}{10}a^2\nabla^2\right\}\nabla_k J_{ij}(x-x^{\alpha})S_{jk}^{\alpha}\right] \quad 3.28$$

This gives us an approximation for the far field interactions upon truncation.

### 3.6 Faxen's laws

Initially Faxen's laws were used to determine the force and torque on a single sphere. This was derived from Lamb's (1921) general solution. Force and torque on a single sphere can be given as a function of particle and ambient fluid velocity.

$$F = 6\pi\mu a\left\{\left(1+\frac{1}{6}a^2\nabla^2\right)u_i^{\infty}(x)|_{x=0} - U\right\} \quad 3.29a$$

$$T = 8\pi\mu a^3(\Omega^{\infty} - \omega)|_{x=0} \quad 3.29b$$

This gives the relationship between the velocity of particles and the corresponding hydrodynamic force/torque.

The following relationships are obtained.

$$U_i^{\alpha} - u_i^{\infty}(x^{\alpha}) = \frac{F_i^{\alpha}}{6\pi\mu a} + \left(1+\frac{1}{6}a^2\nabla^2\right)u'_i(x^{\alpha})$$

$$\Omega_i^{\alpha} - \Omega_i^{\infty} = \frac{L_i^{\alpha}}{8\pi\mu a^3} + \frac{1}{2}\varepsilon_{ijk}\nabla_j u'_k(x^{\alpha}) \quad 3.30$$

$$-E_{ij}^{\infty} = \frac{S_{ij}^{\alpha}}{\frac{20\pi\mu a^3}{3}} + \left(1+\frac{a^2}{10}\nabla^2\right)e'_{ij}(x^{\alpha})$$

$U_i^{\alpha}$ is the velocity of $\boldsymbol{\alpha}^{\text{th}}$ particle. It is the sum of the velocity of the ambient fluid and the additional velocity as a result of external force acting upon it. Velocity field induced by other particles, $u'_i$. $\Omega_i^{\alpha}$is the particle's rotational velocity. $E_{ij}^{\infty}$is the first moment of velocity on the surface of the particle and is the strain rate. More velocity moments can also be included. Faxen's laws are given in the appendix of this chapter.

### 3.7 Method of reflections

Hydrodynamic interactions among particles occur over long distances for low Reynolds number flow. The presence of other particles within a fluid creates a disturbance field that alters the ambient field incident on each particle. This new ambient field on a selected particle generates another disturbance. This new disturbance in turn alters the ambient field incident on another particle. This method is iterative.

Let us consider two particles A and B. Particle A is fixed and particle B is moving with a constant velocity generating a disturbance field that decays as $r^{-1}$. This disturbance field induces a force on particle A and this force generates another velocity field. Now the velocity field generated by particle A induced a force on B. The force exerted on particle B decays with every reflection. The reflections between A and B continues resulting in an infinite series.

Let $u^{\infty}$ be the ambient field incident on spheres 1 and 2. The induced velocity field upon reflection by the first sphere, $u_1$ is incident on sphere 2.

Velocity field, u upon initial reflections from the two spheres

$$u = u^{\infty} + u_1 + u_2 \qquad 3.31$$

The sphere 1 is moving with translational velocity $U_1$ and rotational velocity $\Omega_1$. Sphere 2 is moving with translational velocity $U_2$ and $\Omega_2$. The velocity field induced from sphere 1 with centre $x_1$ and the velocity field induced by sphere 2 with centre $x_2$ are

$$u_1 = U_1 + \Omega_1 \times (x - x_1) - u^{\infty} \text{ on S1} \qquad 3.32$$

$$u_2 = U_2 + \Omega_2 \times (x - x_2) - u^{\infty} \text{ on S2} \qquad 3.33$$

There will be small errors as a result of reflections and the errors will be as small as 1/R. Such errors are corrected in the next reflection. The boundary conditions at S1 and S2 are

$$u_{21} = -u_2$$

$$u_{12} = -u_1$$

However, an infinite number of reflections would be needed to account for the lubrication interactions between closely spaced particles. All the moments from the multipole expansion can also be obtained from infinite reflections. [Kim & Karrila (1991)]

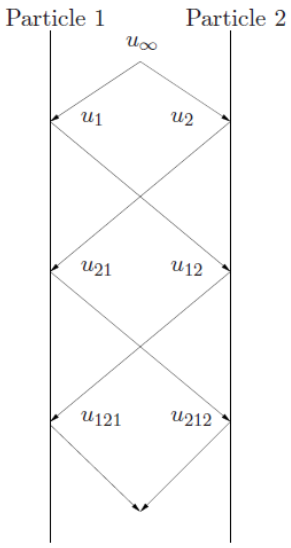



*Figure 3-2: Representation of the method of reflection for two spheres [Brickell (2012)]*

### 3.8 Lubrication forces between two spheres

Distance between two particles with characteristic length, a, can be given by $\varepsilon a$ where $\varepsilon$<<1. When particles come into close contact, $\varepsilon$ decreases and the force required to bring two particles together increases. Spheres moving towards each other undergo squeezing flow and the force and torque acting on the particles scale as $\varepsilon^{-1}$. Shearing flow generated by one sphere moving in a direction across another stationary sphere will result in the force and torque scaling as ln $\varepsilon^{-1}$.

*Steps for calculating lubrication forces and torques*

Navier-Stokes equations govern the flow for a two-sphere problem. Boundary conditions on the surface of the spheres on known. The two spheres can be moving towards each other along the line of centre. The spheres can be rotating along the axis or shearing past each other. No-slip boundary conditions on both the spheres are known. The governing equations for flow are expressed in cylindrical coordinates. The velocity components are expressed as functions of $(r, \theta, z)$ and pressure is also obtained as a function of $(r, \theta, z)$ from the governing equations. $U, V, W$ and P expressed in these components are non-dimensional.

A new expression for the surface of particles in new coordinates are obtained. The non-dimensional variables U, V,W and P are expanded in terms of ε. The leading order term can be obtained from the boundary conditions. The expanded form of the non-dimensional variables is substituted in the governing equations. The PDEs obtained are solved using the boundary conditions to get an expression in terms of pressure. From this equation, force and the torques on the spheres can be obtained. An example of a shearing flow problem is given in Appendix 3F of this chapter.

### 3.9 Resistance matrix

For systems with particles in close proximity, resistance formulation is accurate. Lubrication interactions mean to keep particles from each other. Contact between particles was straight forward in the mobility formulation when the velocities were added pairwise. The grand resistance matrix R contains nine sub-matrices, out of which six are said to be independent. Each of the component matrix contains further sub-matrices, which are particle–particle interaction tensors for all particle pairs [Kim & Mifflin (1985)].

*3.9.1 Symmetric nature of R*

$$\begin{pmatrix} F \\ L \\ S \end{pmatrix} = -\mu \begin{pmatrix} A & \tilde{B} & \tilde{G} \\ B & C & \tilde{H} \\ G & H & M \end{pmatrix} \begin{pmatrix} U^{\infty} - U \\ \Omega^{\infty} - \Omega \\ E^{\infty} \end{pmatrix} \qquad 3.35$$

All the elements in the matrix are not independent. The elements A, B, C and $\tilde{B}$ are rank 2 Elements $\tilde{G}$ and $\tilde{H}$, G and H are rank 3 tensors. The element M is a rank four tensor.

$$A_{ij} = A_{ji}, C_{ij} = C_{ji}, B_{ij} = \tilde{B}_{ji}$$
$$M_{ijkl} = M_{klij}, G_{ijk} = \tilde{G}_{kij}, H_{ijk} = \tilde{H}_{kij} \qquad 3.36$$

The matrix R is also said to be positive definite since it obeys energy dissipation theorem [Kim and Karrila (1991)].

*3.9.2 Solution of two-sphere problems*

Creeping flow problems of two spheres is very important since the aggregates are assumed to breakup into particle pairs. For two rigid spheres, if the velocities of the particles and velocity of ambient fluid are known

$$\begin{pmatrix} F_1 \\ F_2 \\ L_1 \\ L_2 \end{pmatrix} = \mu \begin{pmatrix} A_{11} & A_{12} & \tilde{B}_{11} & \tilde{B}_{12} \\ A_{21} & A_{22} & \tilde{B}_{21} & \tilde{B}_{22} \\ B_{11} & B_{12} & C_{11} & C_{12} \\ B_{21} & B_{22} & C_{21} & C_{22} \end{pmatrix} \begin{pmatrix} U_1 - u^{\infty}(x_{\alpha}) \\ U_2 - u^{\infty}(x_{\beta}) \\ \Omega_1 - \Omega^{\infty} \\ \Omega_2 - \Omega^{\infty} \end{pmatrix} \qquad 3.37$$

The resistance matrix contains rank-2 tensors A, B and C. These tensors are functions of r = $x_2 - x_1, a_1 and\ a_2$. Here $r$ is the distance between the two spheres. $a_1$ *and* $a_2$ are the size of two spheres, sphere 1 and sphere 2. Any tensor P when the spheres are relabelled gives us

$$P_{\alpha\beta}(r, a_1, a_2) = P_{(3-\alpha)(3-\beta)}(-r, a_2, a_1) \quad 3.38$$

Any tensor P can be given as a function of scalar functions [Brenner, (1963)]

Reciprocal theorem proves the resistance matrix is symmetric [Brenner, 1972)]

$$A_{ij}^{(\alpha\beta)} = A_{ji}^{(\beta\alpha)}, B_{ij}^{(\alpha\beta)} = \tilde{B}_{ji}^{(\beta\alpha)}, C_{ij}^{(\alpha\beta)} = C_{ji}^{(\beta\alpha)} \quad 3.39$$

To give the components of force due to translation of particles with symmetry about z-axis. The unit vector along z-axis is **d**=**r**/r.

$$F_x = A^{\perp}(U_X - u_x^{\infty})$$

$$F_y = A^{\perp}(U_y - u_y^{\infty})$$

$$F_z = A''(U_z - u_z^{\infty}) \quad 3.40$$

$$F_i = \left[A'' d_i d_j + A^{\perp}\left(\delta_{ij} - d_i d_j\right)\right](U_i - u_i^{\infty})$$

This gives us our tensor A as a function of two scalar functions, $X^A$ and $Y^A$ .

$$A_{ij}^{(\alpha\beta)} = X_{\alpha\beta}^{A} d_i d_j + Y_{\alpha\beta}^{A}\left(\delta_{ij} - d_i d_j\right)$$

$$B_{ij}^{(\alpha\beta)} = Y_{\alpha\beta}^{B} \epsilon_{ijk} d_k \quad 3.41$$

$$C_{ij}^{(\alpha\beta)} = X_{\alpha\beta}^{C} d_i d_j + Y_{\alpha\beta}^{C}\left(\delta_{ij} - d_i d_j\right)$$

It is known that the dimension of A is unit length, that of B is length$^2$ and that of C is length$^3$ [Jeffrey & Onishi (1984)].

Non-dimensionalizing the tensors and non-dimensional terms can be given by

$$\hat{A}_{\alpha\beta} = \frac{A_{\alpha\beta}}{3\pi\left(a_\alpha + a_\beta\right)} \quad 3.42a$$

$$\hat{B}_{\alpha\beta} = \frac{B_{\alpha\beta}}{\pi\left(a_\alpha + a_\beta\right)^2} \quad 3.42b$$

$$\hat{C}_{\alpha\beta} = \frac{C_{\alpha\beta}}{\pi\left(a_\alpha + a_\beta\right)^3} \quad 3.42c$$

Likewise, the variables, r=$x_2 - x_1, a_1$ *and* $a_2$ are also non-dimensionalized to get two non-dimensional variables [Batchelor et.al, (1982)]

$$s = \frac{2r}{(a_1 + a_2)} \text{ and } \lambda = \frac{a_2}{a_1} \quad 3.43$$

Mobility matrix

$$\begin{pmatrix} U_1 - u^\infty(x_\alpha) \\ U_2 - u^\infty(x_\beta) \\ \Omega_1 - \Omega^\infty \\ \Omega_2 - \Omega^\infty \end{pmatrix} = \mu^{-1} \begin{pmatrix} a_{11} & a_{12} & \tilde{b}_{11} & \tilde{b}_{12} \\ a_{21} & a_{22} & \tilde{b}_{21} & \tilde{b}_{22} \\ b_{11} & b_{12} & c_{11} & c_{12} \\ b_{21} & b_{22} & c_{21} & c_{22} \end{pmatrix} \begin{pmatrix} F_1 \\ F_2 \\ L_1 \\ L_2 \end{pmatrix} \qquad 3.44$$

Symmetricity of the matrix can also be shown using reciprocal theorem.

$$a_{ij}^{(\alpha\beta)} = a_{ji}^{(\beta\alpha)}, b_{ij}^{(\alpha\beta)} = \tilde{b}_{ji}^{(\beta\alpha)}, c_{ij}^{(\alpha\beta)} = c_{ji}^{(\beta\alpha)} \qquad 3.45$$

Mobility matrix is also said to be positive definite. Mobility matrix cannot be inverted if it is not positive definite.

Similar to the resistance matrix, the tensors can be decomposed into scalar functions [Jeffrey & Onishi (1984)]

$$\begin{aligned} a_{ij}^{(\alpha\beta)} &= x_{\alpha\beta}^{a} d_i d_j + y_{\alpha\beta}^{a}\left(\delta_{ij} - d_i d_j\right) \\ b_{ij}^{(\alpha\beta)} &= y_{\alpha\beta}^{b} \epsilon_{ijk} d_k \\ c_{ij}^{(\alpha\beta)} &= x_{\alpha\beta}^{c} d_i d_j + y_{\alpha\beta}^{C}\left(\delta_{ij} - d_i d_j\right) \end{aligned} \qquad 3.46$$

Non-dimensionalizing the tensors and non-dimensional terms can be given by

$$\begin{aligned} \hat{a}_{\alpha\beta} &= 3\pi\left(a_\alpha + a_\beta\right) a_{\alpha\beta} \\ \hat{b}_{\alpha\beta} &= \pi\left(a_\alpha + a_\beta\right)^2 b_{\alpha\beta} \\ \hat{c}_{\alpha\beta} &= \pi\left(a_\alpha + a_\beta\right)^3 c_{\alpha\beta} \end{aligned} \qquad 3.47$$

Mobility and resistance matrices can be related as

$$\begin{pmatrix} a_{11} & a_{12} & \tilde{b}_{11} & \tilde{b}_{12} \\ a_{21} & a_{22} & \tilde{b}_{21} & \tilde{b}_{22} \\ b_{11} & b_{12} & c_{11} & c_{12} \\ b_{21} & b_{22} & c_{21} & c_{22} \end{pmatrix} \begin{pmatrix} A_{11} & A_{12} & \tilde{B}_{11} & \tilde{B}_{12} \\ A_{21} & A_{22} & \tilde{B}_{21} & \tilde{B}_{22} \\ B_{11} & B_{12} & C_{11} & C_{12} \\ B_{21} & B_{22} & C_{21} & C_{22} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \qquad 3.48$$

Relation between scalar functions:

$$\begin{pmatrix} x_{11}^{a} & \dfrac{2}{1+\lambda} x_{12}^{a} \\ \dfrac{2}{1+\lambda} x_{21}^{a} & \dfrac{1}{\lambda} x_{22}^{a} \end{pmatrix} \begin{pmatrix} X_{11}^{A} & \dfrac{(1+\lambda)}{2} X_{12}^{A} \\ \dfrac{(1+\lambda)}{2} X_{21}^{A} & \lambda X_{22}^{A} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \qquad 3.49$$

Scalar functions Y and y can also be related.

$$\begin{pmatrix} y_{11}^{a} & \frac{2}{1+\lambda}y_{12}^{a} & \frac{3}{2}y_{11}^{b} & \frac{6}{(1+\lambda)^2}y_{21}^{b} \\ \frac{2}{1+\lambda}y_{21}^{a} & \frac{1}{\lambda}y_{22}^{a} & \frac{6}{(1+\lambda)^2}y_{21}^{b} & \frac{3}{2\lambda^2}y_{22}^{b} \\ \frac{3}{2}y_{11}^{b} & \frac{6}{(1+\lambda)^2}y_{12}^{b} & \frac{3}{4}y_{11}^{c} & \frac{6}{(1+\lambda)^3}y_{\alpha 12}^{c} \\ \frac{6}{(1+\lambda)^2}y_{21}^{b} & \frac{3}{2\lambda^2}y_{22}^{b} & \frac{6}{(1+\lambda)^3}y_{21}^{c} & \frac{3}{4\lambda^3}y_{22}^{c} \end{pmatrix}$$

$$= \begin{pmatrix} Y_{11}^{A} & \frac{(1+\lambda)}{2}Y_{12}^{A} & \frac{2}{3}Y_{11}^{B} & \frac{(1+\lambda)^2}{6}Y_{21}^{B} \\ \frac{(1+\lambda)}{2}Y_{21}^{A} & \lambda Y_{22}^{A} & \frac{(1+\lambda)^2}{6}Y_{21}^{B} & \frac{2\lambda^2}{3}Y_{22}^{B} \\ \frac{2}{3}Y_{11}^{B} & \frac{(1+\lambda)^2}{6}Y_{12}^{B} & \frac{4}{3}Y_{11}^{C} & \frac{(1+\lambda)^3}{6}Y_{\alpha 12}^{C} \\ \frac{(1+\lambda)^2}{6}Y_{21}^{B} & \frac{2\lambda^2}{3}Y_{22}^{B} & \frac{(1+\lambda)^3}{6}Y_{21}^{C} & \frac{4\lambda^3}{3}Y_{22}^{C} \end{pmatrix}^{-1} \qquad 3.50$$

*Property of scalar functions*

$$X_{\alpha\beta}^{A}(s,\lambda) = X_{\beta\alpha}^{A}(s,\lambda) = X_{(3-\alpha)(3-\beta)}^{A}(s,\lambda^{-1})$$
$$Y_{\alpha\beta}^{A}(s,\lambda) = Y_{\beta\alpha}^{A}(s,\lambda) = Y_{(3-\alpha)(3-\beta)}^{A}(s,\lambda^{-1})$$
$$Y_{\alpha\beta}^{B}(s,\lambda) = -Y_{(3-\alpha)(3-\beta)}^{B}(s,\lambda^{-1}) \qquad 3.51$$
$$X_{\alpha\beta}^{C}(s,\lambda) = X_{\beta\alpha}^{C}(s,\lambda) = X_{(3-\alpha)(3-\beta)}^{C}(s,\lambda^{-1})$$
$$Y_{\alpha\beta}^{C}(s,\lambda) = Y_{\beta\alpha}^{C}(s,\lambda) = Y_{(3-\alpha)(3-\beta)}^{C}(s,\lambda^{-1})$$

*Closely spaced particles*

Any scalar function P can be represented as

$$P^{Q} = g_1(\lambda)\xi^{-1} + g_2(\lambda)\ln\xi^{-1} + g_3(\lambda)\xi\ln\xi^{-1} + Q^{P}(\lambda) + L^{P}(\lambda)\xi + O(\xi^2\ln\xi) \qquad 3.52$$

This expression is obtained from the lubrication force expression for two particles.

where $\xi = \frac{r-a_1-a_2}{(a_1+a_2)/2} = s - 2\ is$ the non $-$ dimensional gap.

## Appendix 3A. Green's function derivation

For Re<<1, Stokes equation is

$$-\nabla P + \mu\nabla^2 u = 0 \tag{A.1}$$

Equation for Stokes flow forced with a delta function,

$$-\nabla P + f\,\delta(x - x_o) + \mu\nabla^2 u = 0 \tag{A.2}$$

$f\,\delta(x - x_o)$ is the forcing term where f is an arbitrary constant.

The equation in index notation is given by

$$-\frac{\partial P}{\partial x_i} + f_i\,\delta(x - x_{o,i}) + \mu\frac{\partial^2 u_i}{\partial x_k^2} = 0 \tag{A.3}$$

General solution for velocity to (2) is

$$u_i = \frac{1}{8\pi\mu} J_{ij}\, f_j \tag{A.4}$$

Here, the function $J_{ij}$ is to be determined and known as Green's function. It depends on the current position, x and also the position of point force, xo.

Similarly, solution for pressure,

$$P = \frac{1}{8\pi} p_j\, f_j \tag{A.5}$$

Here, $p_j$ is to be determined.

Substituting in (A.3), we get

$$-\frac{1}{8\pi}\frac{\partial p_j}{\partial x_i} f_j + f_i\,\delta(x - x_{o,i}) + \frac{1}{8\pi}\frac{\partial^2 J_{ij}}{\partial x_k^2} f_j = 0 \tag{A.6}$$

Since $f_i$ can be written as $\delta_{ij} f_j$.

$$-\frac{1}{8\pi}\frac{\partial p_j}{\partial x_i} f_j + \delta_{ij} f_j\,\delta(x - x_{o,i}) + \frac{1}{8\pi}\frac{\partial^2 J_{ij}}{\partial x_k^2} f_j = 0 \tag{A.7}$$

Dividing by $f_j$

$$-\frac{1}{8\pi}\frac{\partial p_j}{\partial x_i} + \frac{1}{8\pi}\frac{\partial^2 J_{ij}}{\partial x_k^2} = -\,\delta(x - x_{o,i}) \tag{A.8}$$

$$-\frac{\partial p_j}{\partial x_i} + \frac{\partial^2 J_{ij}}{\partial x_k^2} = -8\pi\delta(x - x_{o,i}) \tag{A.9}$$

We must ensure both Green's function $J_{ij}$ and pressure vector $p_j$ satisfy this equation.

The delta function,

$$\delta(\hat{x}) = -\nabla^2\left(\frac{1}{4\pi r}\right) \tag{A.10}$$

$$\text{where } \hat{x} = x - x_o \text{ and } r = |x - x_o|.$$

Delta function is obtained as follows:

Free-space Green's function J satisfies

$$\nabla^2 J = \delta(\hat{x}) \tag{A.11}$$

The delta function becomes 0 when $x \neq x_o$

$$\nabla^2 J = 0 \tag{A.12}$$

In spherical polars (independent of φ and ∅)

$$\nabla^2 J = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial J}{\partial r}\right) = 0 \tag{A.13}$$

$$r^2 \frac{\partial J}{\partial r} = A$$

$$\frac{\partial J}{\partial r} = \frac{A}{r^2}$$

$$J = -\frac{A}{r} + B \tag{A.14}$$

where A and B are constants.

To find A and B, we integrate (A.13) over a sphere of volume V having a surface S centered at $x_o$ and the radius of the sphere is r.

$$\text{We know that } \int_V \delta(x - x_o)\, dV = 1 \tag{A.15}$$

$$\int_V \nabla^2 J \; dV = \int_V \delta(x - x_o)\, dV = 1 \tag{A.16}$$

$$\int_V \nabla^2 J \; dV = \int_V \nabla . \nabla J \; dV \tag{A.17}$$

With the divergence theorem, this becomes

$$\int_S \frac{\partial J}{\partial r} dS = 1 \tag{A.18}$$

$\frac{\partial J}{\partial r} = \frac{A}{r^2}$ and surface area of the sphere is $4\pi r^2$.

$$4\pi A = 1 \text{ and } A = \frac{1}{4\pi}$$

$$\text{Thus, we get } J = -\frac{1}{4\pi r}$$

Then

$$\delta(\hat{x}) = \nabla^2 J = -\nabla^2\left(\frac{1}{4\pi r}\right) \qquad \text{A.19}$$

Equation (A.2) becomes

$$-\nabla P + \mu\nabla^2 u = \frac{f}{4\pi}\nabla^2\left(\frac{1}{r}\right) \qquad \text{A.20}$$

Pressure satisfies Laplace equation.

$$\nabla^2 P = 0 \qquad \text{A.21}$$

Balancing the dimensions of the pressure term with those of the delta function.

$$P = -\frac{1}{4\pi} f.\nabla\left(\frac{1}{r}\right) \qquad \text{A.22}$$

Substituting P in (A.16), we get

$$\mu\nabla^2 u = -\frac{1}{4\pi} f.(\nabla\nabla - I\nabla^2)\left(\frac{1}{r}\right) \qquad \text{A.23}$$

The velocity field can be written in the form of

$$u = \frac{1}{\mu} f.(\nabla\nabla - I\nabla^2)H \qquad \text{A.24}$$

Where H is a scalar function and I is the identity matrix.

$$\frac{1}{4\pi} f.\nabla\nabla\left(\frac{1}{r}\right) - \frac{1}{4\pi} f.I\nabla^2\left(\frac{1}{r}\right) + f.(\nabla\nabla - I\nabla^2)\nabla^2 H = 0 \qquad \text{A.25}$$

$$\nabla^2 u = \frac{1}{\mu} f.(\nabla\nabla - I\nabla^2)\nabla^2 H \qquad \text{A.26}$$

Substituting (A.26) in (A.2), we have

$$f.(\nabla\nabla - I\nabla^2)\left(\nabla^2 H + \frac{1}{4\pi r}\right) = 0 \qquad \text{A.27}$$

$$(\nabla\nabla - I\nabla^2)\left(\nabla^2 H + \frac{1}{4\pi r}\right) = 0 \qquad \text{A.28}$$

This can be satisfied when $\nabla^2 H = -\frac{1}{4\pi r}$

Taking $\nabla^2$ on both sides of (A.28)

$$\nabla^4 H = -\nabla^2\left(\frac{1}{4\pi r}\right) = \delta(\hat{x}) \qquad \text{A.29}$$

In spherical polars (independent of φ and ∅)

$$\nabla^2 f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) = 0$$

H satisfies the forced biharmonic equation.

$$\nabla^4 H = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial H}{\partial r}\right)\right] = 0 \qquad \text{A.30}$$

$$\left(r^2 \frac{\partial}{\partial r}\right)\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial H}{\partial r}\right)\right] = A$$

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial H}{\partial r}\right) = -\frac{A}{r} + B$$

$$r^2 \frac{\partial H}{\partial r} = -\frac{A}{2} r^2 + \frac{B}{3} r^3 + C$$

$$H = -\frac{A}{2} r + \frac{B}{6} r^2 - \frac{C}{r} + D \qquad \text{A.31}$$

$$\int_V \nabla^4 H \ dV = \int_V \delta(x - x_o)\, dV = 1$$

$$\int_V \nabla^4 H \ dV = \int_V \nabla.(\nabla\nabla^2 H)\, dV$$

Using divergence theorem, this becomes

$$= \int_V (\nabla\nabla^2 H).n \ dS$$

$$\int_S \frac{\partial}{\partial r}\nabla^2 H dS = 1 \qquad \text{A.32}$$

$$H = -\frac{A}{2} r + \frac{B}{6} r^2 - \frac{C}{r} + D \qquad \text{A.33}$$

Since all points where r≠0 , $\nabla^2\left(\frac{1}{r}\right) = 0$, C is immaterial.

$$\nabla^2 H = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial H}{\partial r}\right) = -\frac{A}{r} + B$$

$$\int_S \frac{\partial}{\partial r}\nabla^2 H dS = \int_S \frac{A}{r^2} dS = \frac{A}{r^2} 4\pi r^2 = 4\pi A = 1$$

Hence A=1/4π and B=C=D=0

$$H = \frac{-r}{8\pi} \qquad \text{A.34}$$

$$u = \frac{1}{\mu} f.(\nabla\nabla - I\nabla^2)\left(\frac{-r}{8\pi}\right) \qquad \text{A.35}$$

$$= \frac{1}{\mu} f.(\nabla\nabla)\left(\frac{-r}{8\pi}\right) + \frac{1}{\mu} f.(I\nabla^2)\left(\frac{r}{8\pi}\right) \qquad \text{A.36}$$

$$= \frac{1}{\mu} f.(\nabla\nabla)\left(\frac{-r}{8\pi}\right) + \frac{1}{\mu} f.I\left(\frac{1}{4\pi r}\right) \qquad \text{A.37}$$

$$u = \frac{1}{8\pi\mu} f.\left[\nabla\nabla(-r) + I\left(\frac{2}{r}\right)\right] \quad A.38$$

$$u_j = \frac{1}{8\pi\mu} f_i \left[\frac{\partial}{\partial \hat{x}_i}\frac{\partial}{\partial \hat{x}_j}(-r) + \delta_{ij}\left(\frac{2}{r}\right)\right] \quad A.39$$

$$= \frac{1}{8\pi\mu} f_i \left[-\frac{\partial}{\partial \hat{x}_i}\left(\frac{\hat{x}_j}{r}\right) + \delta_{ij}\left(\frac{2}{r}\right)\right] \quad A.40$$

$$= \frac{1}{8\pi\mu} f_i \left[-\left(\frac{r\delta_{ij} - \hat{x}_j(\hat{x}_i/r)}{r^2}\right) + \delta_{ij}\left(\frac{2}{r}\right)\right] \quad A.41$$

$$\frac{\partial \hat{x}_j}{\partial \hat{x}_i} = \delta_{ij} \quad A.42$$

$$u_j = \frac{1}{8\pi\mu} f_i \left[-\frac{\delta_{ij}}{r} + \frac{\hat{x}_i\hat{x}_j}{r^3} + \frac{2\delta_{ij}}{r}\right] \quad A.43$$

$$u_j = \frac{1}{8\pi\mu} f_i \left(\frac{\delta_{ij}}{r} + \frac{\hat{x}_i\hat{x}_j}{r^3}\right) \quad A.44$$

When the velocity field is of the form,

$$u_j = \frac{1}{8\pi\mu} J_{ij} f_i \quad A.45$$

Where the Green’s function

$$J_{ij} = \frac{\delta_{ij}}{r} + \frac{\hat{x}_i\hat{x}_j}{r^3} \quad A.46$$

$$P = \frac{1}{8\pi}\, p_j\, f_j \quad A.47$$

This is the associated pressure.

$$P = -\frac{1}{4\pi}\, f.\nabla\left(\frac{1}{r}\right) = -\frac{1}{4\pi}\, f.\left(-\frac{\hat{x}}{r}\right) = \frac{1}{8\pi}\, f.\left(\frac{2\hat{x}}{r^3}\right) \quad A.48$$

$$=\frac{1}{8\pi}\, f_j.\left(\frac{2\hat{x}_j}{r^3}\right)$$

So $p_j = \frac{2\hat{x}_j}{r^3}$ A.49

**Appendix 3B. Multipole expansion**

Velocity field anywhere in the suspension can be given by

$$u_i(x) = u_i^{\infty}(\mathrm{x}) - \sum_{\alpha=1}^{N} \int_{S\alpha} u_i \, dS_y \quad B.1$$

Velocity field induced by a point force

$$u_i = \frac{1}{8\pi\mu}\left(\frac{\delta_{ij}}{r} + \frac{\hat{x}_i\hat{x}_j}{r^3}\right) f_j \qquad \text{B.2}$$

where $f_j$ is the force distribution on surface of each particle.

$$u_i = \frac{1}{8\pi\mu} J_{ij} f_j \qquad \text{B.3}$$

$$u_i(x) = u_i^{\infty}(x) - \frac{1}{8\pi\mu}\sum_{\alpha=1}^{N}\int_{S\alpha} J_{ij}\,(x-y)\, f_j\, dS_y \qquad \text{B.4}$$

Expanding Green's function as a Taylor series around $x^{\alpha}$, the centre of particle $\alpha$:

$$J_{ij}\,(x-y) = J_{ij}\,(x - x^{\alpha} + x^{\alpha} - y) \qquad \text{B.5}$$

$$\begin{aligned} &= J_{ij}\,(x - x^{\alpha}) + (x^{\alpha} - y).\nabla[J_{ij}]\,|_{x=x^{\alpha}} \\ &\quad + \frac{1}{2}(x^{\alpha} - y)(x^{\alpha} - y):\nabla\nabla[J_{ij}]\,|_{x=x^{\alpha}} \\ &\quad + \frac{1}{6}(x^{\alpha} - y)(x^{\alpha} - y)(x^{\alpha} - y) \vdots \nabla\nabla\nabla[J_{ij}]\,|_{x=x^{\alpha}} + \ldots \end{aligned} \qquad \text{B.6}$$

$$\begin{aligned} &\sum_{\alpha=1}^{N}\left(\int_{S\alpha} J_{ij}\,(x-y) f_j\, dS_y\right) \\ &= \sum_{\alpha=1}^{N}\int J_{ij}\,(x - x^{\alpha}) f_j\, dS_y + \int (x_k^{\alpha} - y_k).\nabla_k J_{ij}\,(x - x^{\alpha})\, f_j\, dS_y \\ &\quad + \frac{1}{2}\int (x_k^{\alpha} - y_k)(x_l^{\alpha} - y_l).\nabla_k\nabla_l J_{ij}\,(x - x^{\alpha})\, f_j\, dS_y \\ &\quad + \frac{1}{6}\int (x_k^{\alpha} - y_k)(x_l^{\alpha} - y_l)(x_m^{\alpha} - y_m).\nabla_k\nabla_l\nabla_m J_{ij}\,(x - x^{\alpha})\, f_j\, dS_y \end{aligned} \qquad \text{B.7}$$

Thus, we expand the Green's function as moments of force distribution.

$N^{th}$ moment is given by

$$Q^{n}_{i\ldots j} = \int_{S\alpha} \prod\nolimits^{n}_{i\ldots} (y_i - x_i^{\alpha})\, f_j(y)\, dS_y \qquad \text{B.8}$$

This is a tensor of order n+1.

Some moments are given as following

$$Q_j^{o} = -\int_{S\alpha} f_j\, dS_y = \mathrm{O}(f) \qquad \text{B.9}$$

$$Q_{kj}^{1} = -\int_{S\alpha} (x_k^{\alpha} - y_k) f_j\, dS_y = \mathrm{O}(a\nabla f) \qquad \text{B.10}$$

$$Q_{klj}^{2} = -\int_{S\alpha} (x_k^{\alpha} - y_k)(x_l^{\alpha} - y_l)\, f_j\, dS_y = \mathrm{O}(a^2\nabla^2 f) \text{ when } k \neq l \qquad \text{B.11}$$

$$Q^3_{klmj} = -\int_{S\alpha} (x^\alpha_k - y_k)(x^\alpha_l - y_l)(x^\alpha_m - y_m)\, f_j \, dS_y \qquad \text{B.12}$$

a is the radius of the particle.

$u_i(x)$ can now be given as

$$= u^\infty_i(x) - \frac{1}{8\pi\mu}\sum_{\alpha=1}^{N}\Big[\mathrm{J_{ij}}\,(x - x^\alpha)Q^o_j + \nabla_k \mathrm{J_{ij}}\,(x - x^\alpha)Q^1_{kj} + \frac{1}{2}\nabla_k\nabla_l \mathrm{J_{ij}}\,(x - x^\alpha)Q^2_{klj} + \frac{1}{6}\nabla_k\nabla_l\nabla_m \mathrm{J_{ij}}\,(x - x^\alpha)Q^3_{klmj} + \cdots\Big] \qquad \text{B.13}$$

We can write the second and third moments in terms of lower order moments.

$$Q^2_{klj} = \frac{1}{3}a^2\delta_{kl}Q^0_j + O(a^2\nabla^2 f) \qquad \text{B.14}$$

$$Q^3_{klmj} = \frac{1}{10}a^2\big[\delta_{kl}Q^1_{mj} + \delta_{km}Q^1_{lj} + \delta_{lm}Q^1_{kj}\big] + O(a^3\nabla^3 f) \qquad \text{B.15}$$

Substituting these moments in the previous equation, we get

$u_i(x)$ can be given by

$$u^\infty_i(x) - \frac{1}{8\pi\mu}\sum_{\alpha=1}^{N}\Big[\mathrm{J_{ij}}\,(x - x^\alpha)Q^o_j + \nabla_k \mathrm{J_{ij}}\,(x - x^\alpha)Q^1_{kj} + \frac{1}{6}a^2\nabla^2 \mathrm{J_{ij}}\,(x - x^\alpha)Q^0_j + \frac{1}{10}a^2\nabla^2\big[\nabla_m J_{ij}Q^1_{mj} + \nabla_l J_{ij}Q^1_{lj} + \nabla_k J_{ij}Q^1_{kj}\big]\Big] \qquad \text{B.16}$$

The point force distribution also corresponds to the force, torque and stresslet acting on the particle.

The force acting on the fluid is given by zeroth moment.

$$Q^o_j = -\int_{S\alpha} f_j \, dS_y = -F_j{}^\alpha \qquad \text{B.17}$$

First order moment is split into its symmetric and antisymmetric part.

Antisymmetric part corresponds to the torque exerted by the particle on the fluid.

Symmetric part corresponds to the stresslet. Stresslet does not include the trace elements.

$$Q^1_{kj} = -\int_{S\alpha} (x^\alpha_k - y_k) f_j \, dS_y = P^\alpha\delta_{kj} + S^\alpha_{kj} + \frac{1}{2}\epsilon_{jkl}L^\alpha_j \qquad \text{B.18}$$

We neglect the Pressure term as the effect of pressure is less or negligible for suspensions [Townsend (2017)]

$$u_i(x) = u_i^{\infty}(x) - \frac{1}{8\pi\mu}\sum_{\alpha}\left[-\left\{1+\frac{1}{6}a^2\nabla^2\right\}J_{ij}(x-x^{\alpha})F_j^{\alpha} + \frac{1}{2}\epsilon_{jkl}\nabla_k J_{il}(x - x^{\alpha})L_j^{\alpha} + \left\{1+\frac{1}{10}a^2\nabla^2\right\}\nabla_k J_{ij}(x-x^{\alpha})S_{jk}^{\alpha}\right] \quad \text{B.19}$$

**Appendix 3C. Faxen's laws**

For velocity fields $u_1$ *and* $u_2$ and their corresponding stress fields $\sigma_1$ *and* $\sigma_2$ *where* both the velocities are solutions of Stokes equation. Lorentz reciprocal theorem considering for a sphere with surface $S_p$ and the volume V exterior to the particle within another large sphere of surface $S_\infty$.

$$\int_{S_p} u_1 .(\sigma_2.n)dS - \int_V u_1 .(\nabla.\sigma_2)dV = \int_{S_p} u_2 .(\sigma_1.n)dS - \int_V u_2 .(\nabla.\sigma_1)dV \quad \text{C.1}$$

The velocity field $u_1$ generated by a particle moving with a velocity U and the velocity field $u_2$ generated by a point force F.

$$\nabla.\sigma_1 = 0 \quad \text{C.2}$$

$$\nabla.\sigma_{2=}F\delta(x-y) \quad \text{C.3}$$

On the surface of the particle, velocity $u_1$= U and $u_2 = 0$.

Lorentz reciprocal theorem becomes

$$\int_{S_p} U .\sigma_2.n\, dS - \int_V u_1 .F\delta(x-y)\, dV = 0 \quad \text{C.4}$$

Here F2=$\int_{S_p} \sigma_2.n\, dS$ is the force acting on the particle and $\int_V \delta(x-y)\, dV = 1$

$$U.F_2 = u_1(y).F \quad \text{C.5}$$

For a sphere with centre $x_\alpha$

$$(u_1(y))_j = \frac{1}{8\pi\mu}\left(1+\frac{a^2}{6}\nabla^2\right)J_{ji}(y-x_\alpha)F_i \quad \text{C.6}$$

$$(u_1(y))_j = \frac{1}{8\pi\mu}6\pi\mu a U_i\left(1+\frac{a^2}{6}\nabla^2\right)J_{ji}(y-x_\alpha) \quad \text{C.7}$$

Sub in (C.4)

$$U_i(F_2)_i = \left(\frac{3}{4}\right)aU_i\left(1+\frac{a^2}{6}\nabla^2\right)J_{ji}(y-x_\alpha)F_j \quad \text{C.8}$$

$$(F_2)_i = \left(\frac{3}{4}\right) a \left(1 + \frac{a^2}{6}\nabla^2\right) J_{ji}(y - x_\alpha)F_j = \left(\frac{3}{4}\right) a \left(1 + \frac{a^2}{6}\nabla^2\right) J_{ij}(x_\alpha - y)F_j \qquad \text{C.9}$$

Since $J_{ji}(y - x_\alpha) = J_{ij}(x_\alpha - y)$ C.10

$$(F_2)_i = 6\pi\mu a \left(1 + \frac{a^2}{6}\nabla^2\right) u'_i(x_\alpha) \qquad \text{C.11}$$

Where $u'_i(x_\alpha) = \frac{1}{8\pi\mu} J_{ij}(x_\alpha - y)\, F_j$ will be the velocity field generated of the particle is absent. This is a result of having a point force instead of a particle at $x_\alpha$.

$$F_i = -6\pi\mu a \left(1 + \frac{a^2}{6}\nabla^2\right) u'_i(x_\alpha) \qquad \text{C.12}$$

Negative sign indicates the force is external and acting on the particle.

The velocity $U^\alpha\ of$ particle-α is the sum of the velocity of the ambient fluid and the additional velocity as a result of external force acting upon it.[$Brickell$ (2012)]. C.13

$$U^\alpha = u^\infty(x_\alpha) + V^\alpha$$

$$\text{Total force, } F_i = 6\pi\mu a V_i^{\ \alpha} - 6\pi\mu a \left(1 + \frac{a^2}{6}\nabla^2\right) u'_i(x_\alpha) \qquad \text{C.14}$$

Velocity

$$V_i^\alpha = U_i^\alpha - u_i^\infty(x_\alpha)$$

$$F_i = 6\pi\mu a \left[V_i^\alpha - \left(1 + \frac{a^2}{6}\nabla^2\right) u'_i(x_\alpha)\right] \qquad \text{C.15}$$

$$U_i^\alpha - u_i^\infty(x^\alpha) = \frac{F_i^\alpha}{6\pi\mu a} + \left(1 + \frac{1}{6}a^2\nabla^2\right) u'_i(x^\alpha) \qquad \text{C.16}$$

**Appendix 3D. Symmetric nature of Resistance matrix**

Reciprocal theorem by Lorentz et.al (1906) can be given by

$$\oint_S v_1.(\sigma_2.n)\, dS = \oint_S v_2.(\sigma_1.n)\, dS \qquad \text{D.1}$$

Consider two spheres with surfaces $S^1 and\ S^2$ and S=$S^1 + S^2$.

<u>Relation between H and $\widetilde{H}$[$Kim$ & $Karrila$, (2013)]</u>

Velocity fields on the two spheres are given by

$$v_1 = -E^\infty.x\ on\ S^1 \qquad \text{D.2}$$

$$v_2 = \omega\ x\ x - \Omega^\infty \times x\ on\ S^2$$

Torque and stresslet for the respective fields are

$$L_k = \widetilde{H}_{kij}\, E_{ij}^{\infty} = \oint_{S_p} [x \times (\sigma_2 \cdot n)]_k \, dS \qquad \text{D.3}$$

$$S_{ij} = H_{ijk}(\Omega_k^{\infty} - w_k) = \frac{1}{2} \oint_{S_p} \left[x_i(\sigma_1 \cdot n)_j + (\sigma_1 \cdot n)_i x_j\right] dS \qquad \text{D.4}$$

LHS → $\oint_S (v_1)_i . (\sigma_2 . n)_i \, dS = -\oint_S E_{ij}^{\infty} x_j (\sigma_2 \cdot n)_i dS$ D.5

$$= -E_{ij}^{\infty} \oint_S x_j (\sigma_2 \cdot n)_i \, Ds$$

$$= -E_{ij}^{\infty} \oint_S \tfrac{1}{2}\left[ x_j (\sigma_2 \cdot n)_i + x_i (\sigma_2 \cdot n)_j \right] dS$$

$$= -E_{ij}^{\infty} H_{ijk}(\Omega_k^{\infty} - w_k) \qquad \text{D.6}$$

RHS→ $\oint_S (v_1)_i . (\sigma_2 . n)_i \, dS = \oint_S [(\omega - \Omega^{\infty}) x\, x] . (\sigma_1 \cdot n) dS$ D.7

$$= \oint_S \in_{ikj} \left[(\omega_k - \Omega_k^{\infty})\, x_j\right] . (\sigma_1 \cdot n)_i dS$$

$$= -(\Omega_k^{\infty} - \omega_k) \oint_S \in_{kji} x_j (\sigma_1 \cdot n)_i \, dS$$

$$= -(\Omega_k^{\infty} - \omega_k) \oint_S x\, x\, (\sigma_1 \cdot n)\, dS$$

$$= -(\Omega_k^{\infty} - \omega_k) \widetilde{H}_{kij}\, E_{ij}^{\infty} \qquad \text{D.8}$$

Comparing both, we get

$$-E_{ij}^{\infty} H_{ijk}(\Omega_k^{\infty} - w_k) = -(\Omega_k^{\infty} - \omega_k) \widetilde{H}_{kij}\, E_{ij}^{\infty}$$

$$H_{ijk} = \widetilde{H}_{kij} \qquad \text{D.9}$$

## Appendix 3E: Positive definite nature

This can be shown by energy dissipation theorem.

$$\int_V \varphi_V \, dV = \int_V 2\mu e : \nabla v \, dV \qquad \text{E.1}$$

Where $\varphi_V$ is the rate of viscous energy dissipation per unit volume.

$$= \int_V 2\mu e_{ij} : \nabla_i \mathrm{v}_j dV \qquad \text{E.2}$$

$$= \int_V 2\mu e_{ij} : \frac{\partial v_j}{\partial x_i} dV \qquad \text{E.3}$$

$$= \int_V 2\mu e_{ij}\, e_{ij}\, dV \qquad \text{E.4}$$

$$= \int_V \sigma_{ij}\, e_{ij} dV \qquad \text{E.5}$$

$$\sigma_{ij} e_{ij} = (-P\delta_{ij} + 2e_{ij})e_{ij} \; = \; 2\mu e_{ij} e_{ij} \qquad \text{E.6}$$

$$= \int_V \sigma_{ij} \frac{\partial v_i}{\partial x_j} dV \qquad \text{E.7}$$

$$= \int_S v_i \sigma_{ij} n_j\, dS \qquad \text{E.8}$$

The disturbance velocity is given by the motion of the particle with velocity v = U + $\omega\ x\ x$ when exposed to a fluid

$$v^D = (U - U^\infty) + (\omega - \Omega^\infty) \times x - E^\infty \cdot x \qquad \text{E.9}$$

Disturbance stress is given when the stress applied minus its trace.

$$\sigma^D = \sigma - 2\mu E^\infty \qquad \text{E.10}$$

Energy dissipation rate of the disturbance field can be given by

$$= (U^\infty - U)\int_S \sigma\, .\, n\, dS \; + \; (\Omega^\infty - \omega)\int_S x \times (\sigma\, .\, n)\, dS + E^\infty \int_S x\, .\, (\sigma\, .\, n)\, Ds \qquad \text{E.11}$$

which leads to

$$(U^\infty - U).F + \; (\Omega^\infty - \omega).T \; + E^\infty : S \; \geq \; E^\infty : S^\infty \qquad \text{E.12}$$

The stresslet due to the straining field $S^\infty = 2\mu V_p E^\infty$

The inequality shows that the resistance matrix is positive definite [$Kim$ & $Karrila$ (2013)].

## ***Appendix 3*F*:* Shearing flow problem**

Consider two spheres A and B of radius a and b respectively and separated by a distance aƐ. The sphere A is moving in x-direction with a velocity U and the sphere B is stationary. The line joining the centre of the spheres is along the z-axis.

Boundary conditions in Cartesian coordinates

$$U(Z_A) = U_A \hat{\boldsymbol{e}}_x$$
$$U(Z_B) = \mathbf{0} \quad \text{F.1}$$

In cylindrical coordinates, boundary conditions are

$$v_r = U_A cos\theta$$
$$v_\theta = -U_A sin\theta \quad \text{F.2}$$
$$v_z = 0$$

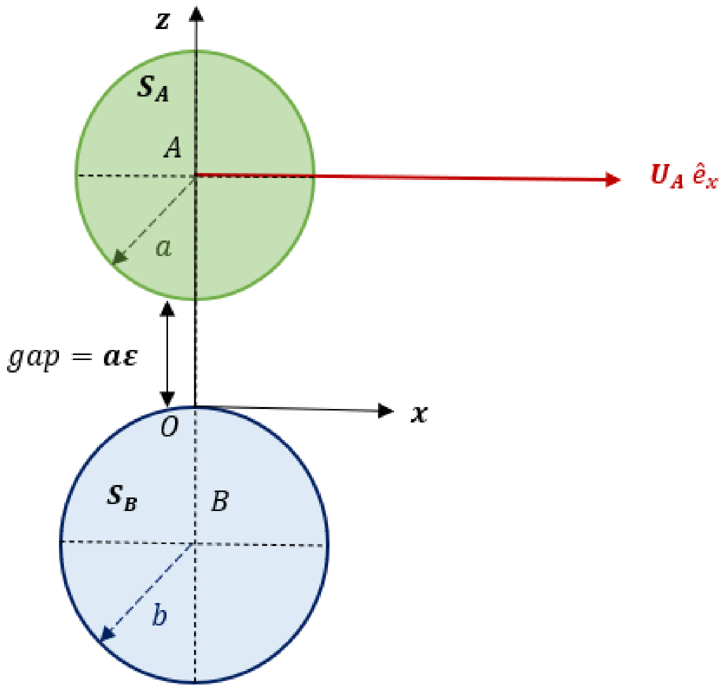



*Figure 2-3:Shearing flow for two spheres*

Navier-Stokes equation in cylindrical coordinates are

$$\frac{\partial p}{\partial r} = \mu\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_r}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 v_r}{\partial \theta^2} + \frac{\partial^2 v_r}{\partial z^2} - \frac{v_r}{r^2} - \frac{2}{r^2}\frac{\partial v_\theta}{\partial \theta}\right)$$

$$\frac{1}{r}\frac{\partial p}{\partial \theta} = \mu\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_\theta}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} - \frac{v_\theta}{r^2} + \frac{2}{r^2}\frac{\partial v_r}{\partial \theta}\right) \quad \text{F.3}$$

$$\frac{\partial p}{\partial z} = \mu\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 v_z}{\partial \theta^2} + \frac{\partial^2 v_z}{\partial z^2}\right)$$

Continuity equation

$$\frac{\partial v_r}{\partial r}+\frac{v_r}{r}+\frac{1}{r}\left(\frac{\partial v_\theta}{\partial \theta}\right)+\frac{\partial v_z}{\partial z}=0 \qquad \text{F.4}$$

Corresponding velocity components are given by

$$v_r(r,\theta,z) = U_A U(r,z)cos\theta$$

$$v_\theta(r,\theta,z) = U_A V(r,z)sin\theta \qquad \text{F.5}$$

Substituting the form of $v_r$ and $v_\theta$ in continuity equation

$$v_z(r,\theta,z) = U_A W(r,z)f(\theta)$$

Here $f(\theta)$= $cos\theta$

$$v_z(r,\theta,z) = U_A W(r,z)cos\theta$$

$$p \;=\; p_o P(r,z)f(\theta). \qquad \text{F.6}$$

U, V,W and P are non-dimensional.

Substituting this in 3c we get $p_o = \frac{\mu U_A}{a}$ $and\ f(\theta)$= $cos\theta$

$$p(r,\theta,z) = \frac{\mu U_A}{a} P(r,z)cos\theta \qquad \text{F.6b}$$

In terms of the non-dimensional variables, governing equations

$$\frac{\partial P}{\partial r}=\left(\frac{\partial^2}{\partial r^2}+\frac{1}{r}\left(\frac{\partial}{\partial r}\right)+\frac{\partial^2}{\partial z^2}\right)U-\frac{2(U+V)}{r^2}$$

$$-\frac{P}{r}=\left(\frac{\partial^2}{\partial r^2}+\frac{1}{r}\left(\frac{\partial}{\partial r}\right)+\frac{\partial^2}{\partial z^2}\right)V-\frac{2(U+V)}{r^2} \qquad \text{F.7}$$

$$\frac{\partial P}{\partial z}=\left(\frac{\partial^2}{\partial r^2}+\frac{1}{r}\left(\frac{\partial}{\partial r}\right)+\frac{\partial^2}{\partial z^2}-\frac{1}{r^2}\right)W$$

Continuity equation F.4 becomes

$$\frac{\partial U}{\partial r}+\frac{U+V}{r}+\frac{\partial W}{\partial z}=0$$

Boundary conditions that satisfy non-dimensional continuity equation

$$U(z_A) = 1$$

$$V(z_A) = -1 \qquad \text{F.8}$$

$$W(z_A) = 0$$

And

$$U(z_B) = 1$$

$$V(z_B) = -1 \qquad \text{F.9}$$

$$W(z_B) = 0$$

The particle surface $z_A$ and $z_B$ from the origin

$$z_A = a(1+\varepsilon) - (a^2 + r^2)^{\frac{1}{2}} \sim a\varepsilon + \frac{r^2}{2a}$$

$$z_B = -\beta a + (\beta^2 a^2 + r^2)^{\frac{1}{2}} \sim -\frac{r^2}{2\beta a} \qquad \text{F.10}$$

Here the ratio of sphere size, $\beta = \frac{b}{a}$

On dividing both sides by 'a' for both equations

$$\frac{z_A}{a} \sim \varepsilon + \frac{r^2}{2a^2} \qquad \text{F.11}$$

$$\frac{z_B}{a} \sim -\frac{r^2}{2\beta a^2}$$

$$O\left(\frac{z_A}{a}\right) = O\left(\frac{r^2}{2a^2}\right) = O(\varepsilon)$$

Two new coordinates Z and R

$$\mathrm{Z} = \frac{\mathrm{z}}{\mathrm{a}\varepsilon} \qquad \text{F.12}$$

$$\mathrm{R} = \frac{\mathrm{r}}{\mathrm{a}\sqrt{\varepsilon}}$$

In the new coordinates, the new surfaces are

$$Z_A = \frac{z_A}{a\varepsilon} = 1 + \frac{r^2}{2a^2\varepsilon} + \cdots = 1 + \frac{\mathrm{R}^2}{2} + \varepsilon\frac{R^4}{8} + O(\varepsilon^2) \qquad \text{F.13}$$

$$Z_B = \frac{z_B}{a\varepsilon} \sim -\frac{r^2}{2\beta a^2\varepsilon} + \cdots = -\frac{R^2}{2\beta} - \varepsilon\frac{R^4}{8\beta^3} - O(\varepsilon^2)$$

Boundary conditions suggest U=O(1), V=O(1) and from the boundary conditions, W~O($\varepsilon^{\frac{1}{2}}$).

U, V, W and P expansions are given by

$$U(r,z) = U_o(R,Z) + \varepsilon U_1(R,Z) + O(\varepsilon^2)$$

$$V(r,z) = V_o(R,Z) + \varepsilon V_1(R,Z) + O(\varepsilon^2)$$

F.14

$$W(r,z) = \varepsilon^{\frac{1}{2}} W_o(R,Z) + \varepsilon^{\frac{3}{2}} W_1(R,Z) + O\left(\varepsilon^{\frac{5}{2}}\right)$$

$$P(r,z) = \varepsilon^{-\frac{3}{2}} P_o(R,Z) + \varepsilon^{-\frac{1}{2}} P_1(R,Z) + O\left(\varepsilon^{\frac{1}{2}}\right)$$

Substituting in governing equations, leading order terms can be obtained from

$$P_o = P_o(R)$$

$$\frac{dP_o}{dR} = \frac{\partial^2 U_o}{\partial Z^2}$$ F.15

$$-\frac{P_o}{R} = \frac{\partial^2 V_o}{\partial Z^2}$$

Integrating gives us an expression for $U_o \; and \; V_o$.

$$U_o = -P'_o(R)\frac{Z^2}{2} - \frac{P'_o(R)}{2}(Z_A + Z_B) - \frac{P'_o(R)}{2}(Z_A Z_B)$$

$$= \frac{P'_o(R)}{2}[Z^2 - (Z_A + Z_B) + Z_A Z_B]$$

$$U_o = \frac{P'_o(R)}{2}(Z - Z_A)(Z - Z_B) + \left(\frac{Z - Z_B}{H}\right)$$ F.16

$$V_o = -\frac{P_o(R)}{2}(Z - Z_A)(Z - Z_B) - \left(\frac{Z - Z_B}{H}\right)$$ F.17

Integrate continuity equation from Z=$Z_B$ and Z=$Z_A$

$$W_o(R, Z_A) - W_o(R, Z_B) = -\int_{Z_B}^{Z_A} \left[\frac{\partial U_o}{\partial R} dZ\right] - \frac{1}{R}\int_{Z_B}^{Z_A} (U_o + V_o)\, dZ$$ F.18

We equate the above equation to get Reynolds equation

$$R^2 P_o'' + \left[R + 3\left(1 + \frac{1}{\beta}\right) R^3 H^{-1}\right] P_o' - P_o = -6\left(1 - \frac{1}{\beta}\right) \qquad \text{F.19}$$

We consider forces and torques acting on a small area $dS$ on the spheres and integrate them to show that the leading order terms in force and torque scale as $ln\frac{1}{\epsilon}[Kim \,\&\, Karrila, (2013)]$

$$\begin{aligned}\frac{F_x}{6\pi\mu a U_A} = &-\frac{4\beta(2 + \beta + 2\beta^2)}{15(1 + \beta)^3} ln\frac{1}{\epsilon} + A(\beta) \\ &- \frac{4(16 - 45\beta + 58\beta^2 - 45\beta^3 + 16\beta^4)}{375(1 + \beta)^4} \epsilon ln\frac{1}{\epsilon} + O(\epsilon)\end{aligned} \qquad \text{F.20}$$

$$\begin{aligned}\frac{T_y}{8\pi\mu a^2 U_A} = &\frac{\beta(4 + \beta)}{10(1 + \beta)^2} ln\frac{1}{\epsilon} + B(\beta) \\ &- \frac{(32 - 33\beta + 83\beta^2 + 43\beta^3)}{250(1 + \beta)^3} \epsilon ln\frac{1}{\epsilon} + O(\epsilon)\end{aligned} \qquad \text{F.21}$$

# Chapter 4

# Simulation setup and verification

## 4.1 Introduction

Codes obtained from Townsend (2017) is initially used to reproduce the results by Durlofsky (1987) for simulating aggregates and chain of particles sedimenting in a fluid. There are three input files position_setups.py, input_setups.py and inputs.py. The configuration of particles are given in position_setups.py. External force and torque applied on the particles are given in input_setups.py. The other input file contains parameters for timestep, number of such timesteps (frame no), version of SD and changes in the resistance matrix.

There are few options of how resistance matrix can be used:

1. Inverting the far field mobility matrix.
2. Using only the lubrication interactions.
3. Lubrication Hydrodynamics: Effecting adding the far field mobility matrix with the lubrication interactions for suspensions of high concentration.

Townsend's code has F-T-E and F-T-S versions with respect to spheres. In the F-T-E version, force, torque and strain rate are the inputs and the output obtained will be the translational velocity, rotational velocity and stresslet. In the F-T-S version, force, torque and stresslet are the inputs and the output obtained will be the translational velocity, rotational velocity and the strain rate. Townsend's work is predominantly on particles dispersed in non-Newtonian fluid, the behaviour of which he characterizes using bead-and-spring dumbbells for which rest of the versions of his codes are used. For example, one of the versions is called U-F-T-E. In order to understand the rheology of a viscoelastic fluid, the dumbbells are placed randomly between wall of spheres and each side of the wall moves in opposite horizontal directions. This movement of the wall made of spheres is given by a fixed velocity. In our simulations, only F-T-S version is used.

This study involves only particles of the same size. Hence, all length scales are made non-dimensionalized by particle radius $a$ and time is made dimensionless by $F/6\pi\eta a^2$ where $\eta$ is the solvent viscosity. This would imply velocity can be made dimensionless by a/time. For a single particle experiencing a unit force in one direction, its velocity is also unity in the same direction. Resistance matrix used by Townsend is multiplied by a factor of 6π and hence it is divided by 6π for unit dimensionless velocity to result in a unit dimensionless force. The non-dimensionliazation scheme employed is the same as Swan et al. (2011).

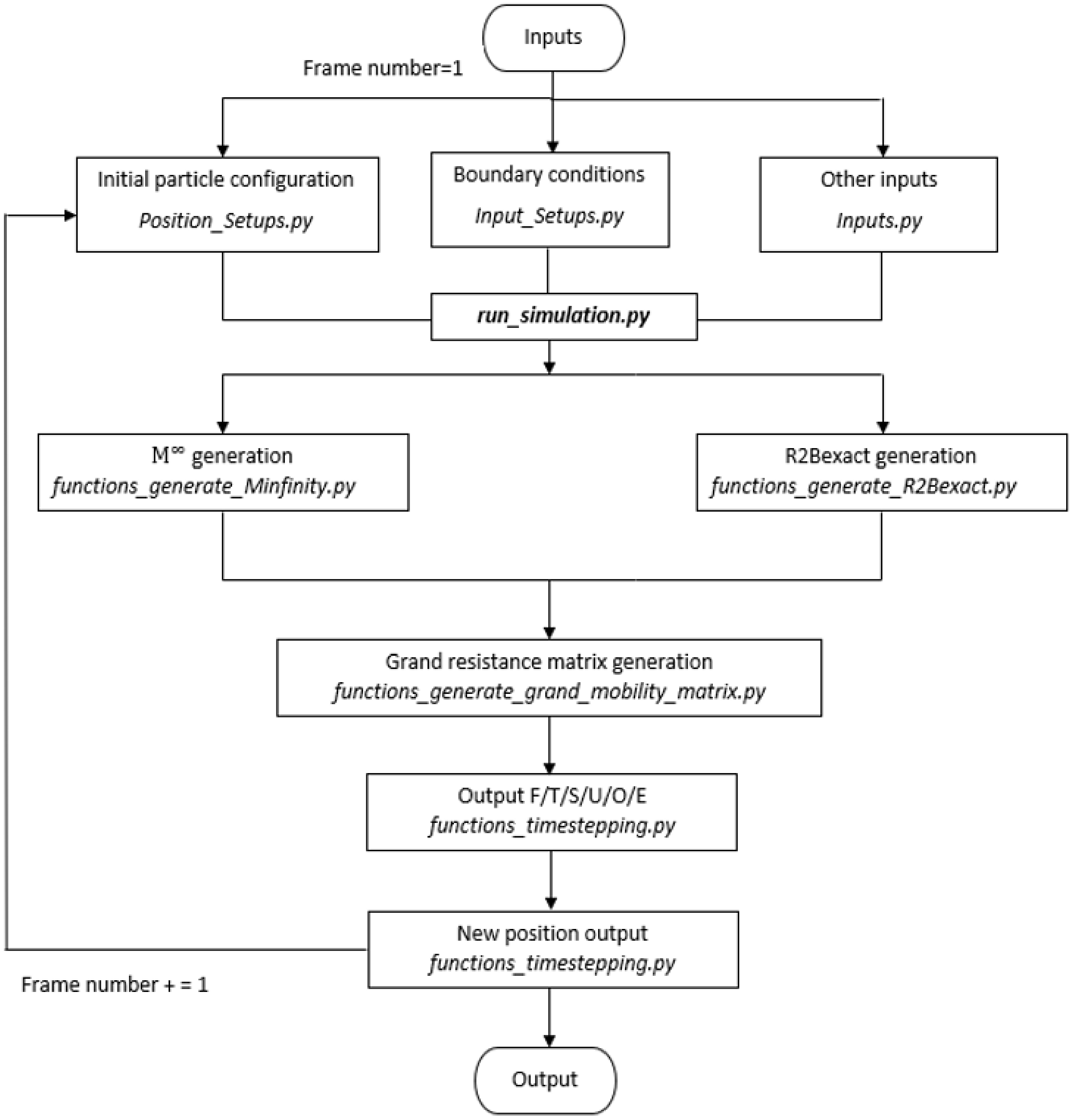



*Figure 4-1: Flowchart involved in Stokesian Dynamics Algorithm*

The steps involved in the algorithm are

1. Distance between particle pairs is computed. The inter-particle distance should be more than the two radii.
2. Far-field mobility matrix is generated.
3. Lubrication interactions are added to the mobility matrix.
4. Velocity of the particles are determined by $U = R^{-1} F$. (functions_timestepping.py)
5. New position of the particles is obtained by integrating the velocities over a time-step. This position is obtained either by Euler time stepping or by RK4 method.

### 4.2 Sedimenting spheres

Simulation for particles sedimenting in a fluid at rest is carried out and compared to the results of Durlofsky et al. (1987). The first case being considered consists of horizontal chains of spheres having equal inter-particle spacing of 4 and falling vertically. The chain of spheres consists of 5,9 and 15 spheres and since the drag acting on this chain will be symmetric, only one half of the chain is shown. The drag coefficient on each particle is determined by $\lambda = F/6\pi\mu Ua$ where a is the particle radius which is 1. Sphere zero is nothing but the central sphere. Values obtained using Townsend's almost match those obtained by Durlofsky et al. (1987). It can be observed that the spheres at the end of the chain experience the most drag.

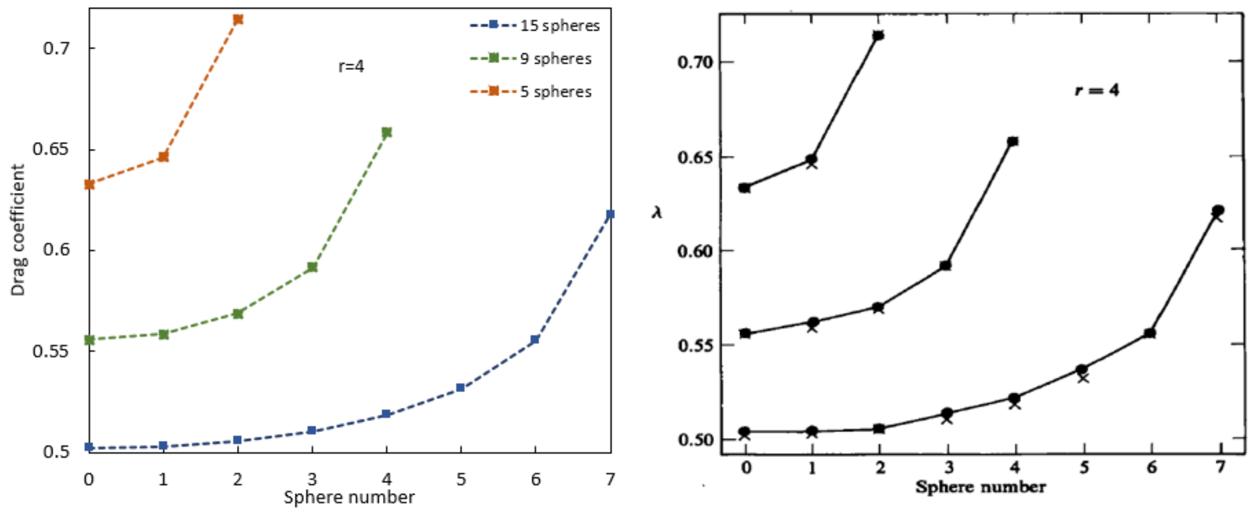



*Figure 4-2:Simulation results using Townsend's codes (Left) for 5(■), 9(■), 15(■) spheres. Results obtained by Durlofsky is given on the Right (×)*

The difference between the two results are so small and given by error %.

| **Sphere No** | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| **Error %** | 0.044 | 0.015 | 0.079 | 0.12 | 0.1 | 0.019 | 0.054 | 0.15 |

*Table 4-1: Error percentage observed between the two results for each sphere number(15 spheres)*

Next, we consider linear chain of seven spheres spaced uniformly sedimenting in a direction perpendicular to the axis joining the centres. Drag coefficient is symmetric about the centre of the chain of spheres when the numbers of spheres are odd and hence only half the spheres in the chain are considered.

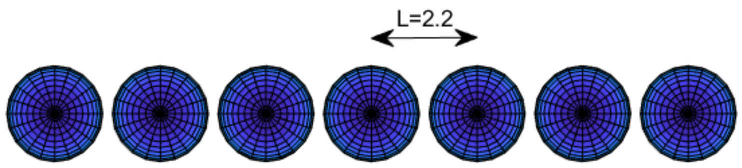



*Figure 4-3:Linear chain of 7 spheres with r=2.2*

Distance between the spheres is varied as r=2.005,2.2 and 2.6. Figure in the left represents the results obtained using codes of Townsend (2017). This result is obtained when lubrication interactions are not included. Figure in the right represents results of Durlofsky et al. (1987).

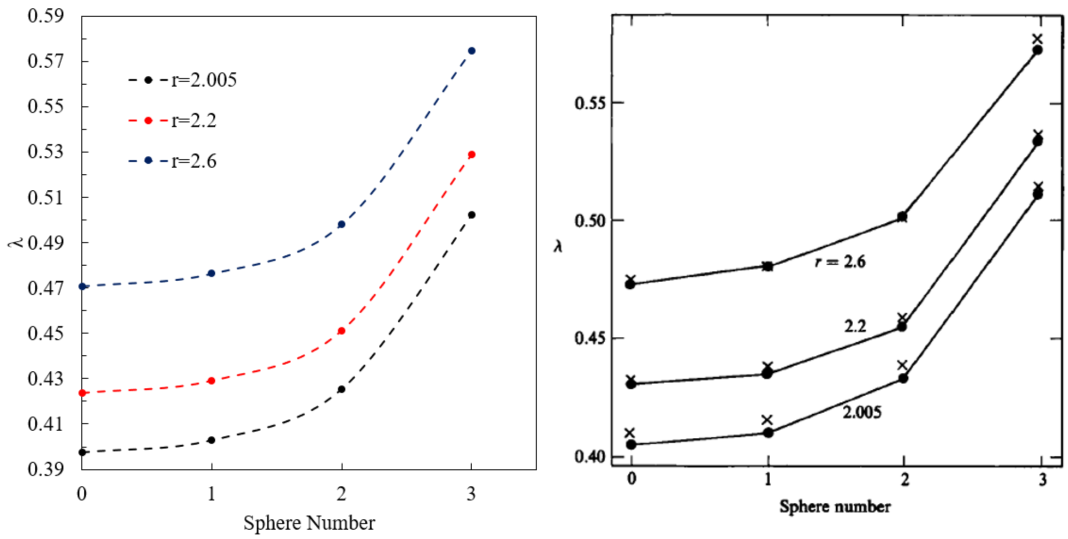



*Figure 4-4:Drag coefficient for a linear chain of spheres by simulation(left) and Durlofsky's results (right)*

From the results, it can be observed that the sphere at the centre translates at a velocity greater than the other spheres. Velocity for the particles at ends of the chain are relatively lower and settles slower compared to the particles at the centre. As the spacing between the spheres increases, spheres fall slowly. As the number of particles increases, they fall faster.

Also, we consider the case where spheres of unit radii are placed at the corners and vertices of a square and equilateral triangle of side L and compare the results obtained from Durlofsky et al. (1987).

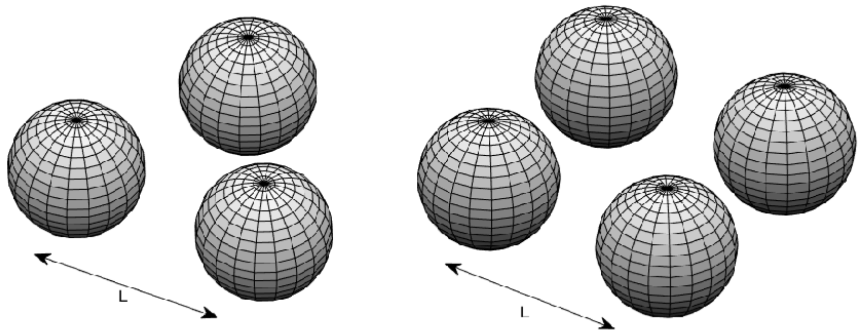



*Figure 4-5:Spheres placed at the vertices and corners of a triangle and square*

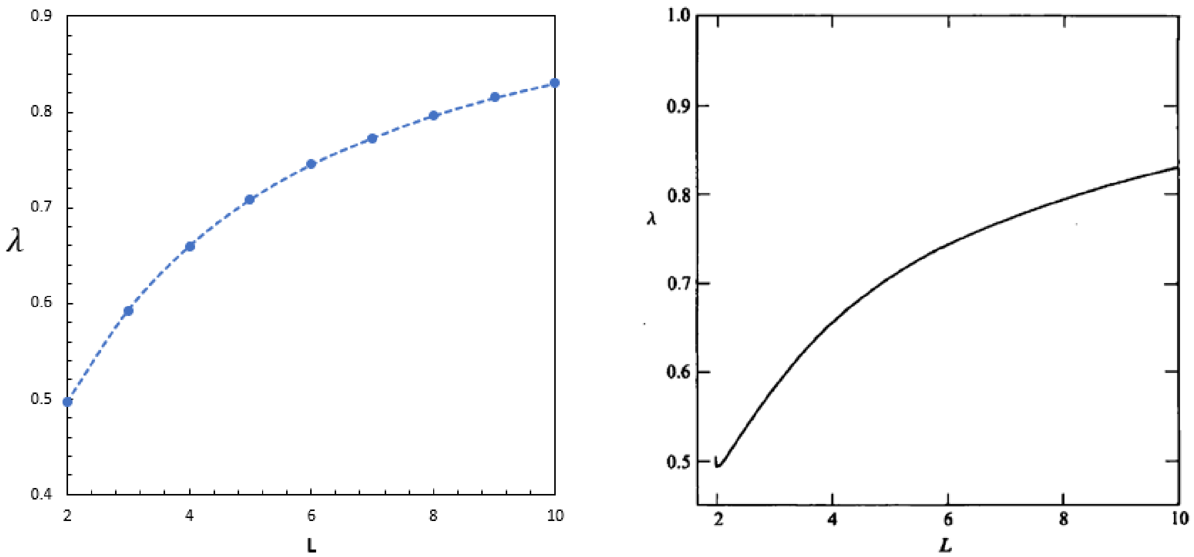



*Figure 4-6:Drag coefficient for four spheres at the corners of a square of side L in a direction perpendicular to the plane of the square by simulation(left) and Durlofsky's results(right)*

Error % of results obtained using different codes are shown and it can be observed that the ones used to obtain this figure has less error than the other one.

| **Side Length L** | | 2 | 4 | 6 | 8 | 10 |
|---|---|---|---|---|---|---|
| **Error %** | *Swan's* | 0.38 | 0.69 | 0.31 | 0.40 | 0.10 |
| | *Townsend's* | 0.28 | 0.01 | 0.15 | 0.34 | 0.09 |

*Table 4-2:Error % as side length increases (vertices of a triangle) using Swan [2011] (top) and Townsend [2017] (bottom) codes*

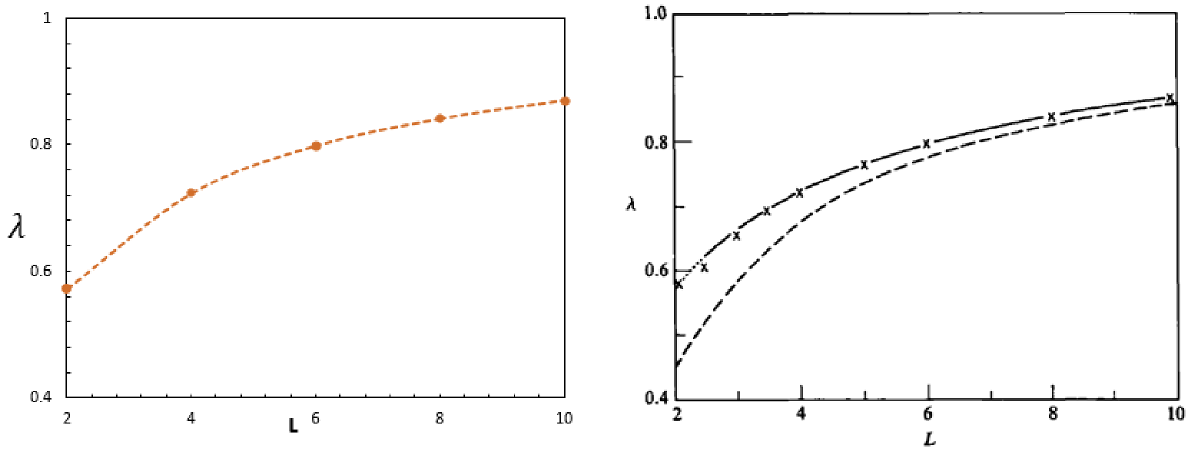



*Figure 4-7:Drag coefficient for three spheres at the vertices of an equilateral triangle of side L in a direction perpendicular to the plane of the triangle using Townsend's(left) and Durlofsky's results (right)*

By comparing Figure 19 and 20, it is clear that $\lambda$ is smaller for four spheres than three. The distance between the spheres is varied from L=2 to L=10 and as the spacing increases, $\lambda$ increases. This means that the velocity of a particle decreases as particle-particle spacing increases.

| **Side Length L** | | 2 | 4 | 6 | 8 | 10 |
|---|---|---|---|---|---|---|
| **Error %** | Swan's | 1.05 | 0.3 | 0.22 | 0.34 | 0.11 |
| | Townsend's | 1.05 | 0.29 | 0.08 | 0.31 | 0.09 |

*Table 4-3:Error % as side length increases (sides of a square) using Swan[2011](top) and Townsend [2017](bottom)*

All the calculations are done by including lubrication interactions except that of the case of 7 spheres.

Three particles initially at x = -5,0,7 and z=0, y=0 experiences a unit force and the trajectory till t≈600 is observed. Position is obtained every 10 dimensionless time units.

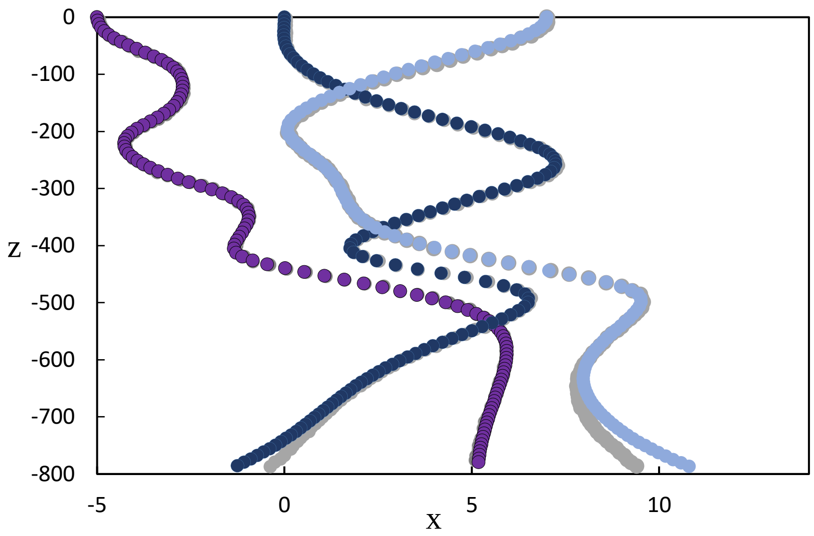



*Figure 4-8:Trajectory of three spheres till t=600 using the codes (●●●) and F-T-S version of Durlofsky[1987](●). Positions are shown for every 10-time units (dimensionless)*

The difference in positions near z≈-600 is observed and the difference increases as we go further. This tells how the simulations are sensitive to particle configuration. This particular change near z≈-600 is said to have caused by the manner by resistance scalars are obtained by interpolation as particles approach each other. [Townsend, (2017)]

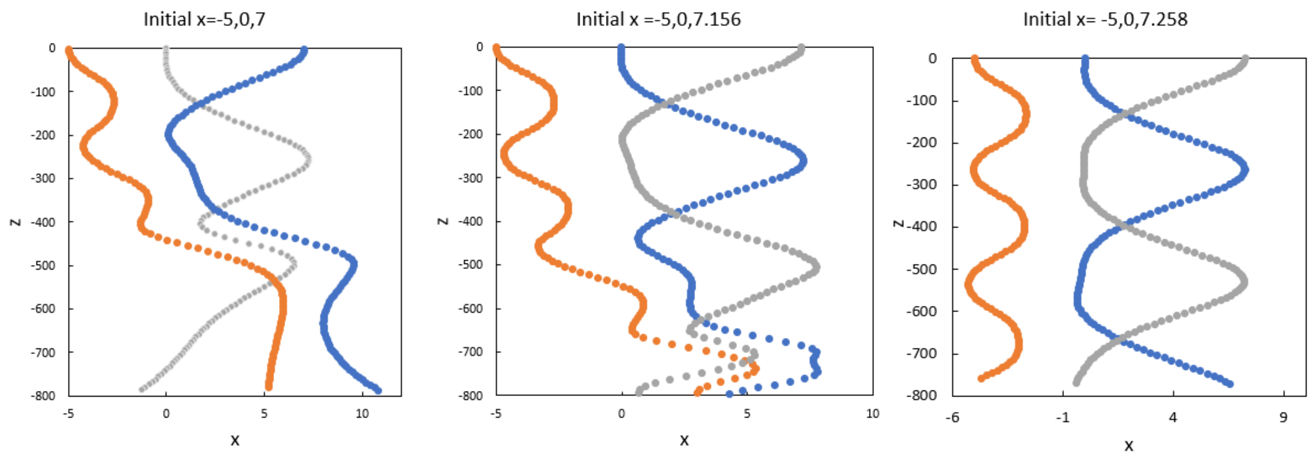



Figure 4-9:Trajectory of three spheres when the third sphere is slightly moved to the right. Particle 1(●), Particle 2(●) and Particle 3(●). Position is plotted every 10 dimensionless time units.

The trajectory changes can be shown when the position of the third particle is slightly changed from 7, 7.156 and 7.258. It can be seen that in the third setup, particle one initially at -5 is not moving towards the right near z≈ -500, as the second sphere is not producing a pulling effect on the first one unlike the first two setups.

### 4.3 Rigid body matrix

The original algorithm for SD used in [Swan (2011)] was to find the velocity of micro-swimmers constrained as an rigid assembly. Velocity of an individual particle from an aggregate is determined from the velocity of the aggregate.

$$\mathrm{U}=\Sigma^{T}.\,\widehat{U} \tag{4.1}$$

$\Sigma$ is the rigid body matrix.

$$\Sigma_{(A\alpha)} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & \Delta r_3^{(\alpha A)} & -\Delta r_3^{(\alpha A)} & 1 & 0 & 0 \\ -\Delta r_3^{(\alpha A)} & 0 & \Delta r_3^{(\alpha A)} & 0 & 1 & 0 \\ \Delta r_3^{(\alpha A)} & -\Delta r_3^{(\alpha A)} & 0 & 0 & 0 & 1 \end{pmatrix} \tag{4.2}$$

where $\Delta r^{(\alpha A)} = x^{(\alpha)} - x^{(A)}$

$\Delta r$ denotes the distance between the centre of a particle $\alpha$ from the centre of the sphere A. This form can be extended for a large aggregate with many particles. For N particles in an aggregate, the size of rigid body matrix is 6 x 6N. Velocity of an aggregate when subject to an external force $F^{P}$.

$$\widehat{U} = (\Sigma.\,R_{FU}.\,\Sigma^{T})^{-1}.\,\Sigma.\,F^{P} \tag{4.3}$$

where $R_{FU}$ is the component of resistance matrix that couples the force and the velocity moments. $\Sigma.\,R_{FU}.\,\Sigma^{T}$ is the resistance tensor that govern the hydrodynamic interactions for the aggregates rather than the individual particles. [Swan, (2011)]

Aggregate velocity, $\widehat{U} = \left[\widehat{U}_x \quad \widehat{U}_y \quad \widehat{U}_z \quad \widehat{O}_x \quad \widehat{O}_y \quad \widehat{O}_z\right]^T$

The first three elements are the translational velocity of the aggregate in x,y and z direction whereas the last three are the rotational velocity of the aggregate for the three directions.

External force acting on the aggregate, $\Sigma.\mathrm{F}^{\mathrm{P}} = [F_X \quad F_Y \quad F_Z \quad T_X \quad T_Y \quad T_Z]^T$

The first three elements are the forces acting on the aggregate in x, y and z direction whereas the last three are the torques acting on the aggregate for the three directions.

It is known that for the body to be rigidly connected, all the particles should have the same rotational velocity which is possible only after the inclusion of the rigid body matrix. But as we might think, the translational velocity might not be the same for all particles after all.

For two particles, equation 4.1 written in the form of vectors used in Swan can be written by

$$\begin{pmatrix} U_{1x} \\ U_{1y} \\ U_{1z} \\ O_{1x} \\ O_{1y} \\ O_{1z} \\ U_{2x} \\ U_{2y} \\ U_{2z} \\ O_{2x} \\ O_{2y} \\ O_{2z} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 & -(\Delta r_3)_1 & (\Delta r_2)_1 \\ 0 & 1 & 0 & (\Delta r_3)_1 & 0 & -(\Delta r_1)_1 \\ 0 & 0 & 1 & -(\Delta r_2)_1 & (\Delta r_1)_1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & -(\Delta r_3)_2 & (\Delta r_2)_2 \\ 0 & 1 & 0 & (\Delta r_3)_2 & 0 & -(\Delta r_1)_2 \\ 0 & 0 & 1 & -(\Delta r_2)_2 & (\Delta r_1)_2 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} \hat{U}_x \\ \hat{U}_y \\ \hat{U}_z \\ \hat{O}_x \\ \hat{O}_y \\ \hat{O}_z \end{pmatrix} \tag{4.4}$$

$$\begin{pmatrix} U_{1x} \\ U_{1y} \\ U_{1z} \\ O_{1x} \\ O_{1y} \\ O_{1z} \\ U_{2x} \\ U_{2y} \\ U_{2z} \\ O_{2x} \\ O_{2y} \\ O_{2z} \end{pmatrix} = \begin{pmatrix} \hat{U}_x - (\Delta r_3)_1 \hat{O}_y + (\Delta r_2)_1 \hat{O}_z \\ \hat{U}_y + (\Delta r_3)_1 \hat{O}_y - (\Delta r_1)_1 \hat{O}_z \\ \hat{U}_z - (\Delta r_2)_1 \hat{O}_y + (\Delta r_1)_1 \hat{O}_z \\ \hat{O}_x \\ \hat{O}_y \\ \hat{O}_z \\ \hat{U}_x - (\Delta r_3)_2 \hat{O}_y + (\Delta r_2)_2 \hat{O}_z \\ \hat{U}_y + (\Delta r_3)_2 \hat{O}_x - (\Delta r_1)_2 \hat{O}_z \\ \hat{U}_z - (\Delta r_2)_2 \hat{O}_x + (\Delta r_2)_2 \hat{O}_y \\ \hat{O}_x \\ \hat{O}_y \\ \hat{O}_z \end{pmatrix} \tag{4.5}$$

Velocity vector used by Swan (2011) consists of translational(U) and rotational(O) components of all the particles in three directions. Translational velocity of particle 1 in three direction is followed by rotational velocity of particle 1 in three directions. This is repeated for N number of particles. $\begin{pmatrix} U_{(1)} & O_{(1)} & U_{(2)} & O_{(2)} & \cdots & U_{(N)} & O_{(N)} \end{pmatrix}^T$

The translational velocity of individual particles depends on rotation of the body. Particles in perfectly symmetric fractals will have the same translational velocity as rotational velocity is zero.

Velocity vector used by Townsend (2017) is different from that of Swan's. Translational velocity of particle 1 in three direction is followed by translational velocity of the next particle and so on. After the first 3N elements, the last 3N elements consists of rotational velocity of N particles in the same pattern. $\begin{pmatrix} U_{(1)} & U_{(2)} & \ldots & U_{(N)} & O_{(1)} & O_{(2)} & \cdots & O_{(N)} \end{pmatrix}^T$

Hence, the rigid body matrix has to be altered a bit to adapt to the velocity vector used by Townsend. First three columns in the rigid body matrix (equation 4.2) which contains the distance between particles and centre of aggregate are repeated for N times for N particles and then followed by the last three columns (equation 4.2) repeated N times. Previously, the six columns of rigid body matrix were repeated N times for N particles.

Similarly, the effect of rigid body matrix on force acting on two particles are shown.

$$\Sigma.\begin{bmatrix} F_{1x} \\ F_{1y} \\ F_{1z} \\ T_{1x} \\ T_{1y} \\ T_{1z} \\ F_{2x} \\ F_{2y} \\ F_{2z} \\ T_{2x} \\ T_{2y} \\ T_{2z} \end{bmatrix} = \begin{bmatrix} F_{1x} + F_{2x} \\ F_{1y} + F_{2y} \\ F_{1z} + F_{2z} \\ T_{1x} + T_{2x} + (\Delta r_3)_1.F_{1y} - (\Delta r_2)_1 F_{1z} + (\Delta r_3)_2.F_{2y} - (\Delta r_2)_2 F_{2z} \\ T_{1y} + T_{2y} + (\Delta r_1)_1.F_{1z} - (\Delta r_3)_2 F_{1x} + (\Delta r_1)_2.F_{2z} - (\Delta r_3)_2 F_{2x} \\ T_{1z} + T_{2z} + (\Delta r_2)_1.F_{1x} - (\Delta r_1)_1 F_{1y} + (\Delta r_2)_2.F_{2x} - (\Delta r_1)_2 F_{2y} \end{bmatrix} \quad 4.6$$

If unit force is given on all particles in one particular direction, the rigid body matrix sums the force on individual particles and applies N times the force on the aggregate. This ensures that the force is distributed such that the body stay rigid.

### 4.4 Force distribution

When unit force is applied on all the particles on a fractal, it does not necessarily mean that all the particles experience unit force. Some particles are shielded by the presence of other particles and hence experiences less force than others. Rigid body matrix ensures that the rigid body as a whole experience a force which is the sum of all the forces applied on the particles. In figure, when a unit force was intended to apply on 384 particles and the force acting on the aggregate is $\sum_{i=1}^{384} F_i$=384. The forces acting on individual particles are obtained by multiplying the resistance matrix by the velocity of individual particles obtained by equation 4.1. Here the aggregate settles in vertical z direction and hence the area or the particles projected in that direction experiences more force. Also, the particles in the core of the aggregate experience very less force.

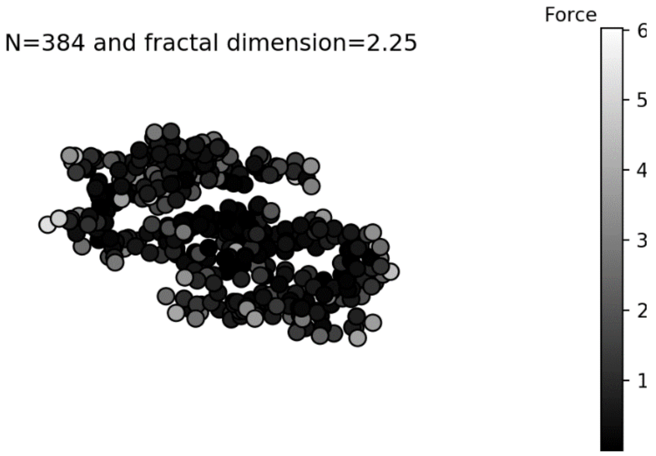



*Figure 4-10:Force distribution on Tunable Monte Carlo clusters of fractal dimension 2.23 and N=384*

### 4.5 Check for rigidity

The difference in the relative position of spheres for various times is considered as a check for rigidity. A force of 1/N is given on a set of Vicsek and DLA fractals. Vicsek fractals containing 343 spheres are simulated for unit timestep till t ≈ 50. Relative distance between neighbouring particles is determined and the maximum change in the relative distance is taken as the maximum error that occurs.

$$\text{Error \%} = \left(\frac{\text{relative distance at t} - \text{relative distance at t=0}}{\text{relative distance at t=0}}\right) * 100$$

Error in relative distance is computed at each timestep and the maximum of the error occurring at every timestep is plotted against time. The maximum error % is of the range of $10^{-14}$ for the case of Vicsek fractals containing 343 spheres and $7 \times 10^{-5}$ for DLA containing 100 spheres. Vicsek fractals do not undergo rotation owing to the symmetry and hence the error is very low or nothing when compared to DLA fractals.

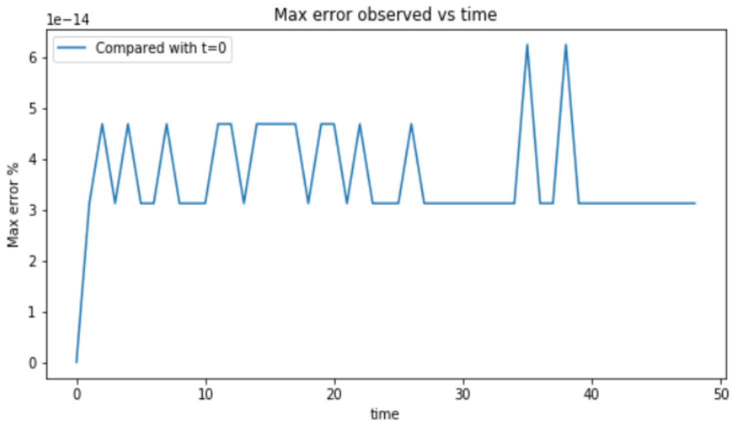



*Figure 4-11:Maximum error % for relative distance between particles as time changes for Vicsek fractal(N=343) when unit force is acting on the aggregate*

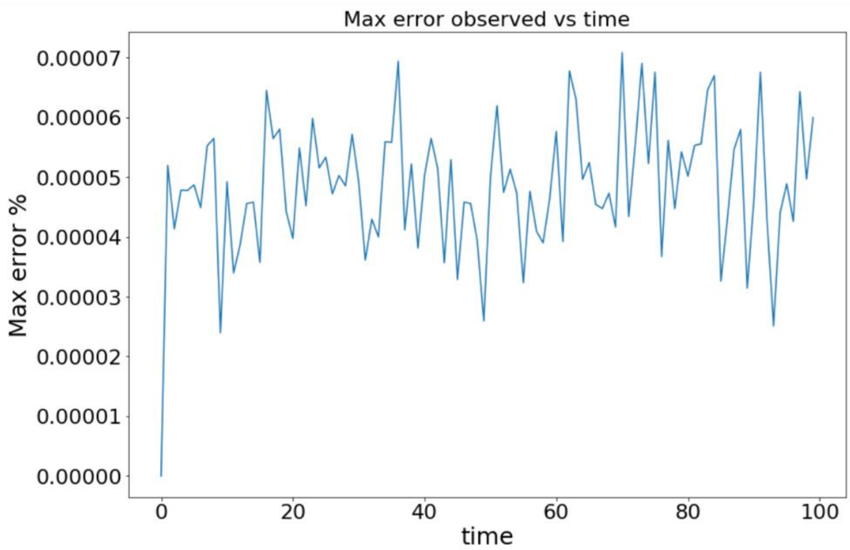



*Figure 4-12:Maximum error % for relative distance between particles as time changes for DLA ( N =100) when unit force is acting on the aggregate*

### 4.6 Time complexity

Time taken for Stokesian Dynamics simulations (F-T version) without lubrication is determined for Tunable Monte Carlo clusters for sizes N=20 to N=10240. It can be observed from Fig 4-13 that the total time taken is approximately of O($N^2$).

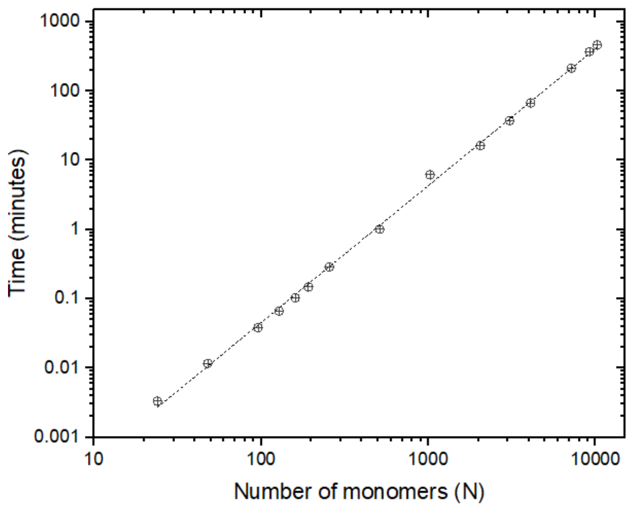



*Figure 4-13:Time taken to simulate Tunable Monte Carlo clusters($D_f$ = 2) for sizes varying from N=20 to N=10240.*

# Chapter 5

# Hydrodynamics of Fractal Aggregates

## 5.1 Introduction

Hydrodynamic behavior plays an important role in various applications in industry and environment. In industries, the hydrodynamics influences the physical features of the final product and also the way the particles are handled. Drag acting on individual spheres are affected by these forces which eventually will decide the mobility with respect to flow conditions.

In Chapter 2, we used two important parameters to characterize fractals: radius of gyration ($R_g$) and fractal dimension ($D_f$). Here, mobility radius ($R_m$) is obtained using Stokesian Dynamics. Radius of gyration depends on the arrangement of mass around the centre of mass whereas mobility radius depends on flow conditions in addition to the distribution of particles.

## 5.2 Mobility Radius with Lubrication Interactions

With the application of Stokesian Dynamics on the generated fractals, mobility radius is obtained using equation 2.5. A dimensionless force, $F^p$ equal to the number of particles, N is applied on the fractals in three directions (x, y and z) and the dimensionless aggregate velocity,$\widehat{U}$ is obtained using

$$\widehat{U} = (\Sigma . R_{FU} . \Sigma^T)^{-1} . \Sigma . F^P \qquad 5.1$$

where $R_{FU}$ is the resistance matrix coupling the force and velocity moments includes both long-range and short-range interactions and $\Sigma$ is the rigid body matrix.

The dimensionless mobility radius, Rm, being the ratio of the total force and the aggregate velocity gives the mobility radius. This is averaged over three Cartesian directions. A linear relationship is expected between mobility radius and gyration radius as predicted by Wiltzus (1987) and De Gennes (1979). However, when lubrication interactions are included, overestimated mobility radius is still a function of radius of gyration i.e., $R_m \sim R_g^y$.

The following table gives the exponents of radius of gyration for the fractals considered.

| Fractals | Tunable Monte Carlo Clusters | | | | Simple cubic unit |
|---|---|---|---|---|---|
| Fractal dimension, $D_f$ | 1.76 | 1.99 | 2.23 | 2.46 | 3 |
| $y$ | 1.07 | 1.24 | 1.38 | 1.40 | 2.41 |

*Table 5-1:Exponents for $R_g$ for various fractal dimension of Tunable Monte Carlo cluster and the simple cubic unit when plotted against mobility radius*

It is observed that as the fractal dimension increases, the relationship moves far away from linearity. Fig 5-1 shows this non-linear relationship between the two radii with the inclusion of close-range interactions.

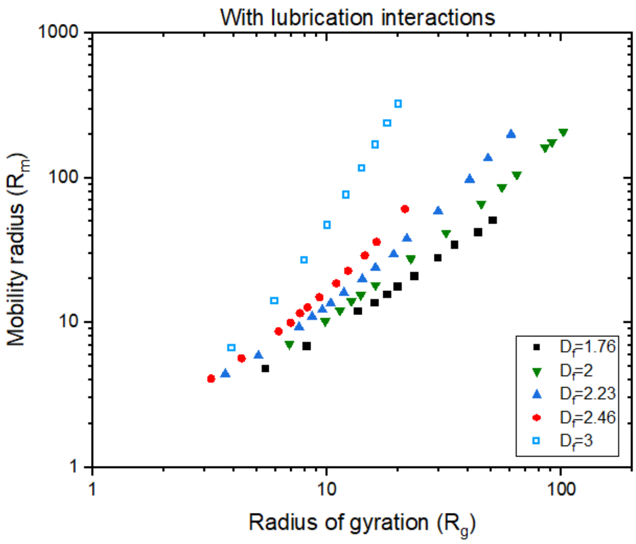



*Figure 5-1: Mobility radius vs radius of gyration for Tunable Monte Carlo Cluster-Cluster aggregates([■ $- D_f = 1.76$,▼ $-D_f = 1.99$,▲ $-D_f = 2.23$,● $- D_f = 2.46$]) and simple cubic unit[ □ $-$ simple cubic unit $- D_f = 3$]*

It is well known that as fractal dimension increases, the aggregate is more compact. The addition of lubrication interaction leads to overestimating the mobility radius. This overestimation can be attributed to the fact that as more and more particles are packed closely, forces due to lubrication forces are too high. This is resulted as the velocity of the aggregate is underestimated and the particles settle more slowly for such cases. The table below shows the change in aggregate velocity (rounded off to 3 significant digits) with and without lubrication interactions for various cluster sizes N for Tunable Monte Carlo cluster of fractal dimension 2.23.

| N | | 512 | 2048 | 6144 | 10240 |
|---|---|---|---|---|---|
| Dimensionless mobility radius | With lubrication | 24 | 58.9 | 136 | 199 |
| | Without lubrication | 17 | 31.8 | 52.5 | 66.8 |

*Table 5-2:Mobility radius (dimensionless) of Tunable Monte Carlo clusters of fractal dimension 2.231 with (top) and without lubrication(bottom) for different cluster sizes*

As the cluster size varies, the estimation of aggregate velocity with lubrication does not differ by much. But it would be practical for 10240 particles to fall more faster when $\sum_{1=1}^{10240} F_i$=10240 is acting on the aggregate owing to the increased mass.

Since there is no relative motion between the particles of a rigid aggregate, lubrication interactions are not necessary. They are appropriate only if particles are approaching each other. The velocity field inside the aggregate is also screened and the short-range interactions misrepresent this screening [Brady & Bossis (1991)]. In rigid clusters, all the particles move at the same velocity.

$$u_i = u_{cm} + \omega_{cm} \times (x_i - x_{cm}) \quad i = 1,2,..,N \qquad 5.2a$$

$$\omega_i = \omega_{cm} \qquad 5.2b$$

The inclusion of lubrication interactions is fundamentally erroneous in the rigid body framework. For a rigid body assembly which is isolated, there are no lubrication interactions. But if the assembly is deforming or there are multiple assemblies, lubrication forces are important [Swan (2011)].

Lattuada et.al (2001) justified the exclusion of lubrication interactions by comparing settling speed ratios obtained using Stokesian Dynamics with the experimental results and numerical analysis of Cichocki and Hinsen (1994). This is the ratio of settling speed of the aggregate with respect to the settling speed of a sphere having a volume equivalent to the aggregate. Lattuada et.al (2001) observed that the settling speed ratio decreases as the number of particles, N increases. These values are a lot less than the ones obtained experimentally by Cichkocki and Hinsen(1994).

### *5.2.1 Mobility Radius without Lubrication Interactions*

The fractals are now simulated using Stokesian Dynamics with only the inverse of the mobility matrix (F-T version). As soon as we remove the close-range interactions, the anomaly disappears and the proportionality between $R_m$ and $R_g$ is restored. The plots obtained below in Fig 5-2 show that the relationship between $R_m$ and $R_g$. Having confirmed the linearity, it now remains to establish the pre-factor in the correlation relation. The following figure is obtained by plotting $R_m$ and $R_g$ values of Tunable Monte Carlo clusters of different fractal dimensions. Error bars are smaller than symbol sizes and are hence ignored.

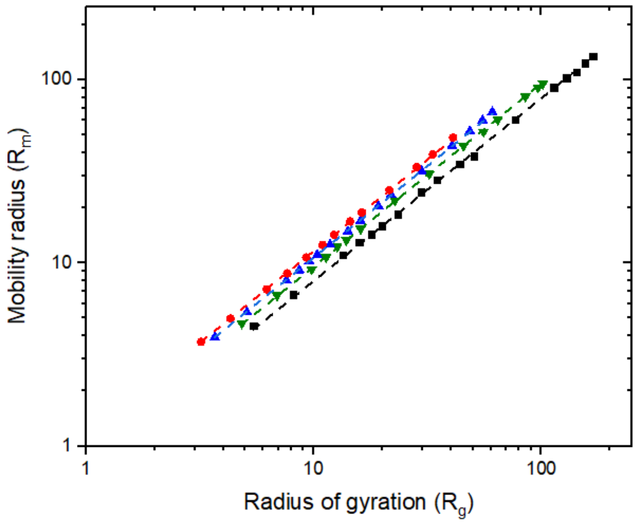



*Figure 5-2:Power-law fits for Mobility radius and radius of gyration for Tunable CCA(left) [* ■ $- D_f = 1.76$, ▼ $-D_f = 1.99$,▲ $-D_f = 2.23$,● $- D_f = 2.46$]

Hydrodynamic ratio is obtained for other fractals by using the slope and the values are shown in the table below. Error being less for the case of "Tunable" clusters indicate the level of fractality exhibited in this structure.

| **Fractal** | | **Fractal dimension** | $R_m/R_g$ |
|---|---|---|---|
| Tunable Monte Carlo clusters | | 1.76 | 0.793 ± 0.003 |
| | | 1.99 | 0.943 ± 0.002 |
| | | 2.23 | 1.084 ± 0.003 |
| | | 2.46 | 1.17 ± 0.003 |
| DLCCA | C = $10^{-3}$ $\gamma = 0$ | 1.84 | 0.94 ± 0.02 |
| CCA | C = $10^{-2}$ $\gamma = -0.5$ | 2.3 | 1.08 ± 0.02 |
| | C = $10^{-2}$ $\gamma = 0$ | 2.2 | 1.09 ±0.01 |
| | C = $10^{-2}$ $\gamma = 0.5$ | 2.12 | 1.10 ± 0.02 |
| DLA | | 2.47 | 1.24 ± 0.01 |
| Vicsek | | 1.75 | 1.01 ± 0.01 |
| Simple Cubic Unit | | 3 | 1.29 ± 0.003 |

*Table 5-3:Fractal dimension and hydrodynamic ratio (3 significant digits) observed for various fractals*

### 5.3 Variation of Mobility Radius on the Size of Cluster

Similar to the scaling relationship of $R_g$ and size of the cluster, aerosol literatures suggest another power law relationship between $R_m$ and size of the cluster given by $N \sim R_m^{D_m}$. This parameter $D_m$ was interpreted as "fractal dimension" by some workers. The above description yielded $D_m = 2.18$ for Schimdt-Ott(1988), whereas the same data analyzed by Wang and Sorenson (1999) determined the fractal dimension to be 1.75. This shows that the scaling relationship of $R_g$ and $R_m$ with the size of the cluster (N) yielded different exponents. Exponent for the former relation is called the "fractal dimension", whereas the exponent for latter relationship is known as "mass-mobility scaling exponent" [Sorenson (2011)].

A correlation to show the dependency of mobility radius with the cluster size N is given

$$R_m/a = R_o N^x \qquad 5.3$$

Here $R_o$ is the pre-factor and $R_o$ tends to unity in N=1 limit. $x = 1/D_m$ is the mobility-mass scaling exponent. Equation 5.3 is similar to the one employed by Sorenson (2011) and the expression is fitted for Tunable Monte Carlo clusters and the fit is shown in Fig 5-3.

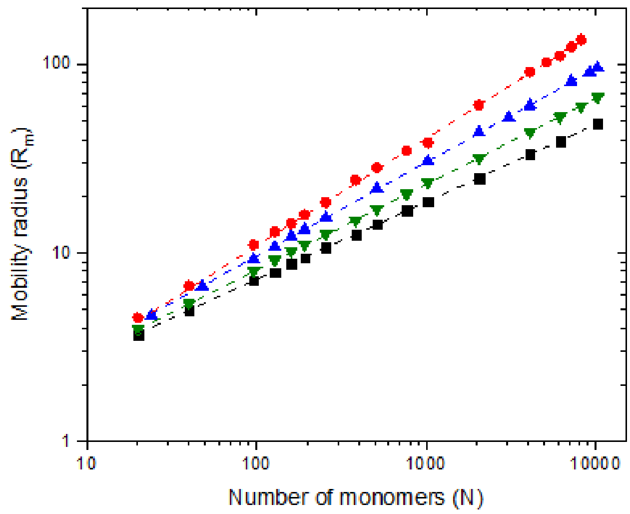



*Figure 5-3:Power-law fits of Mobility radius observed for Tunable CCA fractals of different fractal dimension from N=20 to N=10240 (* ■ $- D_f = 1.76$, ▼ $-D_f = 1.99$, ▲ $-D_f = 2.23$, ● $- D_f = 2.46$)

| **Fractal** | **Fractal dimension** | **Mobility mass exponent, $x$** | **Pre-factor, $R_o$** |
|---|---|---|---|
| Tunable Monte Carlo clusters | 1.76 | 0.561 ± 0.003 | 0.812 ± 0.047 |
| | 1.99 | 0.498 ± 0.001 | 0.966 ± 0.008 |
| | 2.23 | 0.452 ± 0.001 | 1.02 ± 0.01 |

| | | | | |
|---|---|---|---|---|
| | | 2.46 | 0.411 ± 0.001 | 1.09 ± 0.01 |
| DLCCA | C = $10^{-3}$ $\gamma = 0$ | 1.84 | 0.53 ± 0.01 | 0.85 ± 0.06 |
| CCA | C = $10^{-2}$ $\gamma = 0$ | 2.20 | 0.48 ± 0.01 | 1.11 ± 0.06 |
| DLA | | 2.47 | 0.438 ± 0.003 | 1.25 ± 0.03 |
| Vicsek | | 1.75 | 0.53± 0.01 | 0.82 ± 0.03 |
| Simple Cubic Unit | | 3 | 0.33 | 1.24 ± 0.01 |

*Table 5-4: Mobility-mass exponent and pre-factor obtained from power-law relationship between Mobility radius and size of the cluster.*

As fractal dimension increases, mass-mobility exponent decreases and the pre-factor $R_o$ increases. Note that for the dense aggregate ($D_f = 3$), mobility-mass exponent $x = \frac{1}{3}$ which also suggests $D_f = D_m$ for this particular case. We also prove for the other fractals that $D_f \neq D_m$.

To compare with literature, the equation obtained $R_m = 0.852\, N^{0.53}$ for DLCCA ($D_f = 1.78$) is consistent with the ones suggested by Dastanpour et al. (2016) and Sorenson (2011).

### 5.4 Dependency of the hydrodynamic ratio on fractal dimension

The hydrodynamic ratio $R_m/R_g$ depends on the fractal dimension of the aggregate as indicated by various studies [Van Sarloos (1987), Gmachowski (1996)]. Here, it is found that fractals with higher fractal dimension appear to have a hydrodynamic ratio relatively larger compared to the fractals with lower fractal dimension. This is mainly because of the radius of gyration is small for fractals with large $D_f$. These results are also consistent with general observations by Fellay et al. (2013), Nyugen (2007) ,Gmachowski (1996) and Lattuada et al. (2011). We have now extended the results for the hydrodynamic ratio across large N and wider $D_f$. The following graph shows the dependency of hydrodynamic ratio with respect to the fractal dimension for all the fractals.

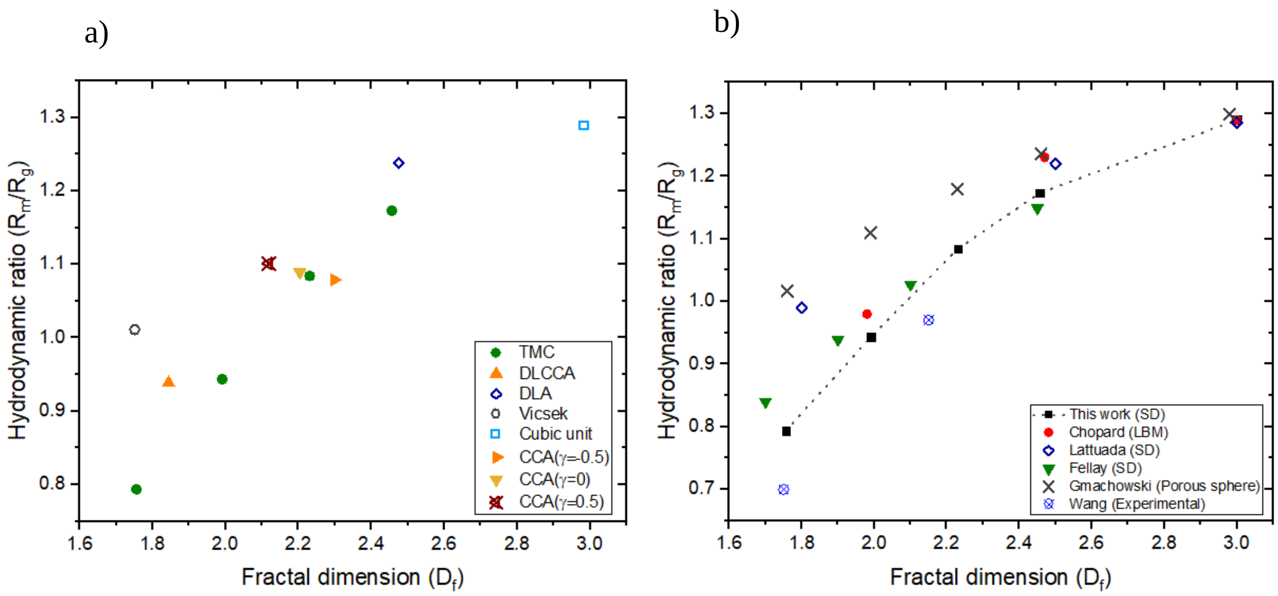



*Figure 5-4 a: Hydrodynamic ratio for different fractal dimensions(left)[* □ $-simple\ cubic\ unit,\ \circ -Vicsek, \diamond - DLA,$▲ $-DLCCA\ (\gamma = 0),$▶ $-DLCCA\ (\gamma = -0.5),$◀ $-DLCCA\ (\gamma = 0.5),$● $- Tunable\ Monte\ Carlo$]; *b: Comparison of different* $D_f$ *of Tunable Monte Carlo clusters with previous works on CCA(right)*

Maximum value of $R_m/R_g$ observed to be 1.29 for simple cubic unit having a fractal dimension of 3. This matches with the analytical expression of ratio $R_m/R_g = \sqrt{(5/3)}$ for a solid sphere

On the other hand, for fractals with $D_f$<2, $R_m/R_g$ turns out to be less than unity. This is related to the transparency of the object for fluid flow in the interior regions and the consequent to resistance. The hydrodynamic ratio for Vicsek is an exception where it is approximately 1.011. But a similar behavior was observed by Filippov (2000) and the $R_m/R_g$ obtained matches the values obtained by Seto et al. (2011) as shown in Table 5-8.

$R_m/R_g$ is compared with different set of results.

*a. Numerical simulations*

Chopard (2006) employed LBM for $100 \leq N \leq 20000$ while Lattuada (2010) and Fellay (2013) used Stokesian Dynamics for small N. All these simulations were conducted on Tunable CCA. Chopard et al. (2006) is the only study that we know to cover a wide range of N.

| **Chopard et.al(2006)** | **Lattuada(2010)** | **Fellay(2013)** | **This study -$R_m/R_g$** |
|---|---|---|---|
| - | 0.99($D_f$ = 1.8) | 0.84($D_f$ =1.7) | 0.79($D_f$ = 1.76) |

| | | | |
|---|---|---|---|
| 0.98($D_f = 1.98$) | - | 1.03($D_f = 2.1$) | 0.94($D_f = 1.99$) |
| 1.23($D_f = 2.47$) | 1.22($D_f = 2.5$) | 1.15($D_f = 2.45$) | 1.17($D_f = 2.46$) |
| 1.29($D_f = 1.29$) | 1.28($D_f = 3$) | - | 1.29($D_f = 3$) |

*Table 5-5: Comparison  for Tunable CCA*

Different methodologies yield same values for compact aggregates while $R_m/R_g$ for an arbitrary $D_f$ is not exactly the same but within close range.

b. *Kirkwood-Riseman theory*

Chen (1988) used Kirkwood-Riseman (1948) theory with RPY tensor and determined hydrodynamic ratio for three types of fractals while we compare with two of those.

| **Fractal type** | $\boldsymbol{D_f}$ | **N range** | **Chen(1988)- $\boldsymbol{R_m/R_g}$** | **This study -$\boldsymbol{R_m/R_g}$** |
|---|---|---|---|---|
| DLA | 2.44 | $100 \le N \le 900$ | 1.14 | 1.15±0.01($100 \le N \le 900$) |
| RLCCA | 2.1 | $100 \le N \le 400$ | 0.97 | - |
| DLCCA | 1.8 | $50 \le N \le 350$ | 0.87 | 0.90 ±0.02(64$\le N \le 389$) |

*Table 5-6 : Comparison with Chen (1988) for DLA and DLCCA*

c. *Porous sphere model* – Gmachowski (1996) uses Brinkman equation on Cluster-cluster aggregates to determine hydrodynamic ratio which is a function of fractal dimension and structure factor obtained by scaling relations The values obtained for $D_f = 1.75$ itself is greater than 1, which is not the case with any other method. As fractal dimension increases(at $D_f = 2.45$), the values match with this study.

d. *Experiment*

While studying the hydrodynamics of colloidal silica aggregates with $D_f$=2.1, Wiltzus(1987) obtained $R_m/R_g$ of 0.72 experimentally. Pusey et al.(1987) later recorrected this to 0.98. At $D_f$=2.15 with 0.95, Wang (1999) obtains $R_m/R_g$ value that is similar to the current investigation.  However, Wang (1999) $\frac{R_m}{R_g}$ of 0.7 at $D_f = 1.75$ is low in comparison to other results of similar fractal dimension.

### 5.5 Dependency of the hydrodynamic ratio on the size of the cluster

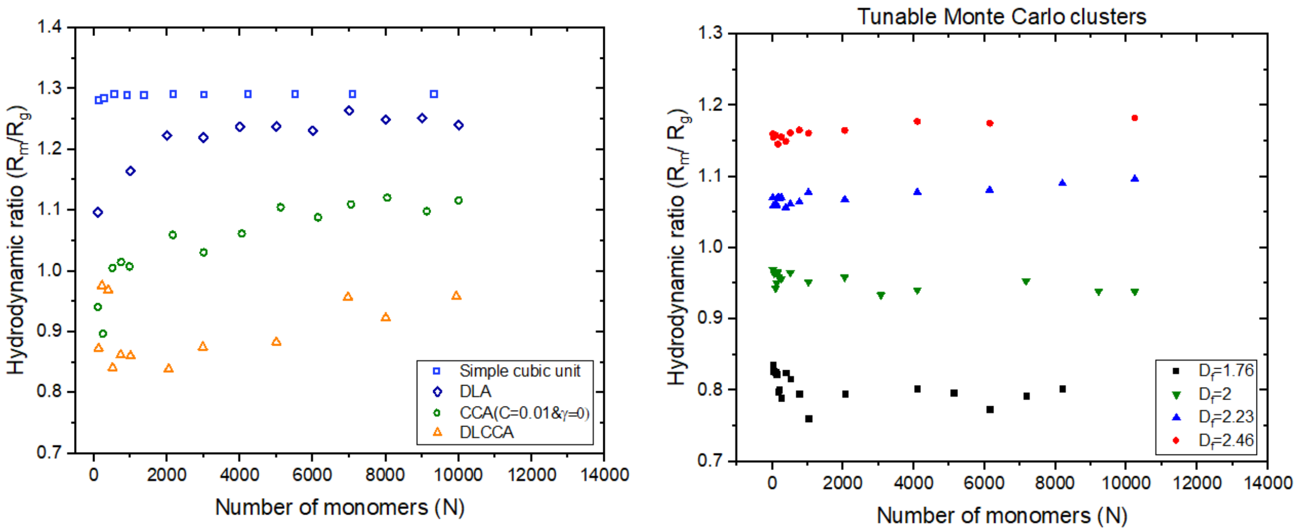



*Figure 5-5:Dependency of hydrodynamic ratio with the aggregate size N for Diffusion Limited Clusters (Left) and Tunable Monte Carlo Clusters (right)*

Sorenson(2011) fits $R_m/R_g$ using the relation: $\frac{R_m}{R_g} = \alpha N^{\beta}$. This fit is essentially a test of linearity by demonstrating that $\beta \sim 0$ . This is confirmed for our case too as seen in Table ().

| Fractal | Fractal dimension | $\alpha$ | $\beta$ |
|---|---|---|---|
| | 1.76 | 0.85 | $-8 \times 10^{-3}$ |
| Tunable Monte Carlo | 1.99 | 0.98 | $-4 \times 10^{-3}$ |
| clusters | 2.23 | 1.04 | $4.1 \times 10^{-3}$ |
| | 2.46 | 1.13 | $4.4 \times 10^{-3}$ |
| DLCCA | 1.84 | 0.87 | $5.4 \times 10^{-3}$ |
| | 2.2 | 0.80 | $3.54 \times 10^{-2}$ |
| DLA | 2.47 | 0.96 | $2.95 \times 10^{-2}$ |
| Simple cubic unit | 3 | 1.27 | $1.5 \times 10^{-3}$ |

*Table 5-7:Demonstration of N independence of $R_m/R_g$*

$\beta$ values are very small ie. Less than 0.04. This confirms there is a very weak dependency with the cluster size and this is observed only when N is less. These might be the result of anisotropic effects [Nyugen (2007)].

1. *Simple cubic unit*: The ratio is relatively low for small N as they are not completely spherical and approaches an asymptotic value of 1.29 for large N.

2. *DLA*: The increase in $R_m/R_g$ for DLA is exactly the same as observed by Warren(1994), Chen et.al(1988), Nyugen(2007) and Seto et.al(2011). Presence of branches which are more exposed to the fluid is responsible for the hydrodynamic ratio for DLA. There is a slow approach to the asymptotic regime which is obtained when N=2000. It's worth noting that Seto et al. (2011) found that when N is small, $R_m/R_g$ values are small as well. Figure 5-6 compares the results obtained in this work with Seto et.al(2011) and Chen et.al(1988).
3. *DLCCA*: The asymptotic regime approaches faster when compared to DLA. This was also observed by Warren (1994) who compares the structure of both DLCCA and DLA. Warren (1994) also suggested that being centrosymmetric aggregates, asymptotic limit is approached only when N→ ∞ for the case of DLA and the centre of mass of the aggregating clusters does not lie near the centre of mass of the overall aggregate for DLCCA.
4. *Tunable Monte Carlo Clusters*: $R_m/R_g$ shows a decreasing trend for fractals with Df < 2 and an increasing trend for $D_f$>2 at low cluster sizes. Since the aggregation model is such that fractality is ensured, there are no fluctuations with respect to cluster size.
5. *Vicsek*: $R_m/R_g$ is decreasing for Vicsek fractals and the values are not constant. The ratio decreases and approaches 1 as N increases. The values obtained match the $R_m/R_g$ values obtained by Seto(2011).

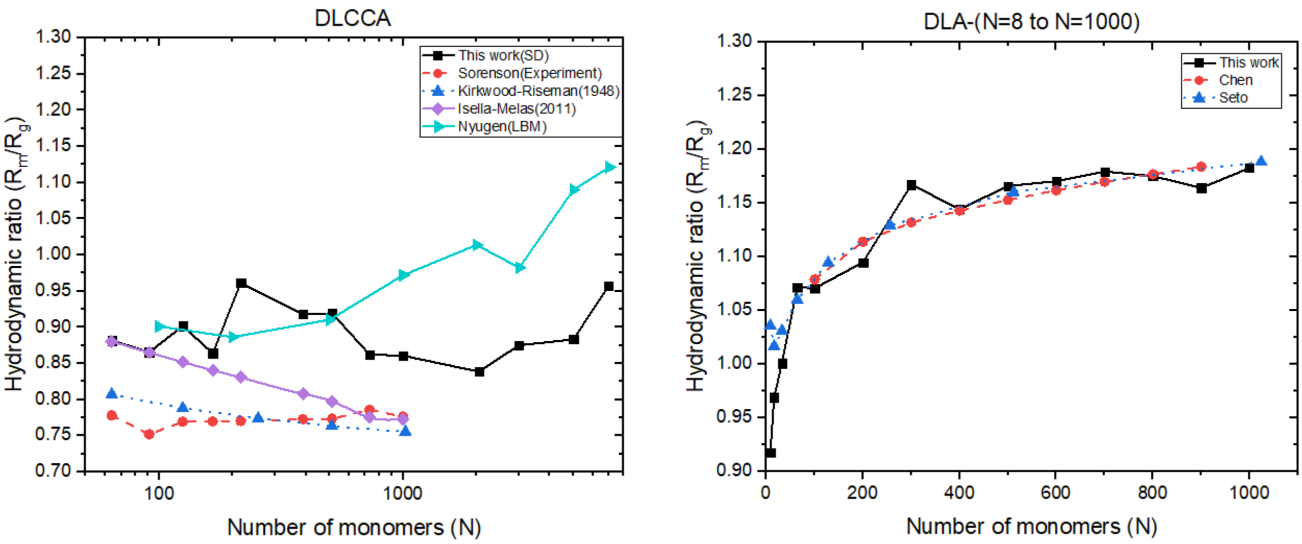



*Figure 5-6:Comparison between different studies on the variation of the hydrodynamic ratio with the aggregate size N for Diffusion Limited Clusters: DLCCA(left) and DLA(right)*

| *N* | Seto et.al(2011) | This work |
|---|---|---|
| *7* | 1.23 | 1.28 |
| *49* | 1.10 | 1.09 |
| *343* | 1.04 | 1.03 |

*Table 5-8: Comparison of $R_m/R_g$ for Vicsek with the work of Seto et al. (2011)*

**5.6 Effect of interparticle distance on the hydrodynamic ratio**

When the interparticle distance between the particles in a rigid aggregate increases one can expect the fractal dimension to be constant and the pre-factor to decrease. Pre-factor is a function of void fraction which increases as the gap increases. DLA with N ranging from 100 to 1000 has been chosen to show the effects of the interparticle distance on the hydrodynamics as well as the geometry.

A fractal dimension of $D_f = 2.43$ is consistent for all the cases. The table below shows the change in pre-factor, $k_f$ and the hydrodynamic ratio for different interparticle distances. This throws some light into the changes in hydrodynamic ratio with the pre-factor. The relation between $R_m/R_g$ with the interparticle distance for N=100 can be given as

$$\frac{R_m}{R_g} = \frac{1.250}{(Interparticle\ distance)^{0.19}}$$

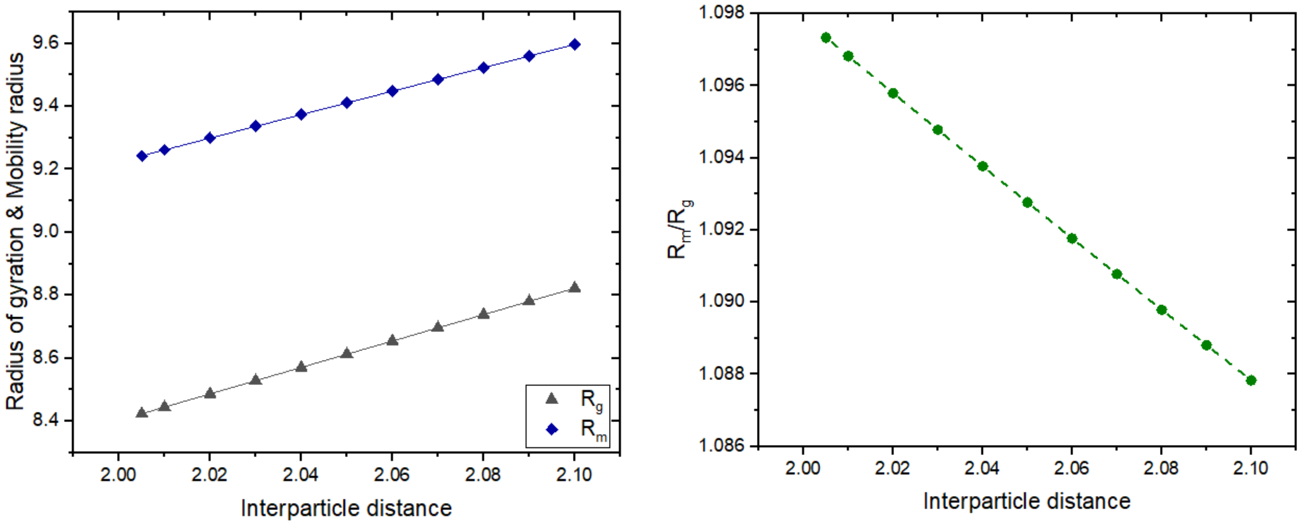



*Figure 5-7:Comparison of the increase in $R_g$ and $R_m$ with respect to changes in interparticle distance for N=100 in a DLA (left). Changes in hydrodynamic ratio with increase in interparticle distance for N=100 in a DLA.*

| Interparticle distance | Pre-factor, $k_f$ | $R_m/R_g$ |
|---|---|---|
| 2.01 | 0.526 | 1.1377 |
| 2.02 | 0.520 | 1.1371 |
| 2.03 | 0.514 | 1.1365 |
| 2.1 | 0.473 | 1.1325 |

*Table 5-9a: Pre-factor and $R_m/R_g$ for different interparticle distances(N=100 to N=1000)*

The hydrodynamic ratio does not change greatly as the interparticle distance increases, as seen in Table 5-9a. If $\frac{R_m}{R_g}|_{r=2.01} = 1.097$ becomes $\frac{R_m}{R_g}|_{r=2.1} = 1.088$ for N=100, it varies from $\frac{R_m}{R_g}|_{r=2.01} = 1.24$ to $\frac{R_m}{R_g}|_{r=2.1}$ =1.238 for N=10000. The change is not appreciable even for the largest cluster size considered. The interparticle distance increases $R_m$and $R_g$ and power-law fit parameters for N=100 is provided below.

| $y = A_o x^{b_1}$ | $A_o$ | $b_1$ |
|---|---|---|
| $R_g = A_o r^{b_1}$ | 4.20 | 1 |
| $R_m = A_o r^{b_1}$ | 5.25 | 0.81 |

*Table 5-9b: Parameters involved in the relation between the radius of gyration and mobility radius with the interparticle distance for N=100*

### 5.7 Coordination number

Coordination number is the average number of monomers connected to a particle in an aggregate and may also affect the properties of a physical structure [Friedlander (1997)]. Nyugen (2017) determined the effect of coordination number on the hydrodynamics of clusters generated using CCA. The hydrodynamic ratio of two separate clusters with the same fractal dimension and pre-factor was different. This was due to the fact that the coordination numbers of the two aggregates were different.

Coordination number for different fractal dimensions of all the fractals generated in this study are shown in the table below.

| **Fractal** | **Coordination number** |
|---|---|
| Tunable CCA, DLA, Vicsek | 2 |
| DLCCA | 2.2 |
| Simple Cubic Unit | 6 |

*Table 5-10: Coordination number observed for different fractals*

Figure 5-8 shows coordination number and forces acting on spheres with a colormap for DLA with N=300. This demonstrates that particles connected to only one neighbouring particle(●) experiences greater force(●).

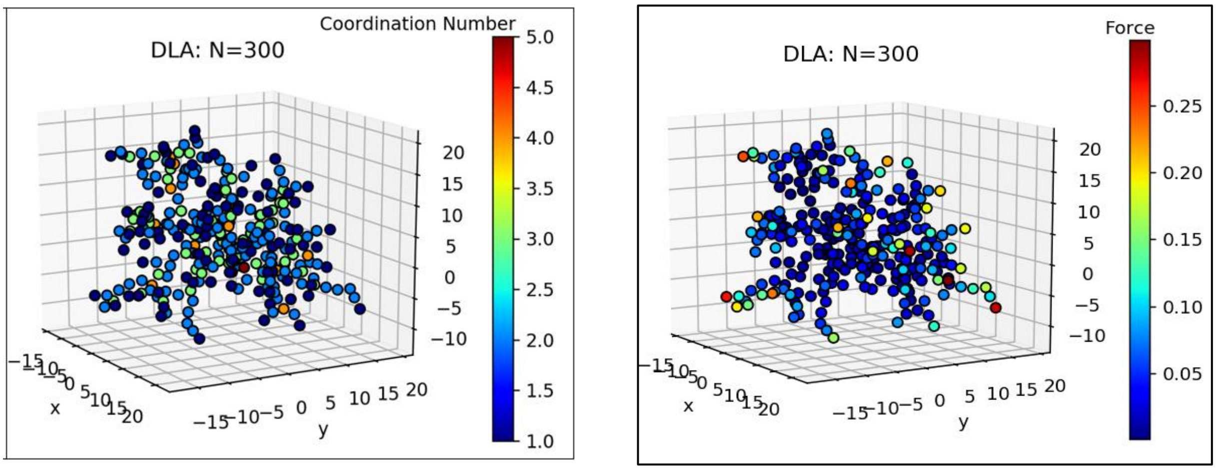



*Figure 5-8: Force distribution(left) and coordination number(right) on DLA fractals with df=2.45 and N=300. Forces are acting in vertical direction.*

Forces and coordination number are averaged from regions $0 < r \leq 4, 4 < r \leq 8$ and so on and the data is represented at the intermediate point (r=6 for regions $4 < r \leq 8$) for all regions in figure 5-9 which shows force distribution for different fractals with N=2000. Fractals with fractal dimension of 2.2 and 2.45 are considered since the lowest fractal dimension does not reveal any connection between the two as they are more open structures. This force distribution will remain the same for the particular fractal and $D_f$ irrespective of the number of spheres, N. Changes in coordination number (●) remains small as the distance from centre changes for all the fractals used in this study. The maximum coordination number for Tunable CCA may be seen at the centre before the drop begins (figure 5-9a&c). DLA is dense towards the centre, and as porosity increases, more branches with only one sphere appear, lowering the coordination number (figure 5-9d).

This force distribution also demonstrates how fractals are screened. Both the plots indicate that average force experienced are minimal in the interior and they gradually increase towards the periphery. Even the maximum forces encountered are small especially around the central regions. This would have happened because the spheres that were exposed to the fluid flow received the majority of the fluid flow while preventing most of it from reaching the interior.

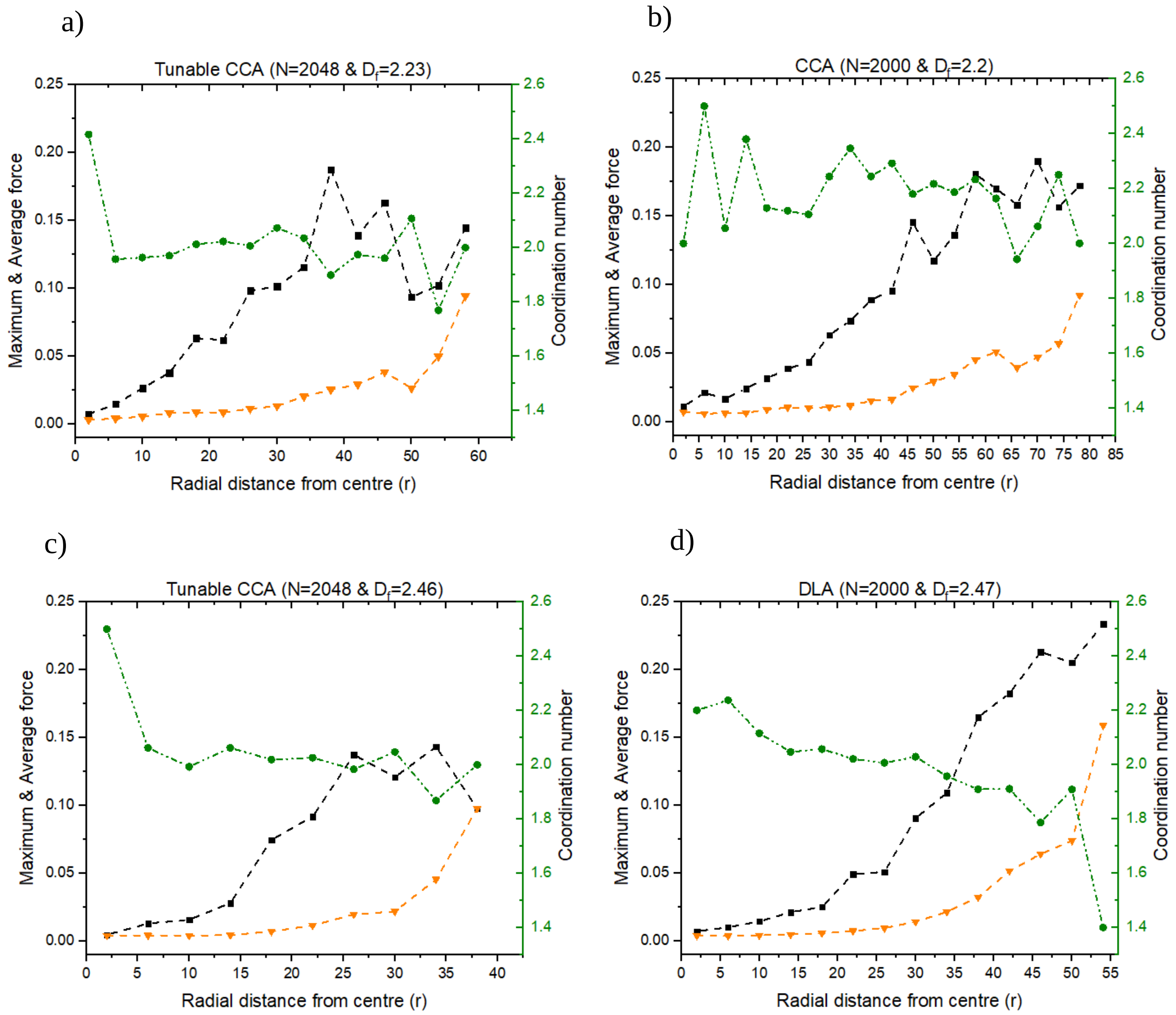



*Figure 5-9: Force distribution for different fractals with N = 2000 a) Tunable Monte Carlo cluster with df = 2.23 and N=2048 (top left). b) DLCCA with df = 2.20 and N=2000 (top right). c) Tunable Monte Carlo cluster with df = 2.46 and N=2048 (bottom left). d) DLA with df = 2.47 and N=2000(bottom right). Maximum force(■), Average force(▼), Coordination number(●)*

For the case of Tunable Monte Carlo cluster ($D_f = 2.46$) and DLA (figure 5-9 c &d), the force distribution illustrates the relationship between the two-coordination number (●) and the forces experienced (■). As we move from the centre, the particles are connected to relatively less number and experience greater forces. This can be observed such that as coordination number decreases, maximum forces increases and vice versa.

For the case of Tunable Monte Carlo cluster ($D_f = 2.23$) (figure 5-9a), the outer regions show a clear connection between the two-coordination number and maximum forces whereas for the case of CCA ($D_f = 2.2$) (figure 5-9b), the relation between the two seems to be changing at different regions. However, all four plots indicate that massive changes in coordination number affects the maximum forces experienced by a large amount. Therefore, this study is concluded having explored an important connection between the number of connecting spheres and forces experienced.

## 5.9 Conclusion

Hydrodynamic ratio, $R_m/R_g$ is clearly a function of fractal dimension which has already been well established. Fractals with higher fractal dimension have high values of hydrodynamic ratio. Maximum limit for $R_m/R_g$ exhibited by Gmachowski (1995) for $D_f = 3$ is 1.29 is obtained for the case of simple cubic unit. Various other fractal dimensions also show agreement with values observed by other researchers. The systematic correlations between hydrodynamic ratio and $D_f$ is being provided, which is useful in various transport problems connected with aggregates. For example, in the problem of gravitation settling or that of particle deposition in the lung, these correlations offer a unique way of expressing the dependencies in terms of experimentally measured parameters. For gravitational settling problems, which involve volume equivalent and the mobility diameters, the present relationship offers a closure by relating all of them to $D_f$(and monomer size).

Hydrodynamic ratio, $R_m/R_g$ is constant for all fractals and does not depend on the size of the clusters except for diffusion limited aggregates which exhibit initial decline or growth before reaching the asymptotic regime. This ratio is expected to be the same for even with $N\approx 100000$. Given the radius of gyration and fractal dimension from the structure, one can obtain the drag force acting on it since mobility radius is known to be associated to the drag acting on the aggregate.

The coordination number has an effect on the forces acting on aggregates with $D_f > 2$. It’s connection with the hydrodynamic ratio or the drag force is not studied in this study.

## Appendix 5A: $R_m$ and $R_g$ plots

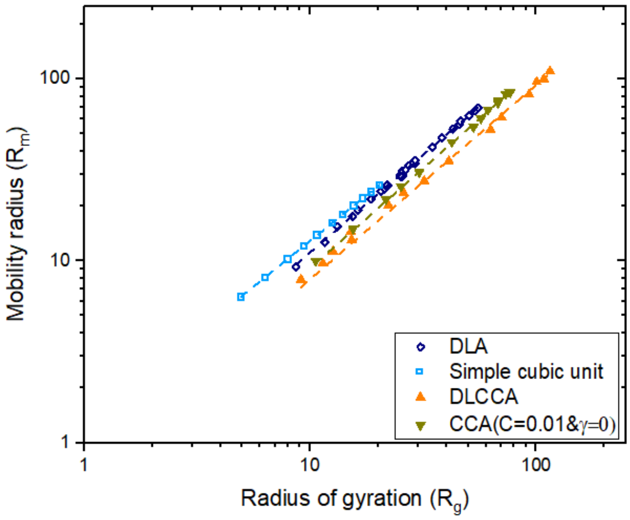



*Figure 5A-1: Power law fits for mobility radius and the radius of gyration for DLA, simple cubic unit and DLCCA*

# Chapter 6

# Screening effects

## 6.1 Introduction

Aggregates are porous and depending on the aggregation phenomena, they may be dense or sparse. The particles which are completely exposed to the fluid generally experience greater forces, hence shielding other particles from bulk of the fluid flow. This also prevents or greatly slows down the fluid penetrating the interior regions of the aggregate. The basic distinguishing feature of hydrodynamics of aggregates from that of spherical objects is the question of fluid penetration into the interior regions of the objects.

Previous chapter dealt with the study of force distribution for different aggregates and different types of aggregates had different flow behaviors. It is well known fluid flows within the body of an aggregate, but the extent of which will give interesting aspects on the relation between how much fluid penetrates through different bodies of fractals. It stands to reason to state that for given aggregate, the higher the fluid penetration higher will be the resistance and correspondingly, higher will be the mobility radius. There must be some correlation between the mobility pre-factors, fractal dimension $D_f$ and the extent of fluid penetration.

## 6.2 Permeability models

Porous sphere models showed the effect of the permeability $\kappa$ to determine the drag acting on them and can also be used to determine 'hydrodynamic ratio'. This permeability affects not only the hydrodynamics, but also the breakup of aggregates either internally or the particles around the edges [Kim et al. (2002)].

$$\phi = N\left(\frac{a}{R_{max}}\right)^3$$

The distance from the centre of mass to the outermost particle in the cluster gave us $R_{max}$, which is used to obtain average volume fraction of particles in the aggregate ($\phi$). The aggregate holding N number of particles are of the same size i.e.unit radius(a)

| Permeability | Source |
|---|---|
| $\boldsymbol{\kappa_{DL} = \frac{2a^2}{9\phi}}$ | Happel and Brenner (1991) |
| $\boldsymbol{\kappa = \kappa_{DL}\frac{6 - 9\phi^{\frac{1}{3}} + 9\phi^{\frac{5}{3}} - 6\phi^2}{6 + 4\phi^{\frac{5}{3}}}}$ | Happel (1958) |

| | |
|---|---|
| $\boldsymbol{\kappa = \kappa_{DL}\left(1+\frac{3}{\sqrt{2}}\phi^{\frac{1}{2}}+\frac{135}{64}\phi\ln\phi + 16.456\phi+\cdots\right)}$ | Howells (1974), Hinch(1977), Kim and Russell(1985) |
| $\boldsymbol{\kappa = \kappa_{DL}\left(1+\frac{3}{4}\phi\left(1-\sqrt{\frac{8}{\phi}-3}\right)\right)}$ | Brinkman (1947) |
| $\boldsymbol{\kappa = \frac{a^2(1-\phi)^3}{9c\phi^2}}$ **where c≈5** | Kozeny-Carman equation |
| $\boldsymbol{\kappa = \frac{a^2}{16\phi^{1.5}(1+56\phi^3)}}$ | Davies (1952) |

*Table 6-1: Permeability expression given in different models*

Permeability, $\kappa$ depends on the solid fraction of particles. The expression of Happel and Brenner is the dilute limit and is obtained for spheres but the hydrodynamic effects are not considered. Other theories obtain the dilute limit values at low solid fractions and account for the effect of other spheres [Kim et al. (2002)].

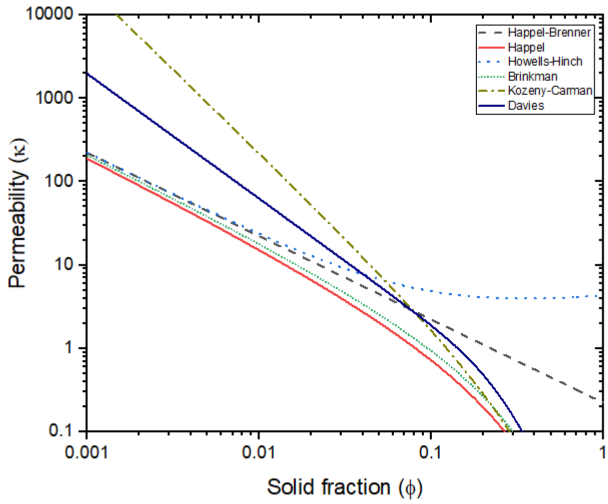



*Figure 6-1:Permeability as a function of solid fraction($\phi$)*

Happel (1958) and Brinkman (1947) are in close agreement for most of the range. Kozeny-Carman equation is used for dense systems and is valid when volume fraction is great than 0.3[Happel & Brenner (1991)]. The relation given by Davies(1953) is used for fibrous porous media and found to be appropriate for highly porous fractal aggregate [Kim et.al (2002)].

Kim et.al (2002) used Stokesian Dynamics to obtain fluid velocity indirectly by the use of “tracer” particle of radius $10^{-3}a$ flowing through the aggregate such that it does not overlap any of the constituent particles. Average velocity was obtained in three different directions and the results obtained shows excellent agreement with Brinkman equation in conjunction with Davies permeability expression for the case of low fractal dimensions. Also, Davies' permeability expression gives velocity matching with Kim et al.(2002) result near the ends of the cluster.

### 6.3 Methodology

The fractal is exposed to an ambient fluid velocity of unity in three different directions (x y and z) and the force moments acting on individual spheres owing to this flow are obtained using Stokesian Dynamics.

$$\begin{pmatrix} F^{\alpha} \\ T^{\alpha} \end{pmatrix} = R \begin{pmatrix} U - u^{\infty} \\ \Omega - \Omega^{\infty} \end{pmatrix}$$

Fluid velocity, $u$ is obtained at various points within using multipole expansion.

$$u_i(x) = u_i^{\infty}(x) - \frac{1}{8\pi\eta} \sum_{\alpha} \left[ -\left\{ 1 + \frac{1}{6} a^2 \nabla^2 \right\} J_{ij}(x - x^{\alpha}) F_j^{\alpha} + \frac{1}{2} \in_{jkl} \nabla_k J_{il}(x - x^{\alpha}) L_j^{\alpha} \right]$$

It can be seen that the expansion includes only the force and torque acting on the spheres. These forces result in the generation of a disturbance field that slows down the flow and reduces the amount of fluid that penetrates the pores of the aggregate. Average of the fluid velocity within different concentric spherical shells from the centre of mass is determined and the changes in fluid velocity for different fractal dimension can be compared.

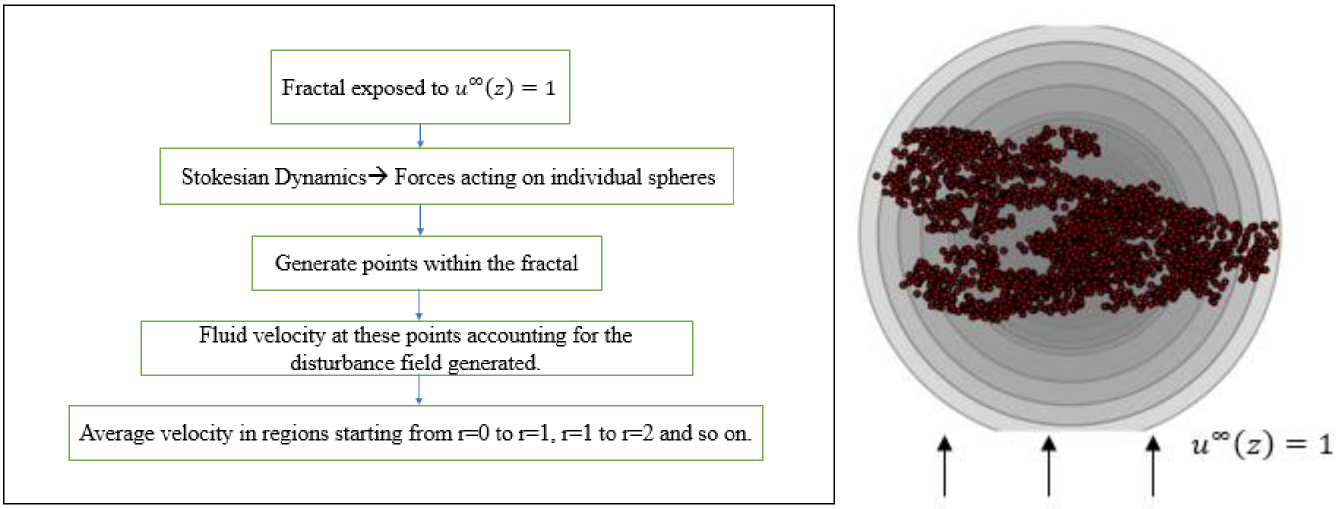



*Figure 6-2:a) Flowchart showing how velocity studies were conducted for flow in z-direction b) Concentric shells considered for velocity in different regions $D_f = 2.25$ and N=2048*

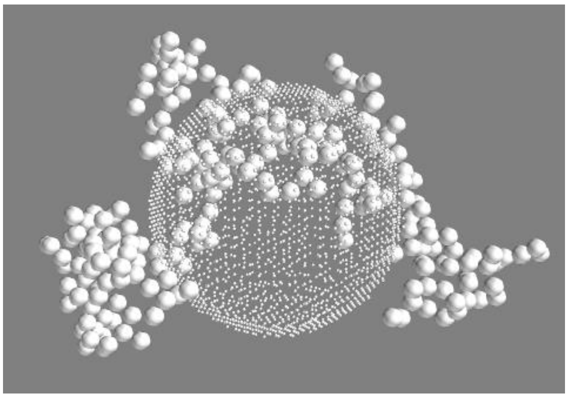
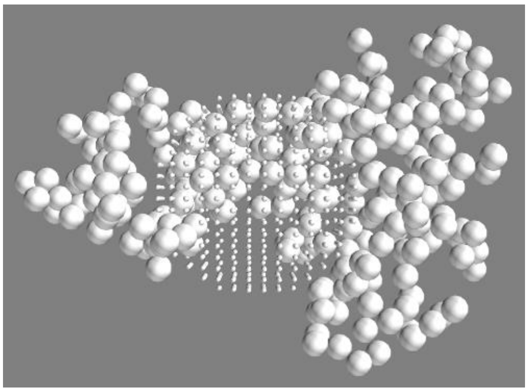
 

*Figure 6-3:Random points chosen for obtaining velocities for Tunable Monte Carlo cluster(N=256) with $D_f$=2(left) and $D_f$=2.25(right). The larger spheres shown are the constituent particles of the aggregate and the smaller ones are the points chosen for r=14 to r=15(left) and r=7 to r=8(right). The points shown here are for picturing how the points are located and are a lot less compared to the actual points chosen.*

Number of points are varied based on fractal dimension and the size of the cluster extending from one end of the cluster to the other end. It covers the entire region till $R_{max}$. Average number of points for different $D_f$ is shown in the table below.

| $D_f$ | N=96 | N=2048 |
|---|---|---|
| 1.76 | $40 \times 40 \times 40$ | $150 \times 150 \times 150$ |
| 2 | $35 \times 35 \times 35$ | $120 \times 120 \times 120$ |
| 2.23 | $30 \times 30 \times 30$ | $100 \times 100 \times 100$ |
| 2.46 | $30 \times 30 \times 30$ | $80 \times 80 \times 80$ |

*Table 6-2: Number of points generated for two extreme cluster sizes chosen for different fractal dimensions*

### 6.4 Single particle in a fluid

Flow around a single sphere shows the disturbance generated by the presence of a particle. Fig 6-4 shows the particle of unit radius moving in the positive z-direction and the resulting velocity disturbances are high near the particle and decays at the rate of $\sim$1/r as we move away. Fluid velocity is negligible at a point far away from the sphere. The origin of the arrow gives the position at which velocity is observed and the length of the arrow shows the magnitude of velocity.

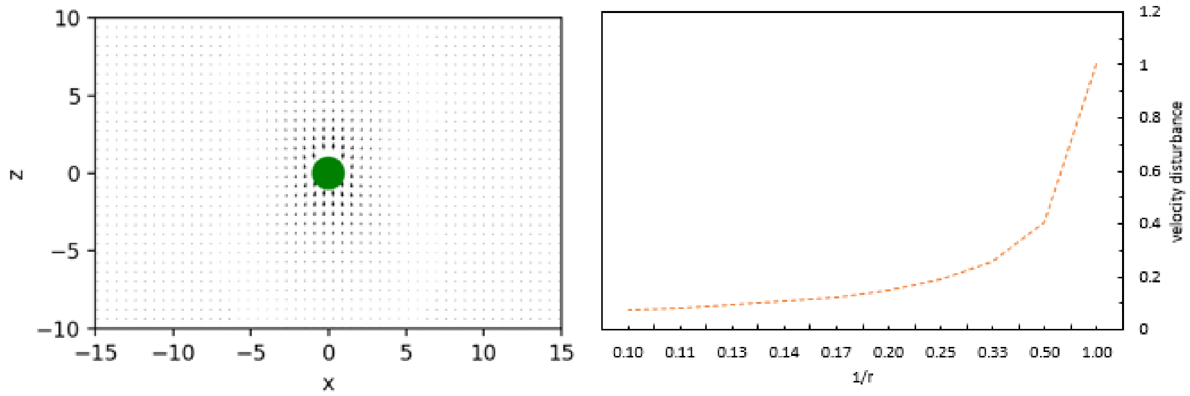



*Figure 6-4: a) Velocity field generated as particle is moving in positive z-direction. b) Velocity disturbances decaying with 1/r for a single stationary sphere in a moving fluid.*

Since Green's function is a function of 1/r, the velocity disturbance also decays as 1/r as shown in the case of a single sphere.

Similarly, Fig 6-5 shows the effect for three different cases for particle of unit radius placed at origin:

a) Particle in a moving fluid.

Fluid is moving with $u^{\infty}(z) = 2$ and the resulting drag force on the particle is 2. Fluid velocity observed on the surface is the same as that of the particle and thus satisfying the boundary conditions.

b) Both particle and fluid moving in opposite directions.

The fluid is moving with $u^{\infty}(z) = 2$ and the particle is moving with $U(z) = 1$. Total drag acting on the particle is 3. Similar to the above case, the fluid velocity on the surface is the same as that of the spherical particle. The fluid in regions closer to the particle has been dragged down together with the particle.

C) Particle moving in a stagnant fluid.

The particle is moving with $U(z) = 2$ and the resulting drag force is 2. The fluid as observed is dragged along with the particle and the velocity of fluid at the surface is same as the particle.

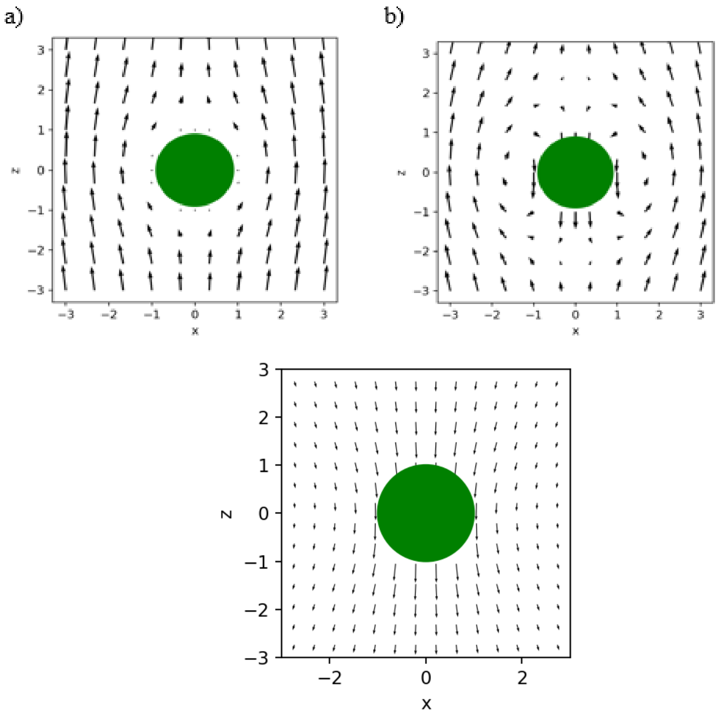
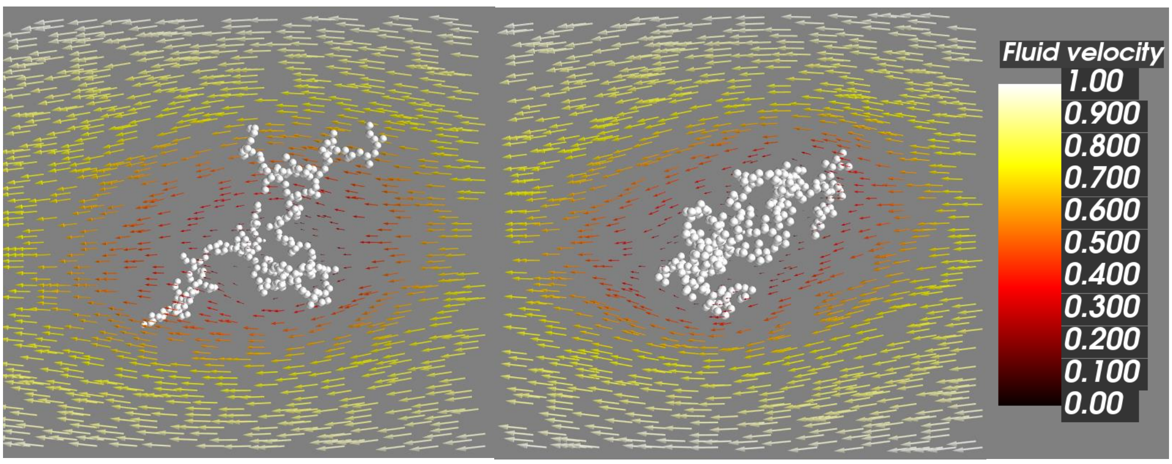



*Figure 6-5:a) Particle in a moving fluid. B) Both particle and fluid moving in opposite directions. c) Particle moving downward in stagnant fluid*

### 6.5 Visualization

Chellam and Weisner (1993) observed that as the fractal dimension decreases, permeability increases and hence the streamlines approach a uniform flow within the aggregate. Figure below shows fluid flowing through fractals of four different fractal dimensions.



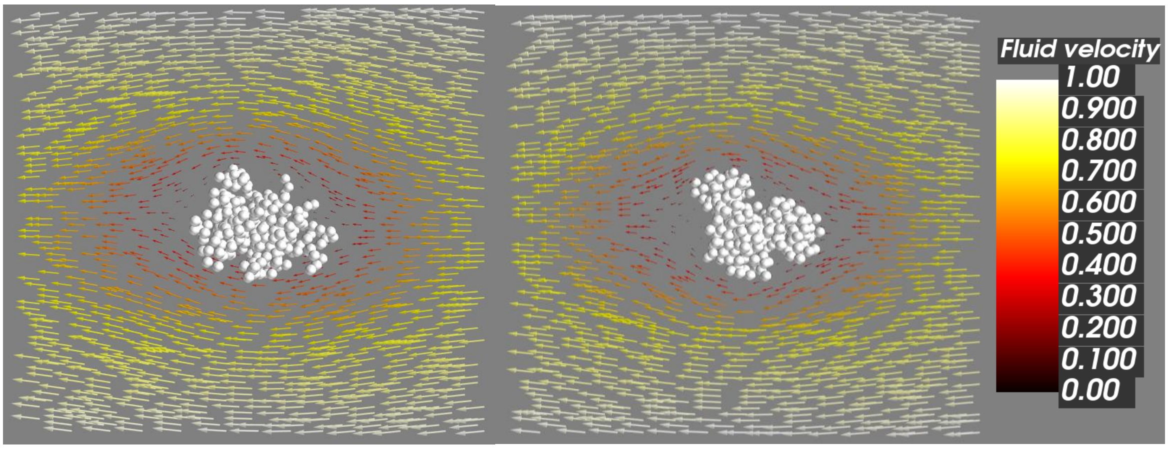



*Figure 6-6:Quiver plots showing the flow of fluid with a velocity of unity in negative x-direction (from right to left) through fractals of* $D_f$ *=1.76(top left) , 2(top right), 2.23(bottom left), 2.46(bottom right)*

Fluid flow is shown at the *y=0* with the intensity reduced due to the presence of aggregate. It can be seen that more fluid flows for cases where $D_f \leq 2$. 3D-Visualization is done by using Mayavi [Prabu Ramachandran (2001)].

### 6.6 Penetration depth

Debye et al. (1948) uses a 'shielding length' to study the exponential decay of velocity for the case of polymer beads. Hwang et al.(2000) obtained 'penetration depth' for measuring the distance that the fluid can penetrate into a porous medium. Similarly, we introduce "penetration depth"- a key parameter to describe the extent of fluid flow within an aggregate. It may be defined as the distance from the cluster surface towards its interior at which the fluid velocity falls by a factor *1/e* and is estimated from the decay profile of the fluid velocity within the aggregate. Formally, it may be written as $u^*(r^*) = u_s \exp\left(-\frac{R_c^* - r^*}{\delta^*}\right)$, where $u^*(r^*)$ is the velocity of the fluid at a dimensionless radial distance $r^* \left(= \frac{r}{R_G}\right)$ within the cluster, $u_s$ is a parameter representing velocity at the aggregate surface ($r^*$=$R_c^*$) and $\delta^* = \frac{\delta}{R_g}$, ($\delta$ is the penetration depth). $R_c^*$ varies from 1.5 to 1.9 for Tunable Monte Carlo clusters. We can absorb $e^{-\frac{Rc*}{\delta*}}$, into the pre-factor and rewrite $u^* = u_i \exp\left(\frac{r^*}{\delta^*}\right)$ where $u_o$ is the core velocity. The decay of fluid velocity as seen from the exterior is now transformed as an exponential increase with respect to radial distance r, as seen from the interior core. A log-linear fit to the simulation data on $u^*$ and $r^*$ would yield $\delta^*$. Figure 6-7 shows the fluid velocity within Tunable Monte Carlo clusters of four different fractal dimensions and N=1024.

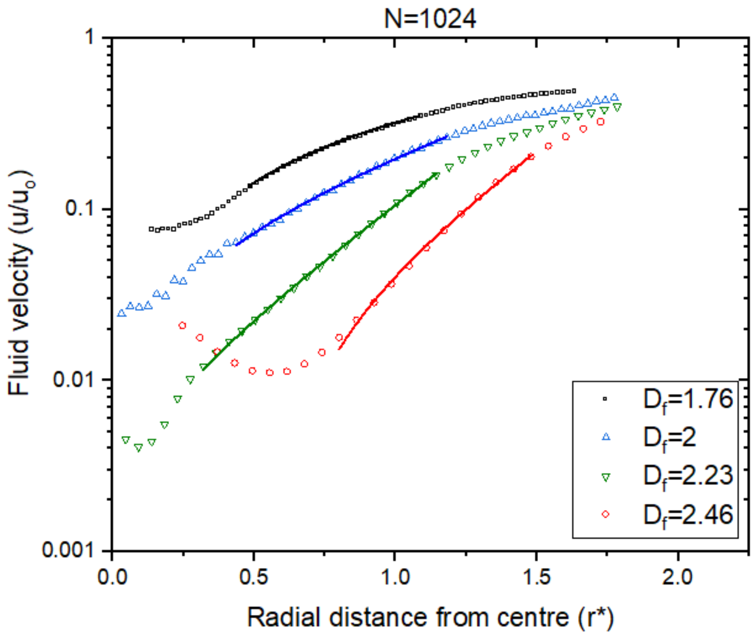



*Figure 6-7:Typical fluid velocity scaled with respect to ambient fluid velocity(z-direction) plotted against non-dimensional radial distance($r^*$) along with the exponential fits to obtain penetration depth(straight lines) for Tunable Monte Carlo clusters with varying fractal dimensions N=1024 [□ $-D_f = 1.76$,△ $-D_f = 1.99$,▽ $-D_f = 2.23$,○ $-D_f = 2.46$].*

| $D_f$ | $\delta^*$ | Region | % aggregate |
|---|---|---|---|
| 1.76 | 0.743 | 0.65< $r^*$<1.15 | 54 |
| 2 | 0.515 | 0.25 < $r^*$<1.15 | 69.8 |
| 2.23 | 0.322 | 0.27 < $r^*$<1.15 | 74.9 |
| 2.46 | 0.279 | 0.62 < $r^*$<1.4 | 89.26 |

*Table 6-2:Penetration depth, the regions considered for the exponential fit and the %particles of the aggregate these regions hold for N=1024 for flow in z-direction for which the plots are shown in figure 6-7*

The results of the penetration depth were obtained by best fit approach for clusters of different sizes and fractal dimensions.

Figure 6-8 gives the velocity profiles for different fractal dimensions of Tunable Monte Carlo cluster. Some of the profiles fit completely, while some others have an S-shaped form. Some of the velocity fits obtained have three regions:

i. Head-section for the screened interior
ii. Mid-section for the bulky regions of the cluster
iii. Tail-section for the regions with sparse particle distribution

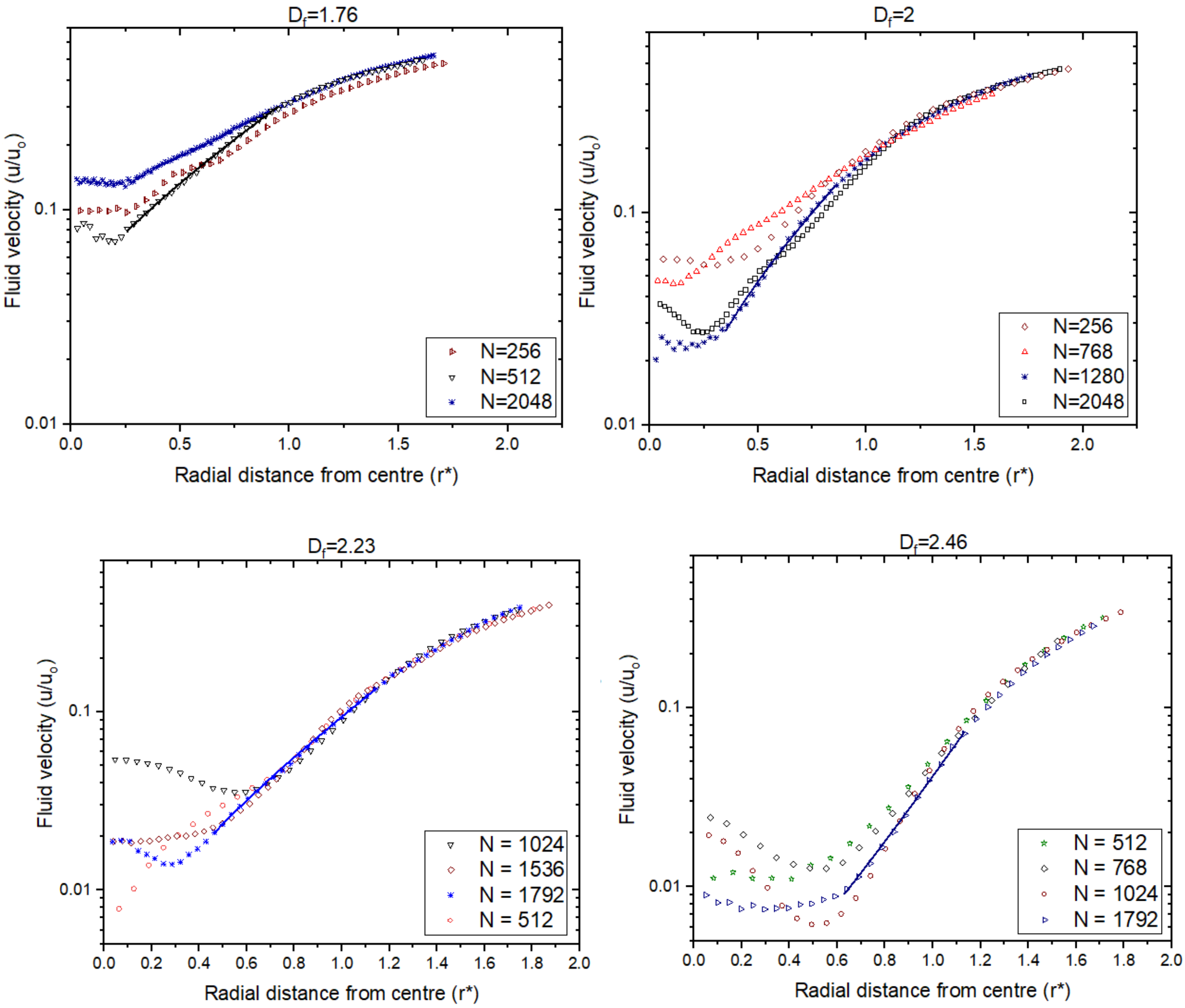



*Figure 6-8:Typical fluid velocity scaled with respect to ambient fluid velocity(z-direction) with non-dimensional radial distance ($r^*$) for Tunable Monte Carlo clusters with varying fractal dimensions [1.76(top left), 2(top right), 2.23(bottom left) and 2.46(bottom right)] and N*

The head-section can always be ignored as it does not provide any useful information. Mid-section holds majority(~ 60 - 80%) of the monomers and also the exponential decay plays a huge part in this region. The tail-section also has an exponential decay but it is very small compared to the decay observed in the mid-section. Also, these are the regions where the fractals are usually depleted of particles.

Comparing the different profiles, it can be seen that the slope keeps increasing as the fractal dimension increases which are of the similar range for every $D_f$. The fluid velocity observed for $D_f$ = 2.46 in the regions $0 < r^* < 0.4$ are proportional to the forces acting on spheres in these regions. Since the fractals of this $D_f$ are more compact, it seems to be valid owing to an internal dense structure. This kind of behavior can be seen for other fractal dimensions, but weaker.

An interesting feature can be seen for the case of spherical aggregates i.e., simple cubic unit. Most of the interior is screened from fluid flow and to be precise fluid flow is screened in the interiors (from $R_g$ to center). Also, the slope keeps on increasing ($R_g < r^* < R_{\max}$) as the number of particles increases suggesting $\delta^*$ to be a function of N for this particular case. Even though the regions for determining $\delta^*$ for simple cubic unit contains >55% particles, number of points remains small i.e., 3-6. In order to compare the results of all the fractals, we are using the most compact aggregate to differentiate the spherical one from irregular structures.

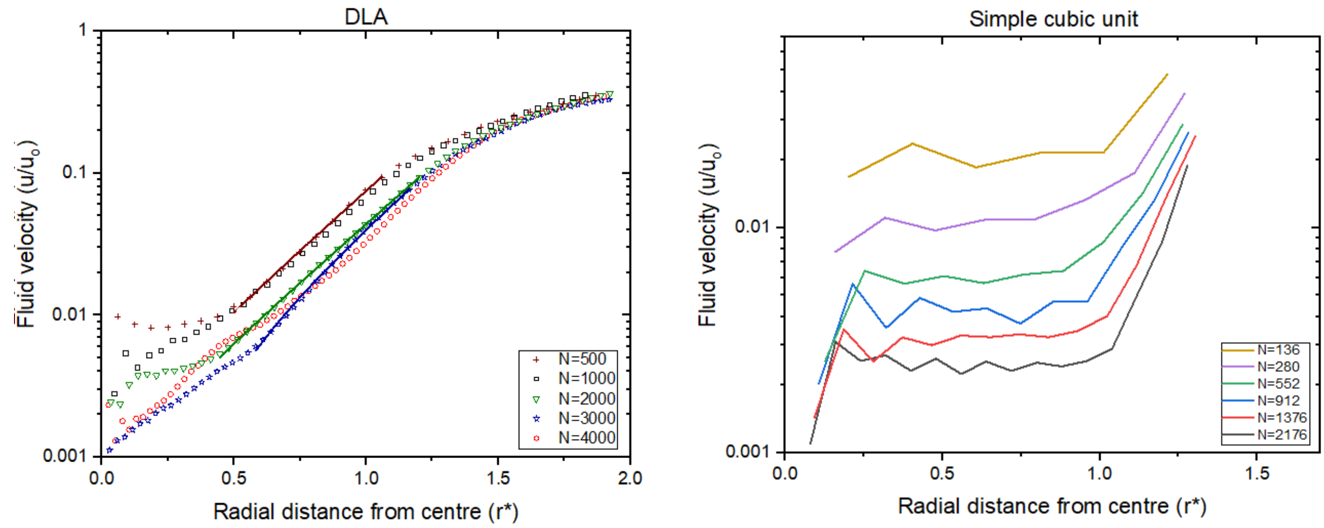



*Figure 6-9:Typical fluid velocity scaled with respect to ambient fluid velocity(z-direction) with non-dimensional radial distance($r^*$) for DLA(left) and Simple cubic unit(right) with varying N*

### 6.6.1 Penetration depth and cluster size

Penetration depth for a single cluster is obtained as an average from fluid flowing in three different directions. Three to four different clusters were used for $D_f$=1.76 and $D_f$=2 and two to three different clusters for the other two fractal dimensions.

In order to understand the cluster size dependency, these results are plotted as a function of N, for various $D_f$, and  are shown in the Fig 6-10 below.

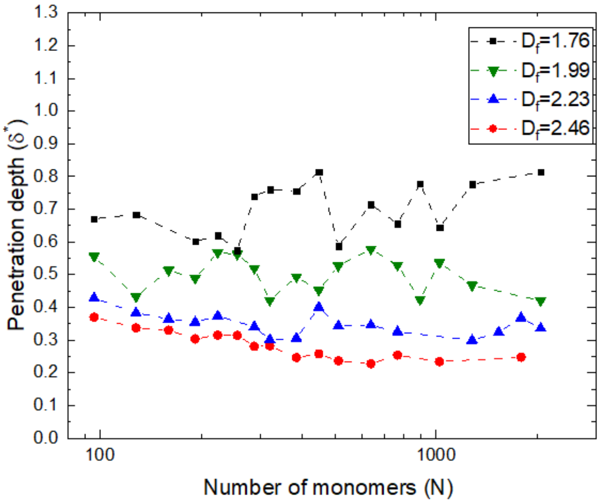



*Figure 6-10:Penetration depth varying with the number of constituent particles for Tunable Monte Carlo clusters of different $D_f$ ( ■ $- D_f = 1.76$, ▼ $-D_f = 1.99$,▲ $-D_f = 2.23$,● $- D_f = 2.46$)*

The first two cases ($D_f$=1.76 and $D_f = 2$) shows some oscillations because of the variations pertaining to the structure in different directions. There is a clear indication that there is a saturation for $\delta^*$ in the region between for N=200 and beyond, thereby suggesting that the relative penetration depth is independent of cluster size as N →∞. This is at variance with the meanfield theory results which predicts significant decay in the same range of cluster sizes. Asymptotic values are reported from the $\delta^*$ values obtained for N=96 to N=2048 in the table below which are obtained by fitting an equation of the form $\delta^* = \delta_e^* + A_o \exp\left(-\frac{N}{N_o}\right)$ where $A_o$ and $N_o$ are constants.

| *Fractal* | $D_f$ | *Average Penetration depth,* $\delta^*$ | $\delta_e^*$ | *Average* $u_i$ |
|---|---|---|---|---|
| *Tunable monte Carlo* | 1.76 | 0.699 ± 0.081 | 0.763 | 0.071 ± 0.011 |
| | 1.99 | 0.484 ± 0.067 | 0.422 | 0.023 ± 0.005 |
| | 2.23 | 0.352 ± 0.046 | 0.336 | 0.061 ± 0.002 |
| | 2.46 | 0.271 ± 0.053 | 0.236 | 0.002 ± 0.001 |
| *DLA* | 2.47 | 0.317 ± 0.052 | 0.286 | 0.003 ± 0.0027 |

*Table 6-3:Penetration depth observed for Tunable Monte Carlo clusters*

There is a monotonic dependence of the penetration depth with the fractal dimension for the case of Tunable Monte Carlo clusters. Larger the penetration depth, smaller the screening effects and a relatively greater penetration of fluid velocity towards the core.

A single cluster of DLA is used for these calculations and the average penetration depth is actually higher than Tunable Monte Carlo cluster of similar $D_f(\delta^*_{DLA(D_f=2.47)} < \delta^*_{D_f=2.46})$. A similar investigation was carried out on a simple cubic unit with $D_f = 3$, but the penetration depth changes with N, therefore it is not discussed here.

### 6.6.2 Comparison with permeability models

Volume fraction can also be scaled with N by using fractal scaling law.

$$\phi_R = N\left(\frac{a}{R_{max}}\right)^3 \rightarrow \phi_R \sim N^{\frac{D_f-3}{D_f}}$$

$$\delta^* = \frac{\sqrt{\kappa(\phi_R)}}{R_g}$$

Permeability($\kappa$) is determined by the volume fraction of solids, from which the "penetration depth" can be calculated. Since most of the models are functions of dilute limit $\kappa_{DL}$, we consider only Happel and Brenner(1991) permeability equation.

| **PERMEABILITY($\boldsymbol{\kappa}$)** | **SOURCE** | **$\boldsymbol{\delta}^*$(N)** | |
|---|---|---|---|
| $\boldsymbol{\kappa_{DL} = \frac{2a^2}{9\phi}}$ | Happel and Brenner (1991) | $C.N^{\frac{1-D_f}{2D_f}}$ | 1 |
| $\boldsymbol{\kappa = \frac{a^2}{16\phi^{1.5}(1+56\phi^3)}}$ | Davies (1952) | $C.N^{\left(\frac{5}{4D_f}-\frac{3}{4}\right)}$ | 2 |

*Table 6-4: Penetration depth obtained from Happel and Brenner (1991) and Davies (1952) permeability equation*

The simplified form of $\delta^*(N)$ in (1) is the same as Van Sarloos (1987) mean field approximation, which states that the constant C that determines the internal density, is assumed to be unity.

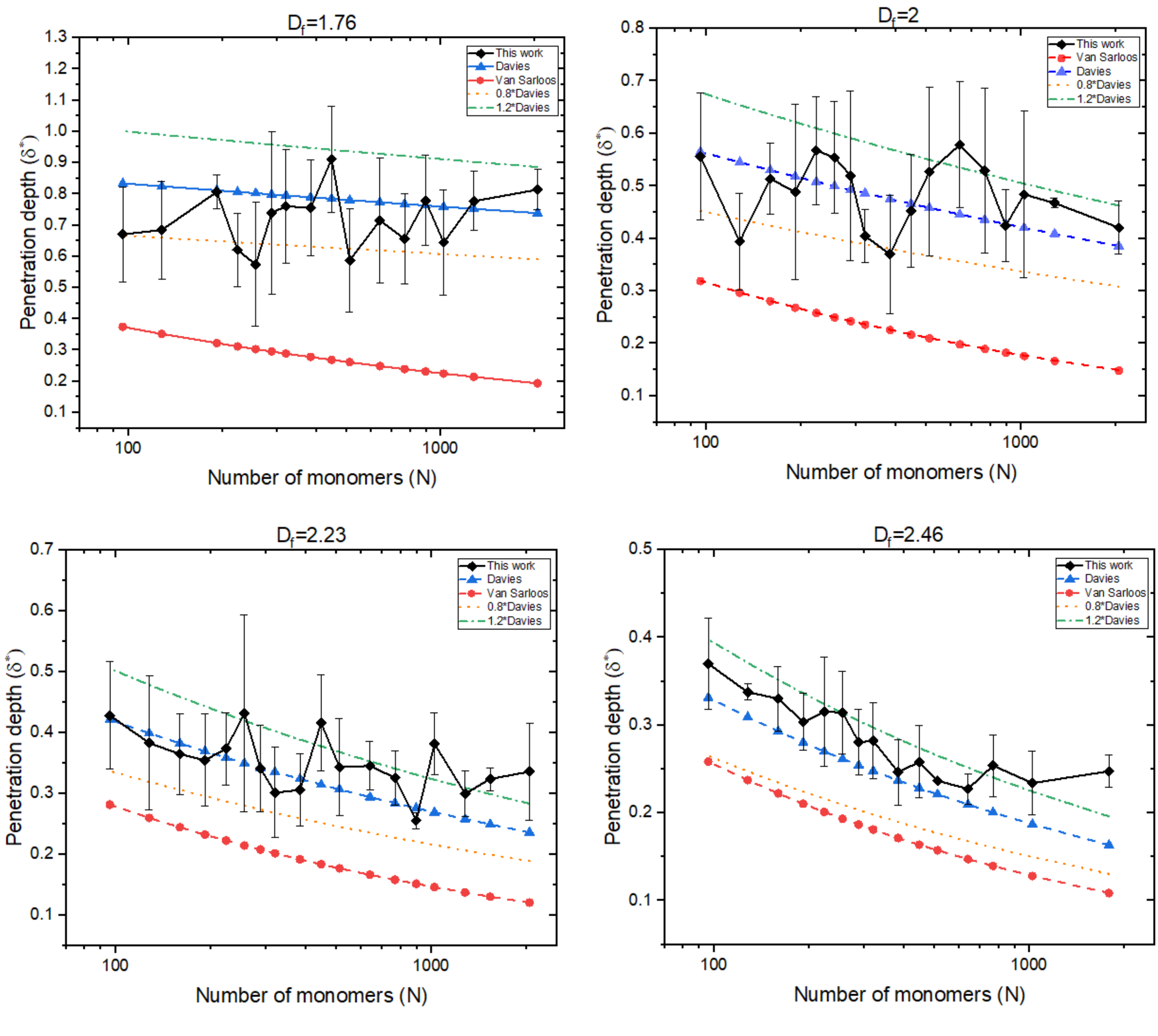



*Figure 6-11:Penetration depth for Tunable Monte Carlo clusters of present work(◆) along with error bars compared with other permeability models.*

The error bars shown indicate the changes of penetration parameter with respect to flow in different direction and also different clusters of same fractal dimension. As the fractal dimension grows larger, the variations in $\delta^*$ diminishes as more variances indicate structural changes i.e., certain fractals elongated more in one direction compared to the other.

The permeability equation of Davies (1952) yields similar penetration depth estimates, especially for porous aggregates. The numbers from Davies (1952) and Van Sarloos (1987) models show a reduction in penetration depth. Although weak, this is still a detectable dependence within the range of N (=20 to 1000) , and predict as much as 30% drop in value between N=200 and N=1000. However, the simulations do not show any concrete evidence of decrease of $\delta^*$ in this range thereby strongly pointing at the limitation of the mean-field approaches.

Both DLA and simple cubic unit show a downward trend, with DLA's % reduction halting while simple cubic unit's decrease continues. $\delta^*$ values are also closer to the mean field approximation by Van Sarloos (1987) for spherical aggregates. This is consistent with the mean-field theory, which scales with N, and our findings match closely with Van Sarloos (1987).

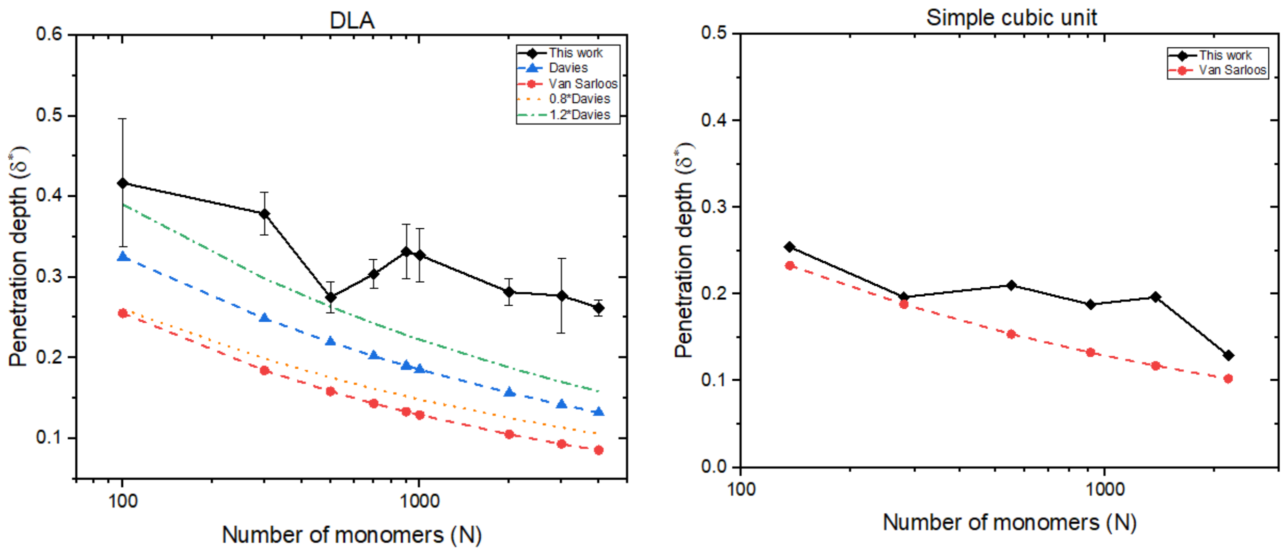



*Figure 6-12:Penetration depth for DLA and simple cubic unit of present work(◆) compared with other permeability models*

$\delta^*$ values of DLA show a significant decline until N=2000 after which the decline has slowed and an asymptotic value has been attained. This reminds us of $R_m/R_g$ values for DLA which increased until they reached a stable value at N=2000. There is definitely a relationship between the two, implying that as one parameter rises($R_m/R_g$) the other falls($\delta^*$).

### 6.6.3 Volume fraction for the clusters studied

Volume fraction of solids justify $\delta^*$ values for DLA in between $D_f$=2.23 and $D_f = 2.46$. Average volume fraction of DLA (=0.015) is in fact less than $D_f = 2.23$(=0.026) and $D_f = 2.46(= 0.063)$ . Fluid penetrating within the aggregate cannot be determined only based on the fractal dimension as it depends on how the particles are distributed within the aggregate [Li and Logan (2001)]

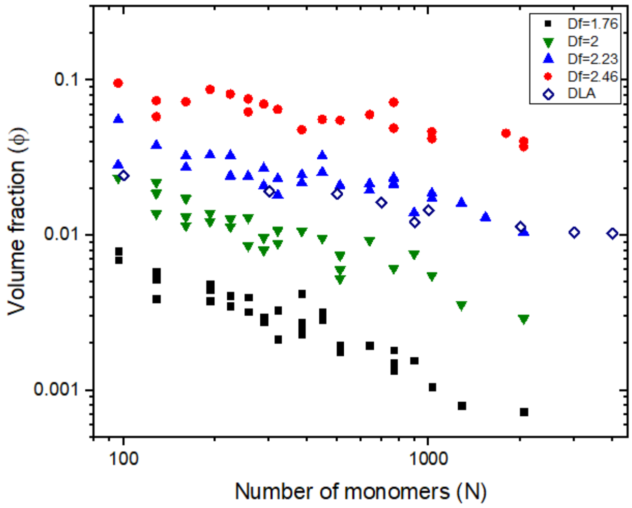



*Figure 6-13:Volume fraction of solids for the different fractals (◇ − DLA) and Tunable Monte Carlo Clusters [ ■ − $D_f$ = 1.76, ▼ −$D_f$ = 1.99,▲ −$D_f$ = 2.23,● − $D_f$ = 2.46] varying with the number of monomers*

This helps us conclude for now that DLA having large $\delta^*$ values despite their $D_f$ is justified with the lower volume fractions. $\delta^*$ also serves as the differentiating factor between cluster-cluster and particle-cluster aggregates. It is known that DLA is dense around the center and since the particles grow around the branches exposed, void fraction keeps on increasing from the center to the exterior. That is not the case with cluster-cluster aggregates as the screening is dominant in regions of the aggregating cluster.

### 6.6.4 Varying permeability for a single cluster

Because the volume fraction of particles is not homogenous, it is critical to consider the aspect of varying permeability of the empirical models. The solid fraction is highest in the center and declines as we move outwards, reaching its lowest point at $R_{max}$. Solid fraction $\phi(r)$ is obtained as radial distance changes for the case of DLA with N=1000 and the corresponding $\kappa(\phi)$ with respect the radial distance/solid fraction. For this scenario, the solid fraction changes from $\phi|_{r=2} = 0.375$ to $\phi|_{r=41} = 0.014$. Fig 6-14 depicts the difference between Davies (1952) model and others towards the cluster's ends in terms of porosity and fluid penetration.

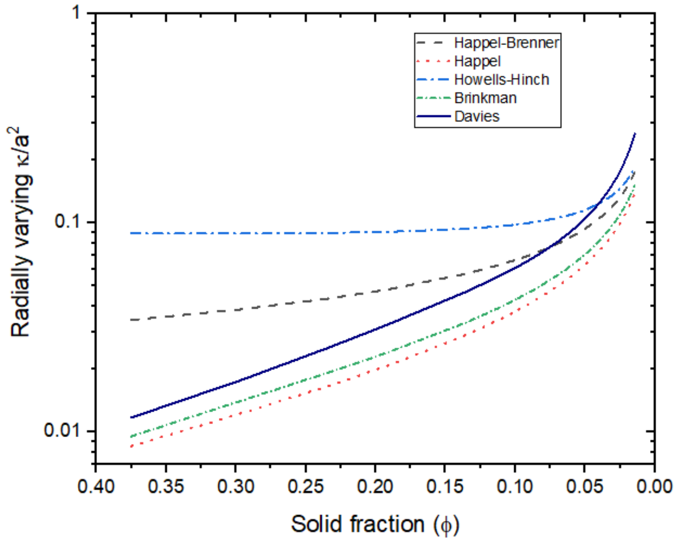



*Figure 6-14:Varying $\frac{\kappa}{a^2}$ as the solid fraction decreases from the center($\phi$=0.37) to the exterior($\phi$=0.014) for the case of DLA: N=1000 for different permeability models*

### 6.7 Role of penetration depth

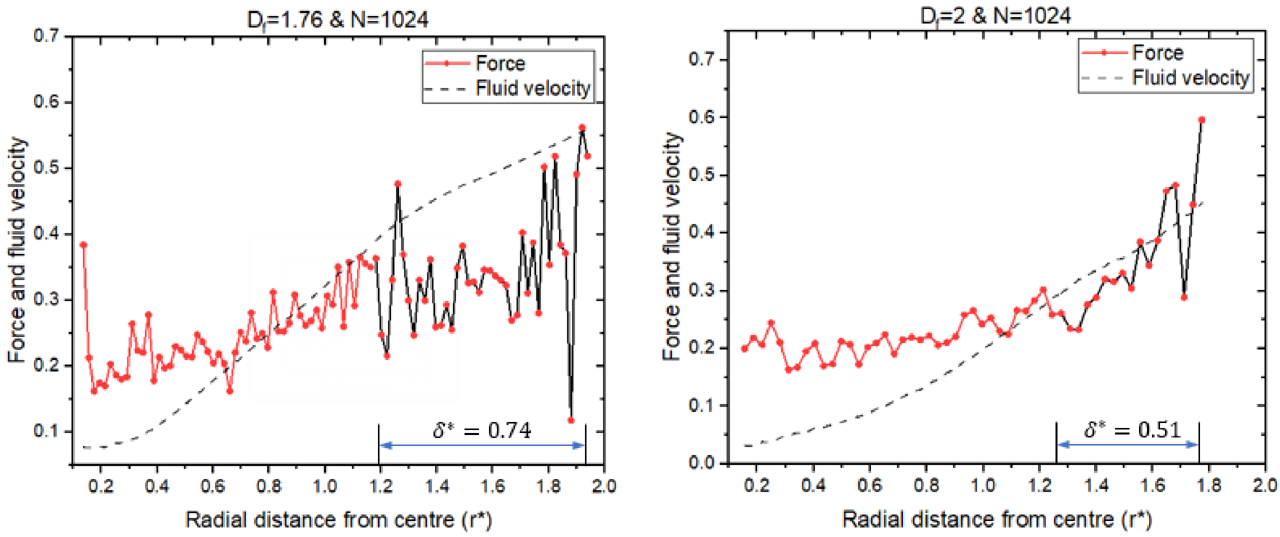



*Figure 6-15:Dimensionless force acting on monomers and the dimensionless fluid velocity in the regions from centre to $R_{max}$ for N=1024 and $D_f = 1.76$(left) and $D_f = 2(right)$*

This $\delta^*$ gives us the regions which experience majority of the flow, thereby shielding the interiors. These regions are also supposedly those that affect the hydrodynamic radius of the aggregate. Average force acting on spherical particles for $D_f = 1.76$(figure 6-15a) are almost in close ranges as a result of the open structure. Even in such case, the above figure suggests the importance of $\delta^*$ as we move from the exterior regions.

Number of particles coming under the $\delta^*$ region also decreases with fractal dimension as shown in Table 6-5. This suggests that as we go towards more compact fractals, only few particles are experiencing more forces while the inner regions are screened. This behavior of $\delta^*$ on the forces experienced has been found to be similar for all the fractals.

Also, for the most compact ones, the average forces and velocities are clearly proportional.

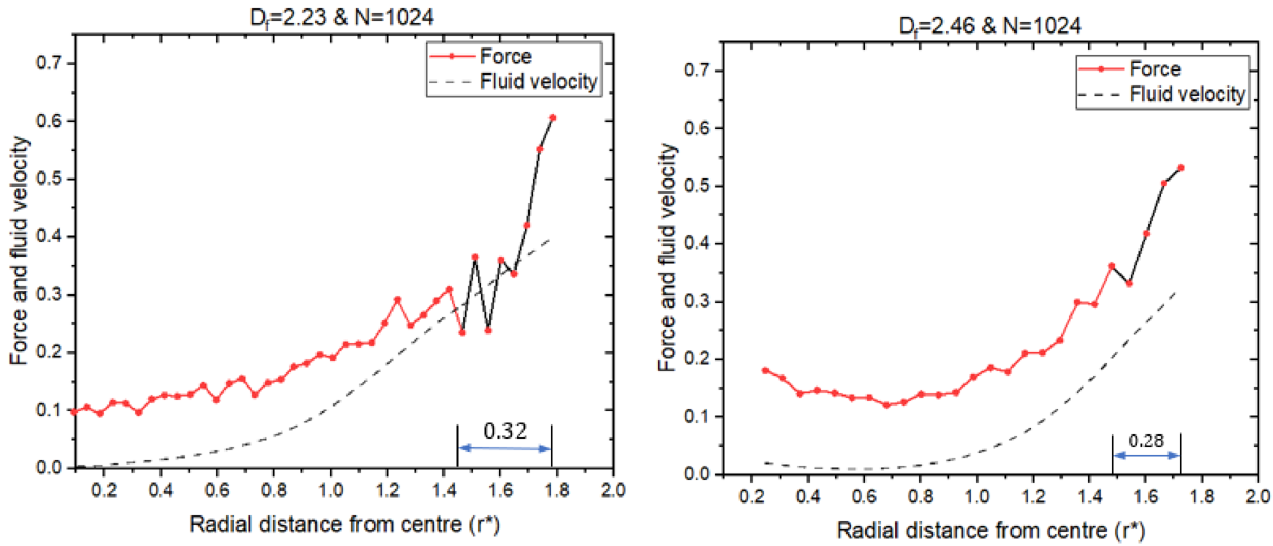



*Figure 6-16: Dimensionless force acting on monomers and the dimensionless fluid velocity in the regions from centre to $R_{max}$ for N=1024 and $D_f = 2.23$(left) and $D_f = 2.46$(right)*

| $D_f$ | N |
|---|---|
| 1.76 | 248 |
| 2 | 208 |
| 2.23 | 73 |
| 2.46 | 38 |

*Table 6-5 : Number of particles within the $\delta^*$ region for four different fractal dimensions*

This study clearly indicates the amount of the structure which are available for the fluid to access using the parameter "penetration depth."

### 6.8 Penetration depth and hydrodynamic ratio

Finally, we examine if the observed trends of $R_m/R_g$ are consistent with those of $\delta^*$ from the perspective of improving our understanding of fluid-aggregate interaction. It stands to reason to argue that an aggregate into which the fluid penetrates more easily will offer more resistance to flow. The monotonic increase of $R_m/R_g$ with $D_f$ and the corresponding decrease of $\delta^*$ strongly justifies this reasoning. This permits us to assume unique one-to-one relationship between $R_m/R_g$ and $\delta^*$as shown in fig 6-17.

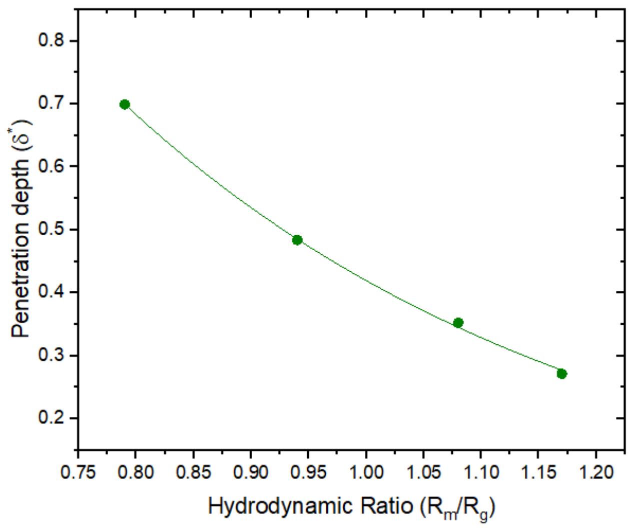



*Figure 6-17:Exponential fit(-) of penetration depth and the hydrodynamic ratio for Tunable Monte Carlo clusters of different fractal dimension (right).*

Clearly the relationship is significantly stronger than a direct linear relationship and as a first attempt we attempt an exponential form of decay within the range of $R_m/R_g$. This results in the fitted form:

$$\delta^* = (4.805 \pm 0.232)\exp\left(-\frac{R_m/R_g}{0.2316}\right) \tag{4}$$

A posterior justification for the choice of the exponential fit is to reflect the observation of a large relative change in $\delta^*$ with respect to, for larger values of $R_m/R_g$. A small change in $R_m/R_g$ results in large change in $\delta^*$. To put it differently, the measured contraction of the cluster span to fluid resistance changes only logarithmically with respect to the measure of its transparency to fluid penetration. This relationship offers a way to estimate the interior properties, such as the velocities at the core region, from the exterior measures such as the $R_m/R_g$. However, this relationship seems valid only for cluster-cluster aggregates.

## 6.9 Dependency on direction of flow and ambient fluid velocity

Fluid penetrates differently depending on the structure exposed to the flow. Figure 6.14 shows the different profiles observed for different flow directions for two extreme cases: One with $D_f = 1.76$ and one with $D_f$=2.46.

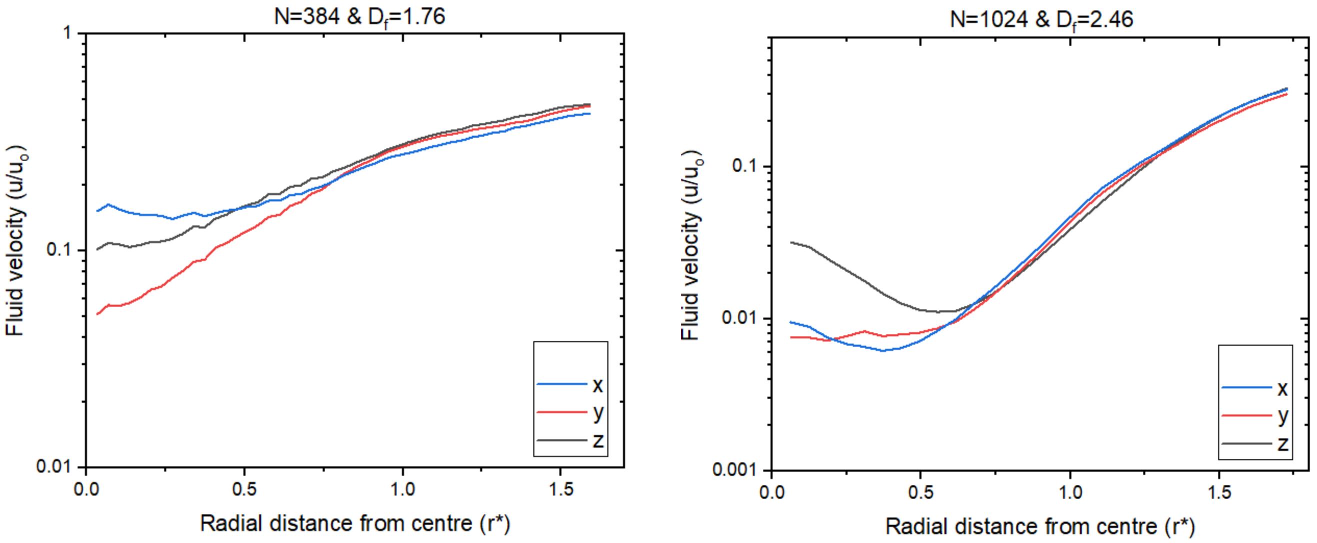



*Figure 6-18:Normalized fluid velocity for Tunable Monte Carlo clusters with the radial distance scaled with $R_g$ for N=384,$D_f = 1.76$(left) and N=1024,$D_f = 2.46$(right). Three curves showing the velocities of fluid when flow is in x(-),y(-) and z(-) directions.*

| | $\delta_x^*$ | $\delta_y^*$ | $\delta_z^*$ | S.E |
|---|---|---|---|---|
| N=384 & $D_f = 1.76$ | 0.84 | 0.49 | 0.65 | 0.177 |
| N=1024 & $D_f = 2.46$ | 0.255 | 0.257 | 0.28 | 0.014 |

*Table 6-6: Difference in penetration values observed for the above two cases*

The above table demonstrates the standard error for both the cases with the $\delta^*$ values. The lowest fractal dimension has an error 10 times larger than the ones with the large fractal dimension. In general, error observed for $D_f$=2.46 has maximum of ± 0.06 while $D_f = 1.76$ has a maximum of ± 0.26.

Figure 6-19 shows the velocity profile when the ambient fluid velocity is varied as $U_o =$ 1,2,3 for one of the dense fractals with $D_f$=2.23 and N=512. The corresponding velocities at different regions are exactly doubled and tripled as $U_o$ is increased. Hence, the slope also remains the same and penetration depth remains constant irrespective of ambient fluid velocity.

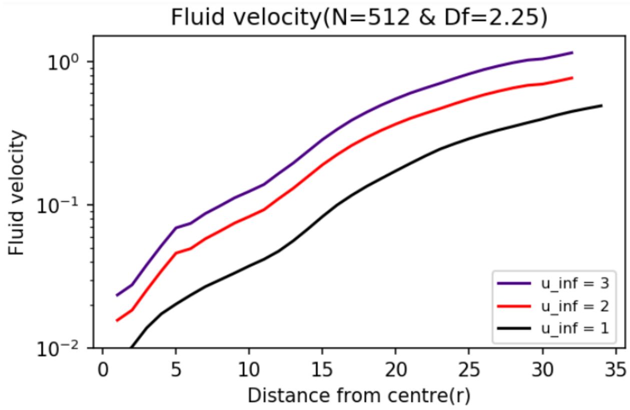



*Figure 6-19:Dimensionless fluid velocity when ambient fluid velocity is varied Tunable Monte Carlo cluster with the radial distance from center for N=512 and* $D_f = 2.23$. $U_o = 1(-), U_o = 2(-), U_o = 3(-)$

## 6.10 Velocity at various locations for different $D_f$

The parameter "penetration depth" was obtained through velocity determined using Stokesian Dynamics and was found to be related with only the hydrodynamic ratio but not the cluster size. We further analyze how fluid velocity changes at certain locations and determine how the changes are with respect to the fractal dimension and type. At the end of this study, it can be seen why spherical aggregates are different from regular fractals.

Fluid velocity$\left(u^* \equiv \frac{u}{u_o}\right)$ at the exterior regions of the fractal(at $rmax^*$), at the radius of gyration($r^* = 1$) and $r^* = 0.5$ is compared for the four different fractal dimensions and N is varied from 96 to 2048. The values shown in figure () are the average values for the different clusters of a particular N and $D_f$ with flow in all three directions.

The observations for the different cases are as follows:

1. At $rmax^*$, velocities for $D_f$=2.46 are almost half of what they are for $D_f$=1.76.
2. Low fluid velocities for $D_f$>2 at r*=1 and r*=0.5 indicate that fluid has only penetrated the outside regions and not the inside.

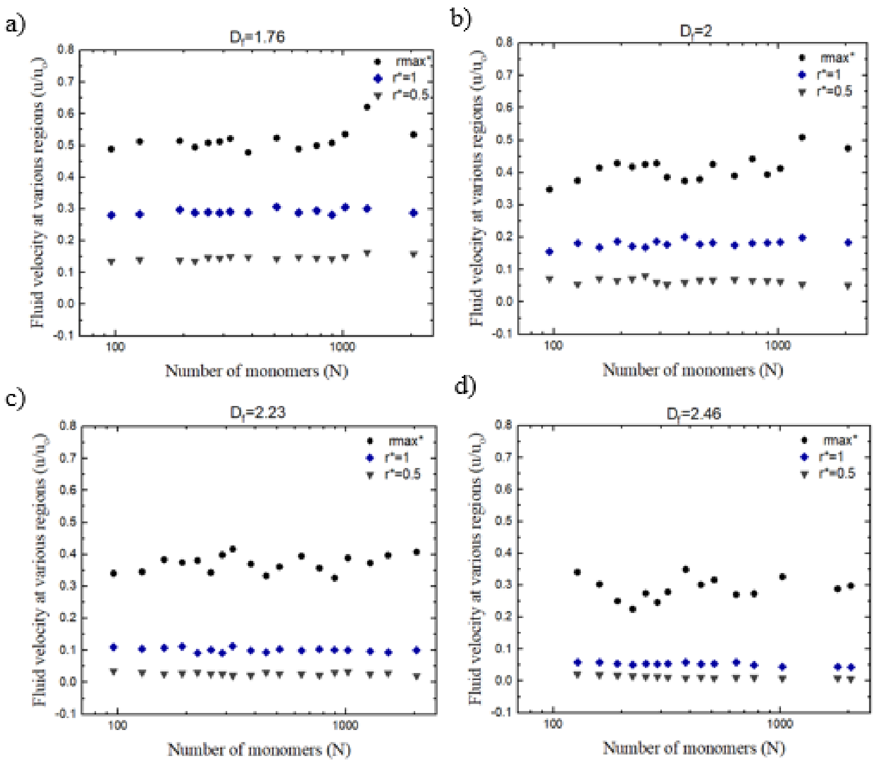



*Figure 6-20:Fluid velocity observed at three different regions of fractals: rmax*(●), r*=1(◆) and r*=0.5(▼) for fractal dimensions of 1.76(top left), 2(top right), 2.23(bottom left) and 2.46(bottom right)*

3. Regardless of the size of the cluster, velocity at r*=1 and r*=0.5 is nearly the same for all $D_f$.
4. At r*=1 and r*=0.5, the velocity decays exponentially with fractal dimension, as shown by the following relationship:

$$u^*(r^*) = u_{pre} \exp\left(-\frac{D_f}{d}\right)$$

| $r^*$ | $u_{pre}$ | **d** |
|---|---|---|
| **1** | 14.013 ± 1.124 | 0.4542 ± 0.0089 |
| **0.5** | 61.894 ± 11.441 | 0.29103 ± 0.0087 |

*Table 6-74:Pre-factor, $u_{pre}$ and exponential factor, d obtained by the above exponential fit for the four different fractal dimensions.*

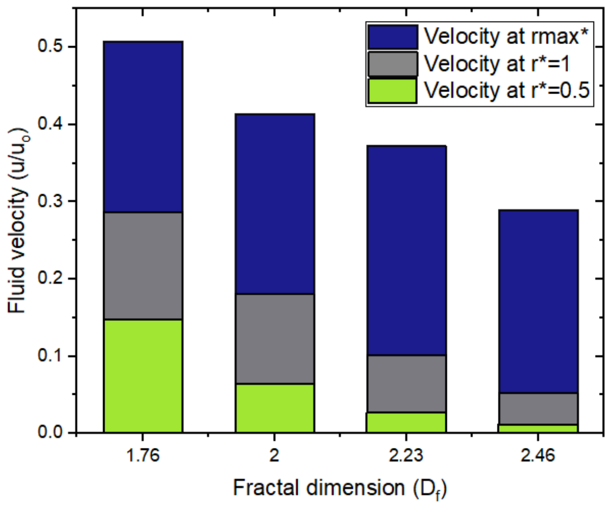



*Figure 6-21:Average Fluid velocity observed at three different regions of fractals: rmax*, r*=1and r*=0.5 for different fractal dimensions(left) and the exponential fit for the velocities at r*=1(○) and r*=0.5(□) for different $D_f$(right)*

5. From r*=1 to r=0.5, the percent drop in velocity is bigger than from rmax* to r*=1 (except $D_f = 2.46$). As the fractal dimension grows, so does the percent drop in velocity. This drop is due to the fluid penetration changing with the fractal dimension.

| $D_f$ | 1.76 | 2 | 2.23 | 2.46 | DLA |
|---|---|---|---|---|---|
| % decrease rmax* to r*=1 | 42.63 ± 3.59 | 56.04 ± 4.5 | 72.13 ± 3.63 | 82.04 ± 2.73 | 83.01 ± 5.06 |
| % decrease r*=1 to r*=0.5 | 48.98 ± 4.59 | 62.61 ± 8.85 | 73.06 ± 7.5 | 76.74 ± 7.71 | 79.61 ± 5.75 |

*Table 6-8:% decrease in the velocity values from rmax* to r*=1 and r*=1 to r*=0.5 for different fractal dimensions.*

6. Velocity of DLA($D_f = 2.47$) at all the three regions are relatively large than Tunable Monte Carlo cluster with similar $D_f$(=2.46). It is to be noted that from chapter 5, the hydrodynamic ratio of DLA($R_m/R_g$ =1.238) is greater than Tunable Monte Carlo cluster with similar $D_f$($R_m/R_g$=1.173).

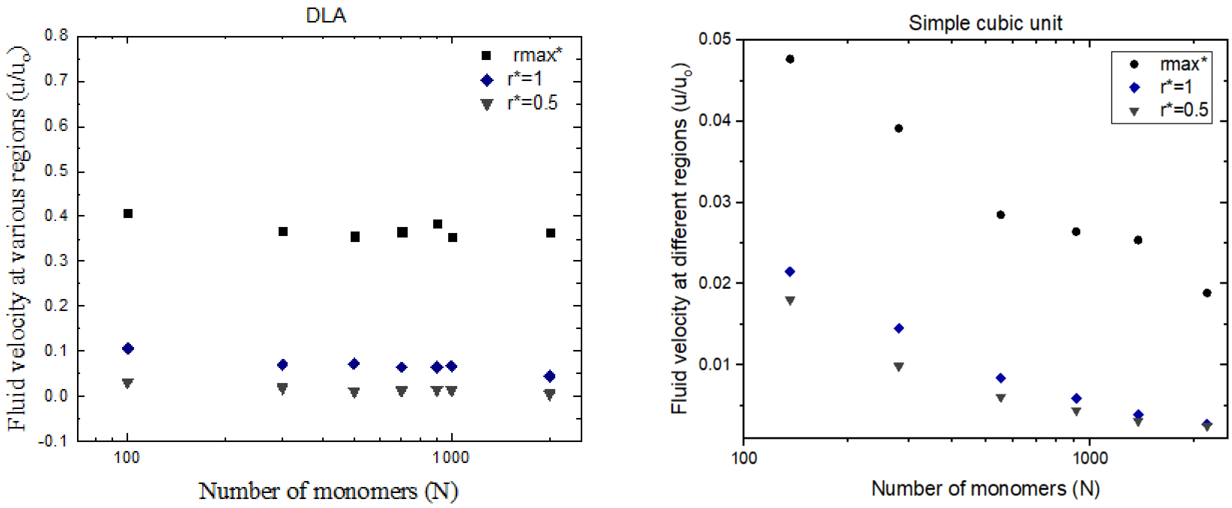



*Figure 6-22:Fluid velocity observed at three different regions of fractals: rmax*(●), r*=1(◆) and r*=0.5(▼) for DLA(left) and Simple Cubic Unit(right)*

Velocity observed for DLA and simple cubic unit are shown in figure 6-22. DLA shows similar behaviour as Tunable Monte Carlo fractals, but with slight reduction in velocities at $r^* = 1$ and $r^* = 0.5$. But all the velocities observed for simple cubic unit falls as N increases. To describe the decrease of fluid velocity, power law fits can be employed for velocity versus N. Even though the fit obtained for the velocity at rmax* isn't good, the exponents give the tale of how the structure behaves.

| $u^* = \frac{a}{N^b}$ | rmax* | r*=1 | r*=0.5 |
|---|---|---|---|
| a | 0.224 ± 0.033 | 0.646 ± 0.132 | 0.746 ± 0.091 |
| b | 0.314 ± 0.025 | 0.688 ± 0.038 | 0.76 ± 0.023 |

*Table 6-9:Pre-factor and exponent, b obtained by the power-law fit for simple cubic unit*

The fit of the form $u^* = \frac{a}{N^b}$ finds that as we go into the fractal, the exponent increases. In the case of spherical aggregates, this leads to the conclusion that as we go deeper into the fractal, velocities fall at a rapid rate as the number of particles grows. This helps to demonstrate how spherical aggregates vary from fractals.

This section helps to understand better about penetration depth. At different locations, all Tunable Monte Carlo clusters had nearly constant velocities for all N, whereas DLA had a modest decrease in velocities with N at the two regions chosen until the asymptotic limit was achieved. Fluid velocities at r*=1 and r*=0.5 will be decreasing if $\delta^*$ is a decreasing function of N (as in the case of the simple cubic unit). As a result, the notion that "penetration depth" is independent of N is strengthened.

## 6.12 Conclusion

In this chapter, velocity was calculated using multipole expansion formulation; and superposition of disturbance velocity fields of all particles in the cluster. This velocity was determined in regions within fractals of different $D_f$. Fluid velocity increases rapidly, nearly in an exponential manner, starting from the interior of the aggregate till the surface. One may define the degree of attenuation as viewed from the outer boundary of the aggregate via a concept of a penetration depth. Under the assumption of an exponential decay of velocity, a length scale referred to as the penetration depth $\delta$ or its dimensionless counterpart $\delta^* = \delta/R_g$ , adequately describes the intensity of velocity attenuation by the cluster. As the cluster size increases, $\delta^*$ in contrast to the prediction of the mean field approximation which indicated a weak power-law decrease with N. It is found that the saturated value of $\delta^*$ decreases as the fractal dimension increases. There exists a systematic inverse relation between $\delta^*$ and the hydrodynamic ratio, thereby implying that larger hydrodynamic ratio is associated with smaller penetration depth. It is found that the fluid penetration $\delta^*$ decreases exponentially with hydrodynamic ratio for the case of Tunable clusters. The results obtained are compared with the permeability equation of Davies and the two are found to be in reasonable agreement with each other. The relationship between fluid penetration and hydrodynamic ratio has wider theoretical and practical implications. Specifically, its implications in the context of aggregate mediated catalysis in which one would prefer deeper penetration of the fluid for more effective performance, need to be explored in the future.

## Appendix 6A: Velocity field with and without lubrication interactions

The forces are usually obtained excluding the lubrication interactions and the reason for which was discussed in Chapter 5. The effect of including lubrication interactions can also be verified by determining the fluid velocity within the fractal. So henceforth this study involves obtaining the forces using Stokesian Dynamics with and without lubrication interactions. DLA fractals of N=100 and N=1000 was chosen for this exercise and the fluid flow in only one direction is considered.

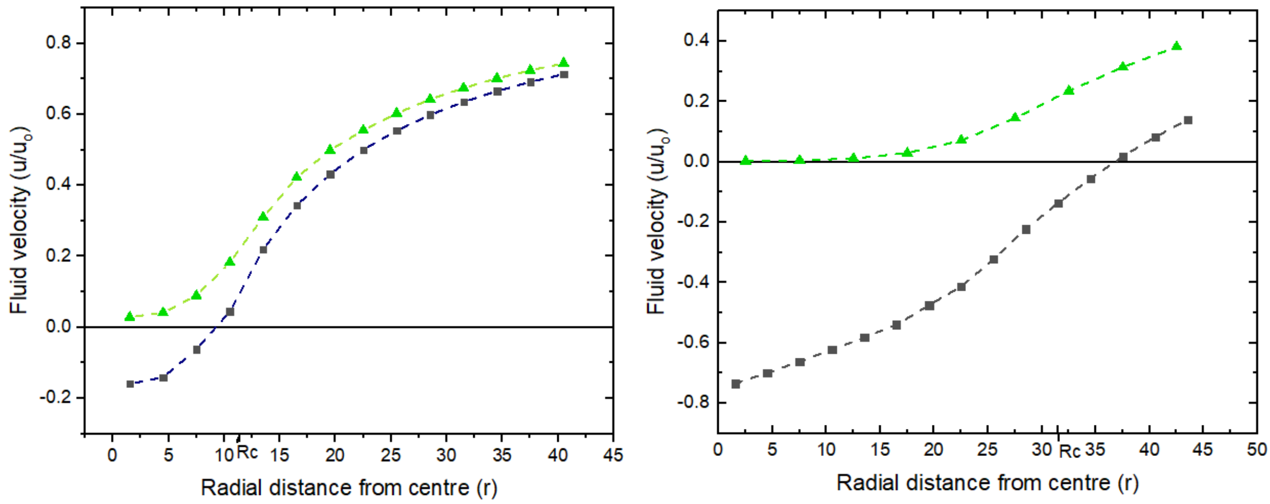



*Figure 6-23:Fluid velocity $u^{\infty}(z) = 1$ with lubrication interactions (■) and without lubrication (▲) interactions for the case of DLA with N = 100(left) and N=1000(right). The outer radius for these clusters is marked on x-axis as $R_c$.*

It can be observed that the intensity of fluid velocity observed with lubrication interactions are negative giving us the reason why aggregate velocity was underestimated. The explanation can be given such as: lubrication forces are too high such that the disturbance field generated does not allow the fluid to pass and worse causing the fluid to move in a direction opposite to its flow direction even though the aggregate is stationary. The effect of lubrication forces worsens as the number of particles increases, causing the fluid velocity to further high values with negative intensity.

# Chapter 7

# Deformable Aggregates

## 7.1 Introduction

Colloidal aggregates studied so far had rigid links between the individual particles. Complex behavior of such aggregates is common when they interact with respect to flow field, such as stirring. They are rigidly connected as they experience the deep van der Waals attractive well [(Brady & Bossis (1991)]. Depending on the type of interparticle interaction or the process of aggregation, the structure can say rigid or deform to break away from the whole structure. Interaction potentials depends on the gap between the particles either center-to-center distance or surface separation distance. Lattuada et al. (2011) used two types of interparticle interactions: van der Waals-Born interaction potential and contact tangential interactions. Use of contact tangential interactions arises to account for the sliding behavior when interparticle distance is very small. When particles are very close together, very small timesteps were need ($10^{-9} - 10^{-8}\, s$) to prevent overlapping. In the following chapter, these interactions are incorporated and tested for small problems.

## 7.2 Methodology

Stokesian Dynamics can be applied to deformable aggregates where the interparticle forces can be added to the external forces directly. Two different types of forces are being considered.

*Spring type forces*

$$F^p = k \sum_{j=1, j\neq i}^{3} \frac{(|r_{ij}| - r_0) r_{ij}}{|r_{ij}|} \qquad 7.1$$

where $k$ is the spring constant, $r_o$ is the equilibrium distance which is a scalar quantity and $r_{ij} = r_j - r_i$ is the interparticle distance. $\left(|r_{ij}| - r_0\right)$ gives the magnitude and $\frac{r_{ij}}{|r_{ij}|}$ gives the direction of the force. Spring constant $k$ is taken as 0.5 for verifying equilibrium and few other cases.

*van der Waals + Born repulsion*

$$F_p = \frac{\left[-\frac{A}{12ah^2} + \frac{A}{360a}\left(\frac{\sigma}{a}\right)^6 \frac{1}{h^8}\right]}{6\pi\eta a U_s} \qquad 7.2$$

where A is the Hamaker's constant,

a is the particle size, h(=$h_s/a$) is the non-dimensional surface-surface separation.

$\sigma$ is the minimum distance at which the potential is zero.

$\eta$ is the fluid viscosity and $U_s$ is the steady state particle velocity.

The parameters used for both van der Waals and Born repulsion are given in table below.

| *Parameters* | *Values* |
|---|---|
| $A$ | $1 \times 10^{-20}$J |
| $a$ | 10 nm |
| $\sigma$ | 0.2 nm |
| $\eta$ | $10^{-5} kg/ms$ |
| $U_s$ | 0.1 m/s |

*Table 7-1:Parameters chosen for the DLVO-type forces*

Hamaker's constant is always in the range of $10^{-19}$ to $10^{-21}$ J. Minimum particle size should be 10 nm < $a$ <100 nm to be in the colloids size range.

New positions are obtained by using

$$U = u_{\infty} + R^{-1}[F_p + F_e] \tag{7.3}$$

Here, lubrication interaction is essential as there will always be relative motion between particles due to their interparticle distance or surface-surface separation. This chapter involves verifying the forces to obtain equilibrium and applying them to certain cases.

Spring-type forces can also be approximated by doing Taylor's expansion using van der Waals and Born repulsion around the minima.

The final form obtained from the approximation gives the spring constant $k$ and can be given as $k = 2.74A\left(\frac{a}{\sigma}\right)^3$ and due to the presence of Hamaker constant(J), this can also be made non-dimensionless by dividing by $6\pi\eta a^2 U_s$. These derivations are shown in the appendix of this chapter.

### 7.2 Two particle system

Two particles are positioned with gap of 2.5 as the line joining the centers is along x-direction. Particle-1 is fixed and particle-2 experiences/given $u^{\infty}$ in x and z-direction respectively.

The interparticle potential drags the particle-1 along with the second one whereas without any interparticle connection, particle-2 moves far away from the first one.

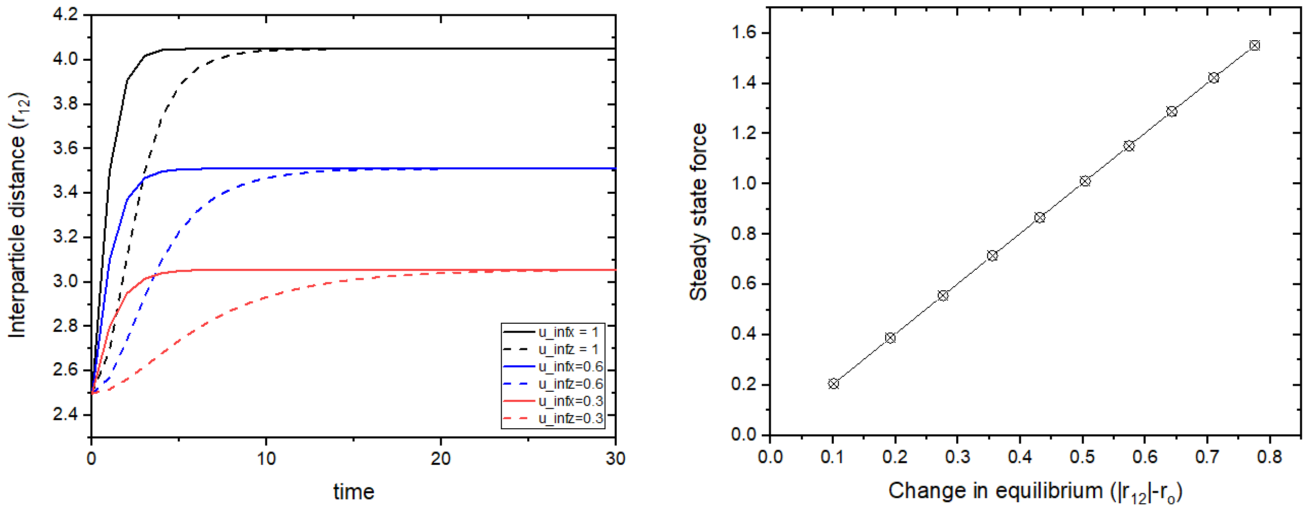



*Figure 7-1:Changes in interparticle distance when one particle is fixed and the other experiencing $u^{\infty}$ in the direction parallel and perpendicular to the line joining the centres(left). Steady state force corresponding to the final state interparticle distances(right)*

Figure 7-1 shows the change in interparticle distance for fluid flow when the line joining the centres is in the horizontal direction ie. *x*-direction. When the flow is in the same direction as the line joining the centres, steady state is reached quickly. Whereas for the other case, when $u^{\infty}$ is in z-direction, it takes more time to reach steady state. However, as the intensity of $u^{\infty}$is increased, time taken to reach steady state decreases and the gap between the two increases. The steady state force corresponding to new interparticle distances are shown in figure, where the relation $F = k(|r_{12}| - r_o)$ is satisfied.

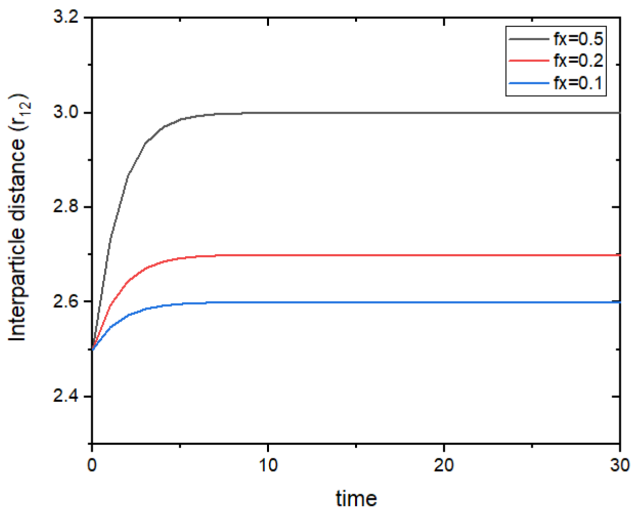



*Figure 7-2:Changes in interparticle distance when one particle is fixed and the other given a force in the direction parallel to the line joining the centres*

Similarly, force is given on the second particle and the corresponding steady interparticle distances are obtained after some time. It increases as the force given is increased.

When $u^{\infty}$ is acting on a particle, the position of that particle is moved by an equivalent distance and forces start acting upon based on the distance between the two. When a force is acting on a particle, hydrodynamics come into picture from t=0. Hence, the distance moved by $u^{\infty}$ is greater than the distance moved by forces.

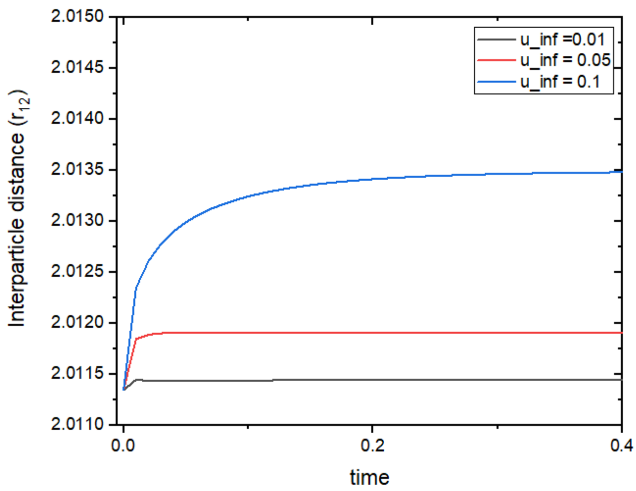



*Figure 7-3:Changes in interparticle distance with time as $u^{\infty}$ is varied as 0.01,0.05 and 0.1 with van der Waals -Born potential*

Particles are placed at an initial equilibrium position of 2.011346 apart for the case of van der Waals and Born repulsion. The van der Waals-Born potential is weak and does not reach a stable state at a critical velocity of 0.11.

Since the effects of $u^{\infty}$ and force is compared, we use the Hooke's law approximation with the spring constant obtained from the DLVO -type forces.

| $u^{\infty}$ | vdW+Born (without lubrication) | vdW+Born (with lubrication) | Hookes' law approximation |
|---|---|---|---|
| 0.01 | 2.011422 | 2.011436 | 2.011433 |
| 0.05 | 2.011797 | 2.011909 | 2.011565 |
| 0.1 | 2.012651 | 2.013492 | 2.012211 |

*Table 7-2:Steady state interparticle distance obtained for when $u^{\infty}$ is applied on particle 2*

The table above shows the difference when lubrication forces are included. With lubrication interactions, the particles move further apart.

## 7.3 Three particle system

Both Hooke's law and DLVO-type forces are applied on a system of three particles arranged at the vertices of the triangle. An equilibrium distance of 2.5 and 2.011346 are for the spring-type forces and DLVO-type forces respectively.

### *7.3.1 Obtaining equilibrium distances*

Three particles are placed at the vertices of a triangle. One of the three particle is placed at a distance slightly away from equilibrium and the interparticle forces are verified without fluid flowing ($u^{\infty} = 0$) or any forces acting. At the end, it is expected that all interparticle distances approach equilibrium.

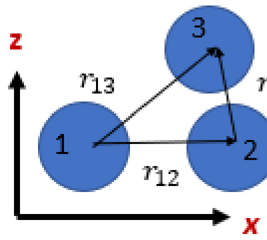



*Figure 7-4:Three particles placed on the vertices of triangle in xz-plane. $r_{12}$ & $r_{13}$ are equal in distance and $r_{23}$ is less compared to that.*

*Case a) Spring-type forces*

Interparticle distances $r_{12} = r_{13}$ =2.5 and $r_{23}$=2.113 with an equilibrium distance of 2.5. Spring constant $k$ for this case is arbitrarily taken as 0.5. This is used just as an indicator for verifying the forces.

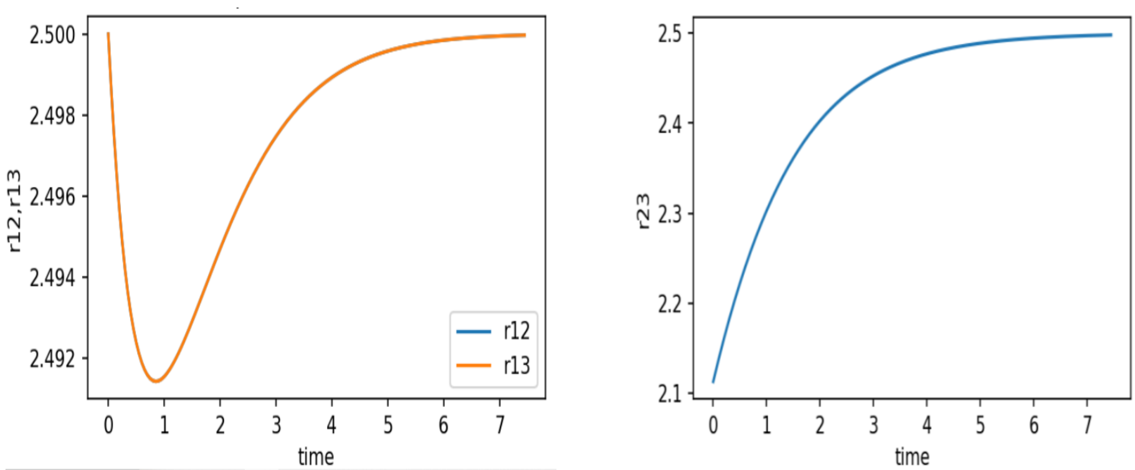



*Figure 7-5:Change in interparticle distance over time till they attain equilibrium with spring-type forces: $r_{12}$&$r_{13}$(left) and $r_{23}$(right)*

One can say that pairs 1-2 and 1-3 come closer before going back to their initial gaps and allowing the other pair 2-3 to reach equilibrium. This could also be applied to gaps of 2.01, but that would be infeasible since particles overlap. This shows that repulsive forces are needed to prevent overlapping of particles.

Case *b) van der Waals and Born repulsion*

$r_{12} = r_{13} = 2.0113$ and $r_{23} = 2.1$ with an equilibrium distance of 2.0113

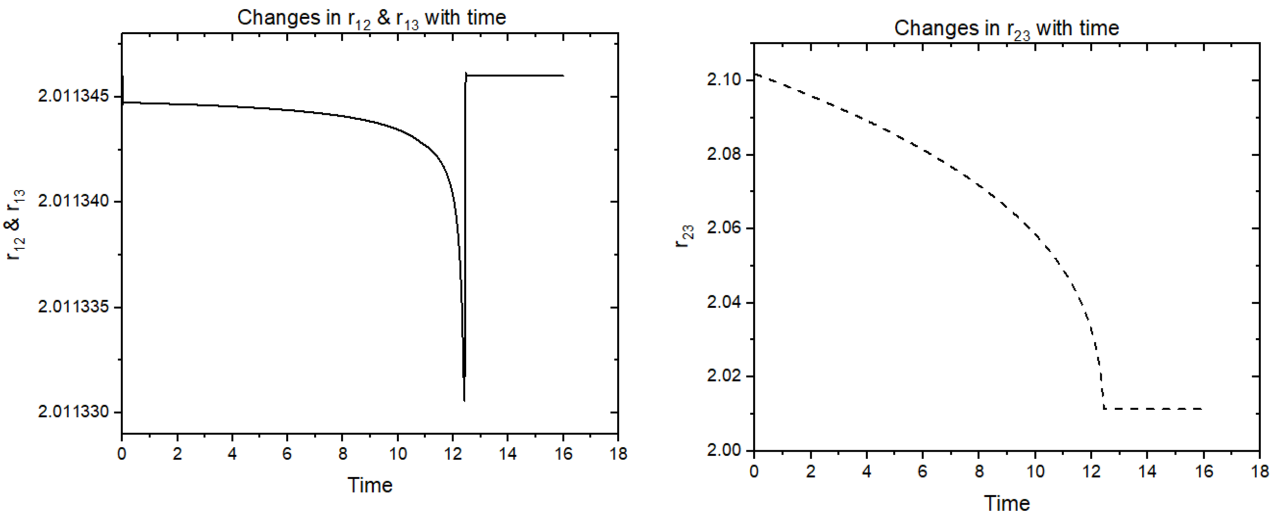



*Figure 7-6:Change in interparticle distance over time till they attain equilibrium with DLVO-type forces:* $r_{12}$&$r_{13}$*(left) and* $r_{23}$*(right)*

## *7.3.2 Particle given a force*

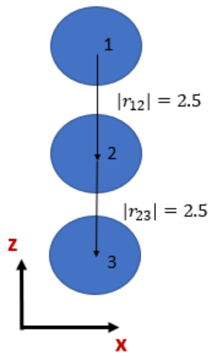



*Figure 7-7:Linear chain of spheres with* $r = 2.5$ *arranged in vertical direction on xz-plane*

Force is given on one of the particles and the resulting effects are observed.

i. For the case of particles arranged on vertices of a triangle, a force is given on particle-2 in the upward z-direction and the steady state pair-pair distances are obtained for the case of spring-type forces.

ii. Force is given on particle-3 in downward z-direction for the linear chain of spheres and the steady state pair-pair distances are obtained for the case of spring-type forces.

The role of spring forces in these two cases are determined.

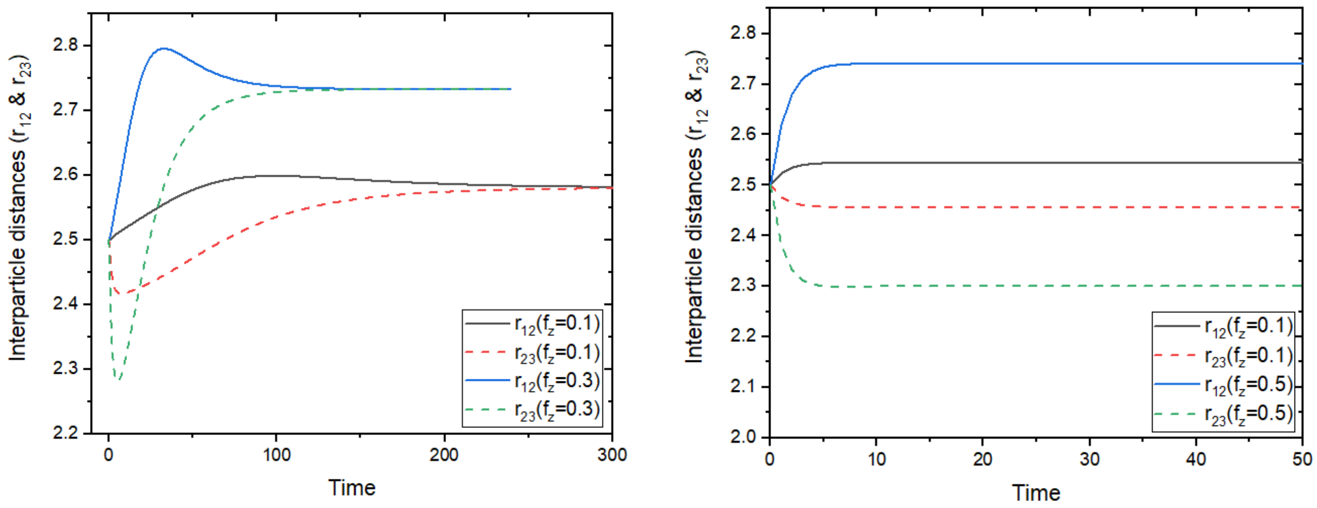



*Figure 7-8:Pair-pair distances observed for case-i: Particles on vertices of triangle with pair 1-2 and 1-3(straight lines) and pair 2-3(dashed line)[left] and case ii. Particle arranged in vertical direction with pair 1-2(straight lines) and pair 2-3(dashed line)[right]*

i. Particle-2 goes in upward z-direction moving closer to particle-3 and away from particle-1. However, these particles rearrange such that particle-2 is at the apex of the triangular form and the other two are pulled in the same direction with the spring forces. As force given is increased, the forced particle goes further close and further away with particle-3 and 1 respectively. Also, steady state is reached quickly when the force is large.

ii. Particle-3 goes in downward z-direction pulling particle-2 closer and increases the gap of pair 1-2. Similar to the above case, the forced particle moves closer to particle-2 and creates further gap to pair 1-2.

When DLVO-type interaction potential is there, the system does not reach steady state or one needs to work with very small timesteps.

If the three particles are considered free (no interaction potential), the particle which is given a force moves far away from the others. Hence various interaction potentials would be needed to model for deformable aggregates.

## 7.4 Conclusion

Critical velocity at which the DLVO-type interaction becomes weak was observed for the case of two particles. Shear flow was also given and tested for the case of 2 and even 5 spheres in a linear chain resulting in a rigid body rotation with the parameters playing a greater role in interparticle distance. Shear flow could be given for the aggregates account for the breakage and restructuring of aggregates as studied by Lattuada et.al (2011). More work is needed on accounting interaction potentials for the case of deformable aggregates.

**Appendix 7A: van der Waals – Born potential**

Potential involved can be given by

$$V(h) = \frac{A}{12}\left\{\frac{a}{h} - \frac{a\,\sigma^6}{210\,h^7}\right\} \qquad \text{A.1}$$

$h = inter - monomer\ distance :\ surface\ to\ surface$ ;

$A = Hamakar\ constant;$

$a = monomer\ radius;$

First term indicates the contribution due to van der Waals and the second term is the repulsion term.

$$\text{Force } F(h) = -\frac{\partial V}{\partial h} = -\frac{Aa}{12h^2} + \frac{Aa}{360}\frac{\sigma^6}{h^8} \qquad \text{A.2}$$

Surface-surface separation,$h$ has been made non-dimensional by dividing by monomer radius, $a$. Similarly, Hamaker's constant can be non-dimensionalised by either $k_B T$ or $6\pi\eta aU$. Since we do not consider Brownian forces, we use $6\pi\eta aU$.

The non-dimensional form for force

$$F = \frac{\left[-\frac{A}{12ah^2} + \frac{A}{360a}\left(\frac{\sigma}{a}\right)^6 \frac{1}{h^8}\right]}{6\pi\eta aU_s} \qquad \text{A.3}$$

Potential can be written in the following term

$$V(h) = \frac{A}{12}\left\{\frac{1}{h} - \frac{1}{210\,h^7}\left(\frac{\sigma}{a}\right)^6\right\} \qquad \text{A.4}$$

Minima of this potential occurs when

$$h_o = 0.5673\,\frac{\sigma}{a} \qquad \text{A.5}$$

which is obtained with $V(h) = 0$.

*Harmonic potential approximation*

Taylor expansion of () about the minima $h_o$ yields

$$V(h) \sim V(h = h_o) + (h - h_o)\, V'(h)|_{h=h_o} + \frac{(h - h_o)^2}{2}\; V''(h)|_{h=h_o} + \cdots$$

$$V(h) \sim 0.13\,\frac{Aa}{\sigma} + \frac{(h - h_o)^2}{2}\left(-2.74\, A\,\left(\frac{a}{\sigma}\right)^3\right)$$

Resulting force

$$F(h) = -\frac{\partial V}{\partial h} = 2.74\, A\left(\frac{a}{\sigma}\right)^3 (h - h_o) \qquad \text{A.6}$$

$$k_i = spring\ constant = 2.74\, A\left(\frac{a}{\sigma}\right)^3$$

This can be assumed equal for identical monomers.

# Chapter 8

# Conclusion

The dependence of hydrodynamic properties on the fractal dimension and the size of the cluster have been determined and compared. This study confirms the linear relation between the hydrodynamic ratio and the fractal dimension. We obtained a maximum of 1.29 for spherical aggregates as expected and the ratio observed for DLA is the same observed by other researchers. Also, this ratio is clearly independent of the size of the cluster irrespective of the aggregate type. This independent nature is essential in estimating the drag acting on large structures only by correlations since simulations are difficult for such structures.

It is well known that there is a dependency on the forces acting on particles with the number of spheres they are connected too. This factor seems to apply for the dense clusters where the maximum forces vary based on average coordination number.

Stokesian Dynamics was also essential for obtaining the fluid velocity within the cluster. This gave us another parameter “penetration depth”, which is related to the amount of fluid that moves through the structure. This parameter is found to be a clear function of fractal dimension where it is also related to the hydrodynamic ratio for the case of cluster-cluster aggregates. This study also revealed the connection between the pre-factor with the penetration depth for the case of DLA. The role of pre-factor in the hydrodynamic ratio is often disregarded which needs to be studied thoroughly.

Aggregates lose the rigid connectivity over time as various parameters play a huge role in the breakup or restructuring of such aggregates. Such potentials have been tried for a few test cases. This study can be taken further with the application of continuous shear and determining the critical stress at which the aggregate breaks.

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